Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```python
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7651
Nombre de lignes df clean	7651
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5355
Taille dataset test	2295
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_12h_fft_f1_cph, r_NH3_fft_12h_fft_period1_h, r_NH3_fft_12h_fft_power1, r_NH3_fft_12h_fft_f2_cph, r_NH3_fft_12h_fft_period2_h, r_NH3_fft_12h_fft_power2, r_NH3_fft_12h_fft_f3_cph, r_NH3_fft_12h_fft_period3_h, r_NH3_fft_12h_fft_power3, r_temperature_fft_12h_fft_f1_cph, r_temperature_fft_12h_fft_period1_h, r_temperature_fft_12h_fft_power1, r_temperature_fft_12h_fft_f2_cph, r_temperature_fft_12h_fft_period2_h, r_temperature_fft_12h_fft_power2, r_temperature_fft_12h_fft_f3_cph, r_temperature_fft_12h_fft_period3_h, r_temperature_fft_12h_fft_power3, r_umidity_fft_12h_fft_f1_cph, r_umidity_fft_12h_fft_period1_h, r_umidity_fft_12h_fft_power1, r_umidity_fft_12h_fft_f2_cph, r_umidity_fft_12h_fft_period2_h, r_umidity_fft_12h_fft_power2, r_umidity_fft_12h_fft_f3_cph, r_umidity_fft_12h_fft_period3_h, r_umidity_fft_12h_fft_power3

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

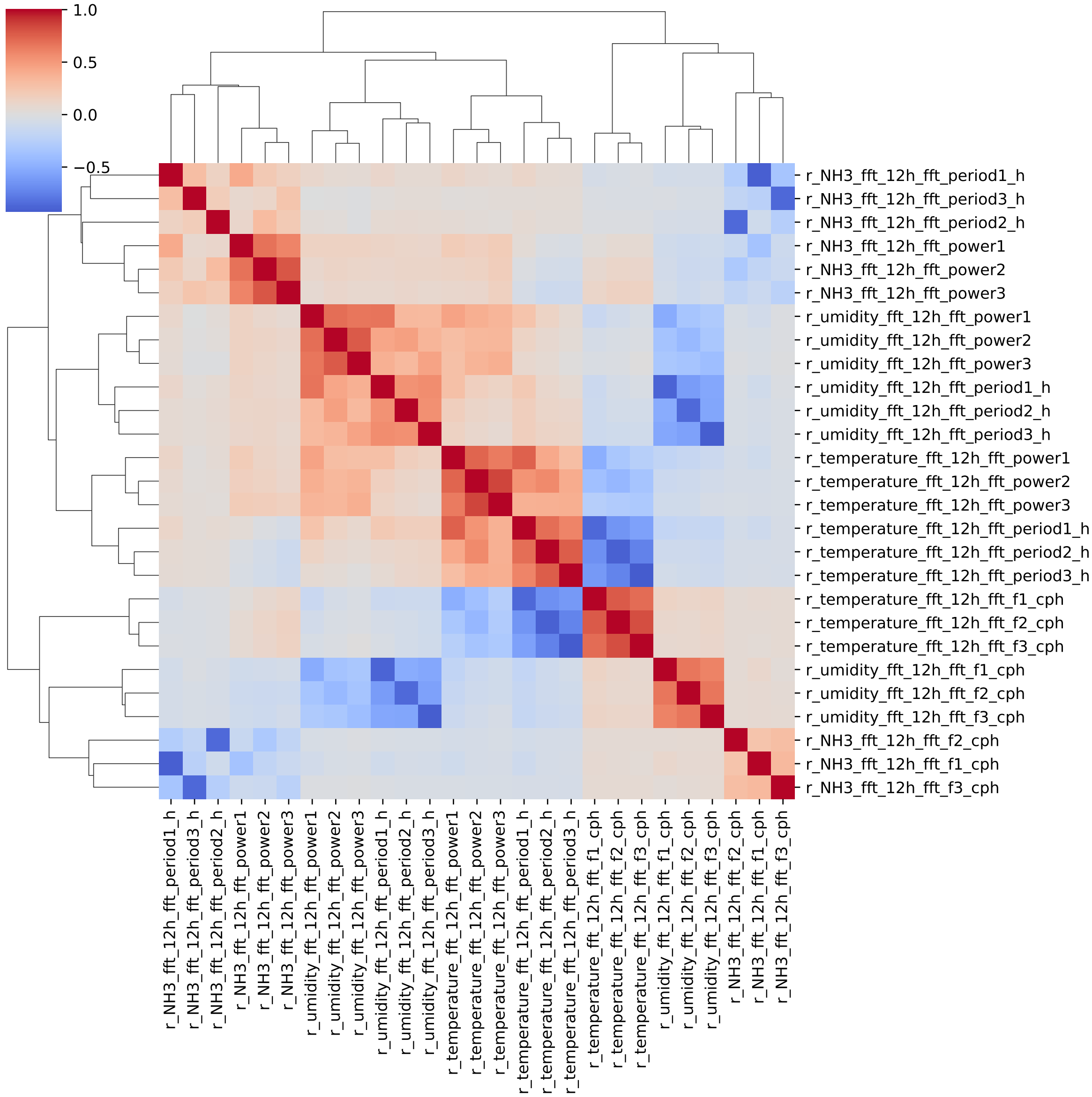
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, *, N, H, 3 ; **cluster 2:** r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ; **cluster 4:** r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_fft_12h_fft_f1_cph, r_NH3_fft_12h_fft_period1_h ; **cluster 7:** r_NH3_fft_12h_fft_power1 ; **cluster 8:** r_NH3_fft_12h_fft_period2_h, r_NH3_fft_12h_fft_f2_cph ; **cluster 9:** r_NH3_fft_12h_fft_power2 ; **cluster 10:** r_NH3_fft_12h_fft_f3_cph, r_NH3_fft_12h_fft_period3_h ; **cluster 11:** r_NH3_fft_12h_fft_power3 ; **cluster 12:** r_temperature_fft_12h_fft_period1_h, r_temperature_fft_12h_fft_f1_cph ; **cluster 13:** r_temperature_fft_12h_fft_power1 ; **cluster 14:** r_temperature_fft_12h_fft_f3_cph, r_temperature_fft_12h_fft_f2_cph, r_temperature_fft_12h_fft_period3_h, r_temperature_fft_12h_fft_period2_h ; **cluster 15:** r_temperature_fft_12h_fft_power3, r_temperature_fft_12h_fft_power2 ; **cluster 16:** r_umidity_fft_12h_fft_period1_h, r_umidity_fft_12h_fft_f1_cph ; **cluster 17:** r_umidity_fft_12h_fft_power1 ; **cluster 18:** r_umidity_fft_12h_fft_f2_cph, r_umidity_fft_12h_fft_period2_h ; **cluster 19:** r_umidity_fft_12h_fft_power2 ; **cluster 20:** r_umidity_fft_12h_fft_f3_cph, r_umidity_fft_12h_fft_period3_h ; **cluster 21:** r_umidity_fft_12h_fft_power3 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_12h_fft_f1_cph, r_NH3_fft_12h_fft_power1, r_NH3_fft_12h_fft_period2_h, r_NH3_fft_12h_fft_power2, r_NH3_fft_12h_fft_f3_cph, r_NH3_fft_12h_fft_power3, r_temperature_fft_12h_fft_period1_h, r_temperature_fft_12h_fft_power1, r_temperature_fft_12h_fft_f3_cph, r_temperature_fft_12h_fft_power3, r_umidity_fft_12h_fft_period1_h, r_umidity_fft_12h_fft_power1, r_umidity_fft_12h_fft_f2_cph, r_umidity_fft_12h_fft_power2, r_umidity_fft_12h_fft_f3_cph, r_umidity_fft_12h_fft_power3



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_12h_fft_f1_cph, r_NH3_fft_12h_fft_power1, r_NH3_fft_12h_fft_period2_h, r_NH3_fft_12h_fft_power2, r_NH3__fft__12h__fft__f3__cph, r_NH3__fft__12h__fft__power3, r_temperature__fft__12h__fft__period1__h, r_temperature__fft__12h__fft__power1, r_temperature__fft__12h__fft__f3__cph, r_temperature__fft__12h__fft__power3, r_umidity__fft__12h__fft__period1__h, r_umidity__fft__12h__fft__power1, r_umidity__fft__12h__fft__f2__cph, r_umidity__fft__12h__fft__power2, r_umidity__fft__12h__fft__f3__cph, r_umidity__fft__12h__fft__power3

Len(Features_X): 21

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

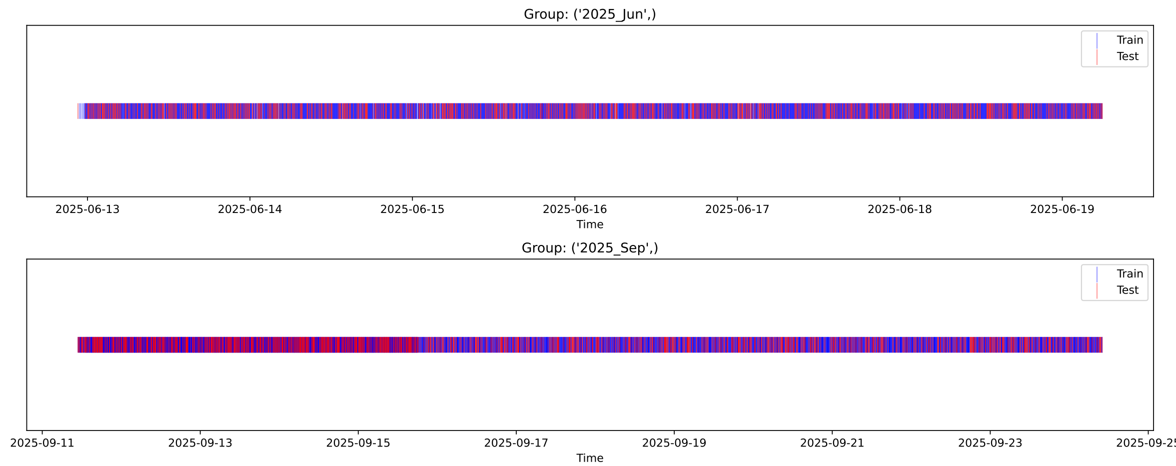
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5355 samples Test: 2296 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3748 samples Test: 1607 samples Excluded: 0 samples Window exclusion: 0h

Fold 2/5

Train: 3748 samples Test: 1607 samples Excluded: 0 samples Window exclusion: 0h

Fold 3/5

Train: 3748 samples Test: 1607 samples Excluded: 0 samples Window exclusion: 0h

Fold 4/5

Train: 3748 samples Test: 1607 samples Excluded: 0 samples Window exclusion: 0h

Fold 5/5

Train: 3748 samples Test: 1607 samples Excluded: 0 samples Window exclusion: 0h

3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.343, std=0.815, min=0.959, max=13.845 y_test: mean=2.357, std=0.767, min=0.973, max=7.277

Fold 0 — y_test: mean=2.32, std=0.77, y_train: mean=2.35, std=0.83

Fold 1 — y_test: mean=2.33, std=0.78, y_train: mean=2.35, std=0.83

Fold 2 — y_test: mean=2.36, std=0.78, y_train: mean=2.34, std=0.83

Fold 3 — y_test: mean=2.37, std=0.86, y_train: mean=2.33, std=0.79

Fold 4 — y_test: mean=2.35, std=0.86, y_train: mean=2.34, std=0.79

train scores CV: [0.40734537 0.39277145 0.39402232 0.41618383 0.42569127]

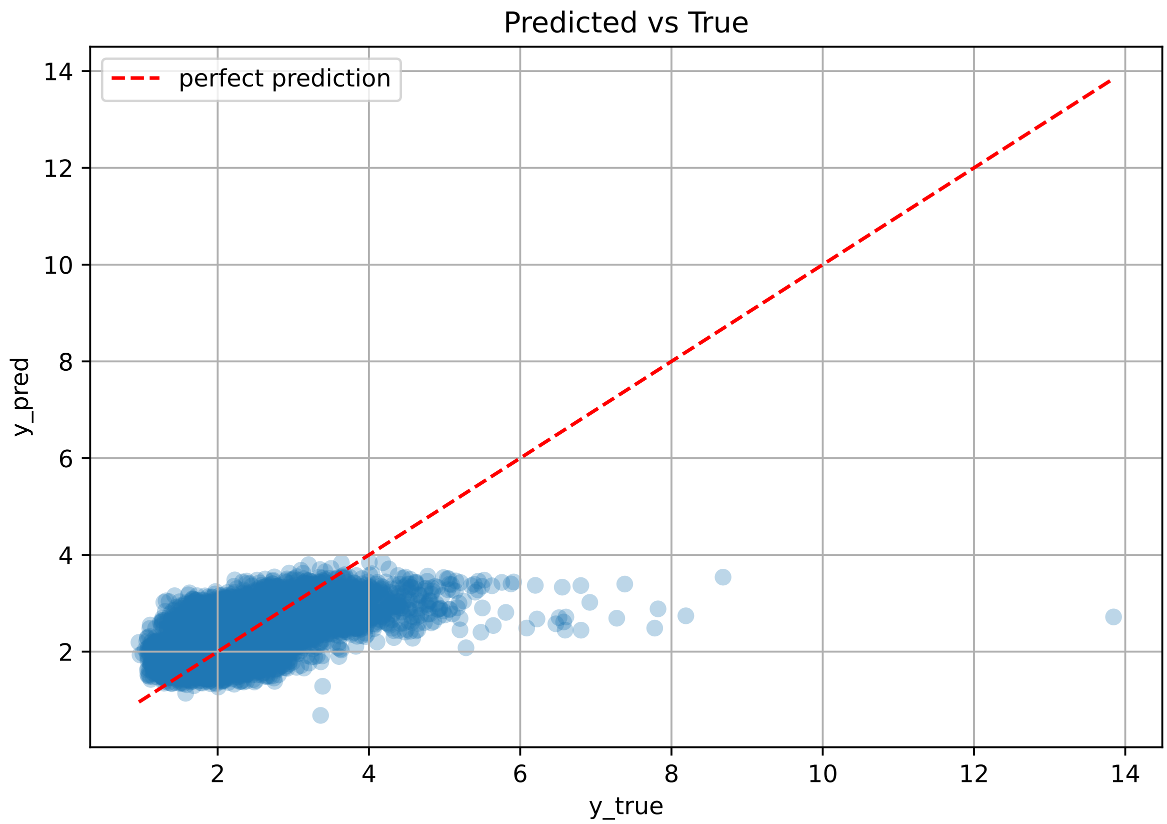
test scores CV: [0.39923042 0.44349918 0.43847217 0.3874125 0.36943834]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4072	0.4076	0.0289	0.4092	0.4105	0.0029	1.0071	0.999

Coefficients:

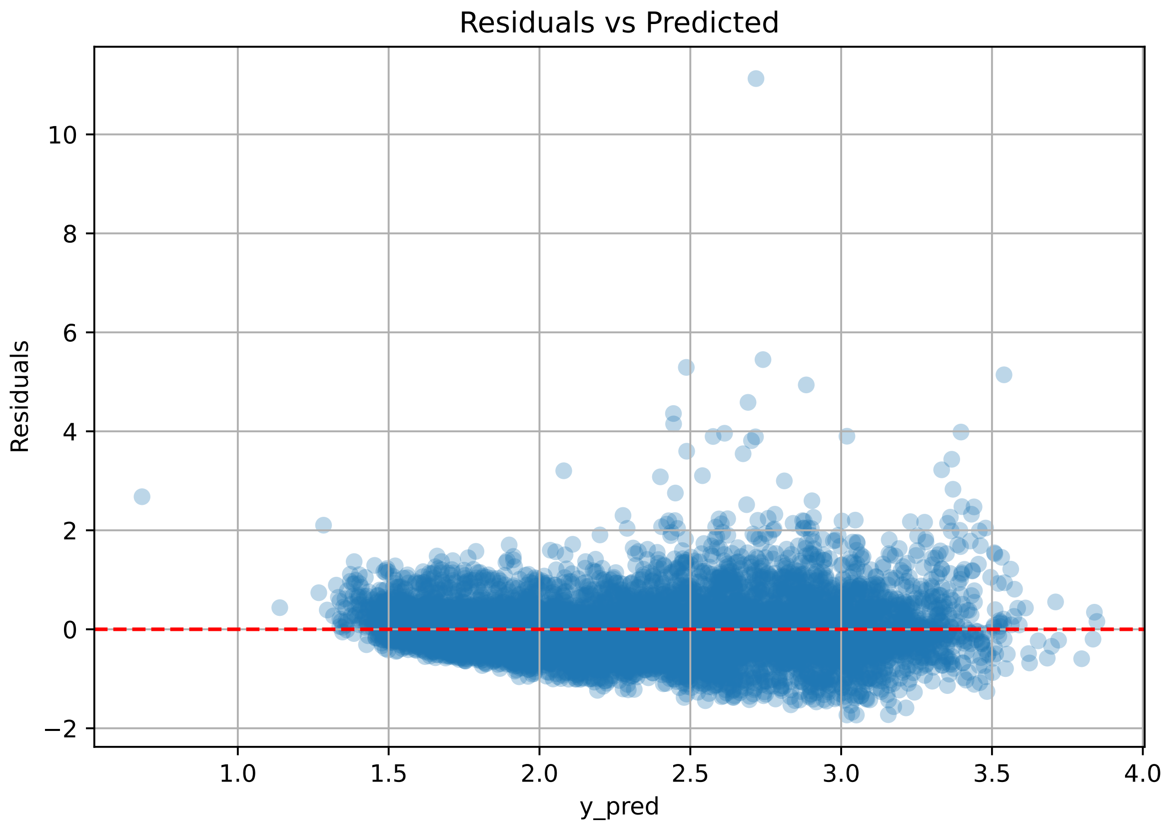
coef_r_CO2 = -0.0362, coef_r_NH3 = -0.0515, coef_r_NH3_fft_12h_fft_f1_cph = 0.0180, coef_r_NH3_fft_12h_fft_f3_cph = -0.0225, coef_r_NH3_fft_12h_fft_period2_h = -0.0337, coef_r_NH3_fft_12h_fft_power1 = 0.0577, coef_r_NH3_fft_12h_fft_power2 = 0.0167, coef_r_NH3_fft_12h_fft_power3 = 0.0148, coef_r_THI = 0.2697, coef_r_temperature = -0.3198, coef_r_temperature_fft_12h_fft_f3_cph = -0.0572, coef_r_temperature_fft_12h_fft_period1_h = 0.0474, coef_r_temperature_fft_12h_fft_power1 = -0.0607, coef_r_temperature_fft_12h_fft_power3 = -0.1260, coef_r_umidity = -0.5914, coef_r_umidity_fft_12h_fft_f2_cph = 0.0045, coef_r_umidity_fft_12h_fft_f3_cph = 0.0320, coef_r_umidity_fft_12h_fft_period1_h = 0.0049, coef_r_umidity_fft_12h_fft_power1 = -0.0422, coef_r_umidity_fft_12h_fft_power2 = -0.0246, coef_r_umidity_fft_12h_fft_power3 = 0.0040, intercept = 2.3431,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

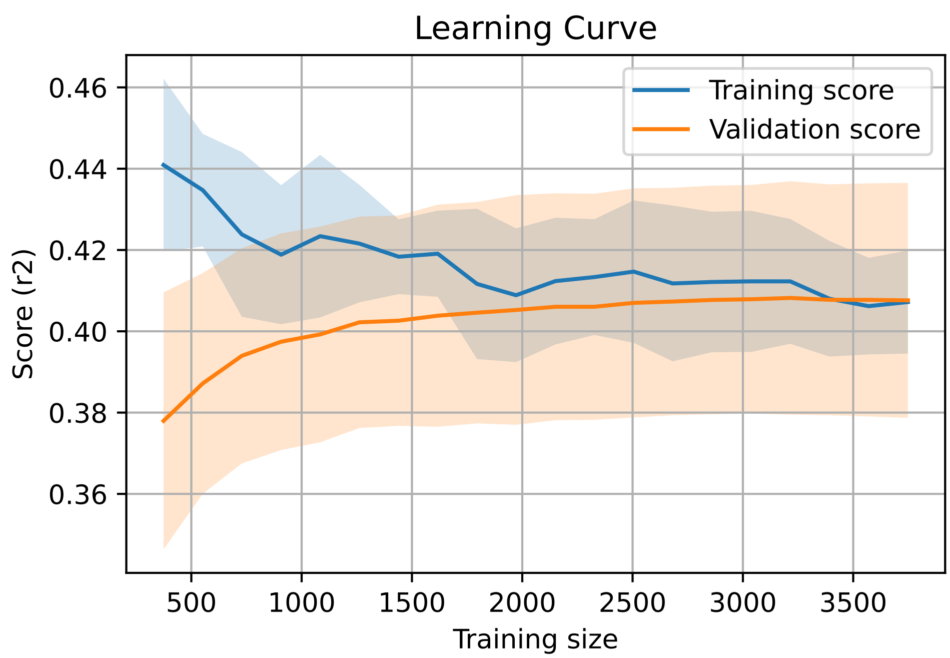
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7651, **Len(training)** : 5356,

Len(training_real) : 3750, **Len(test_of_training)** : 1606, **Len(test)** : 2295

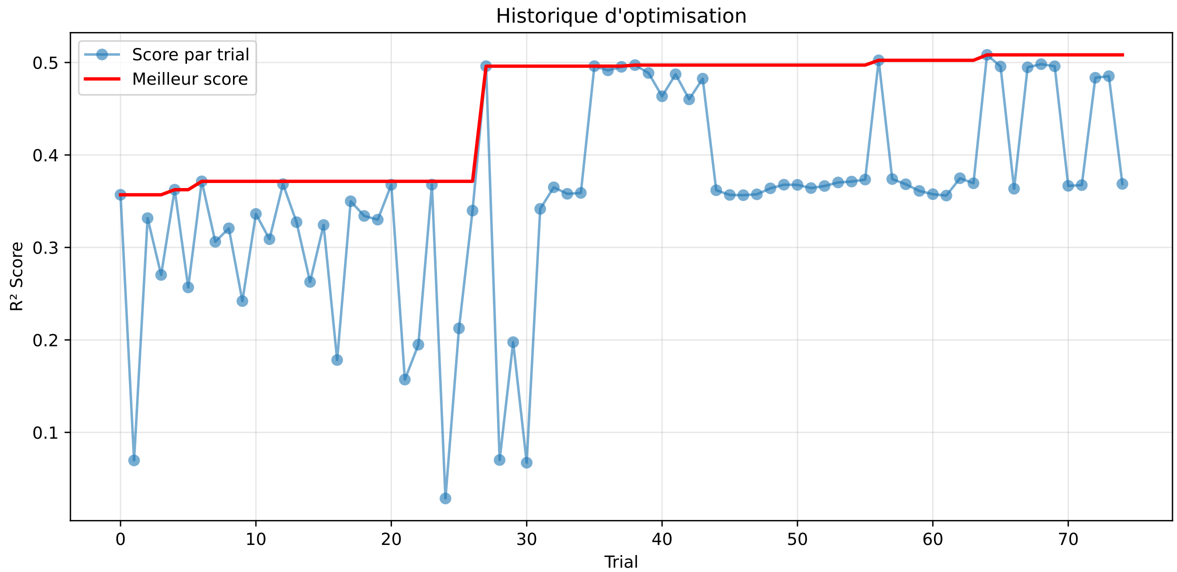


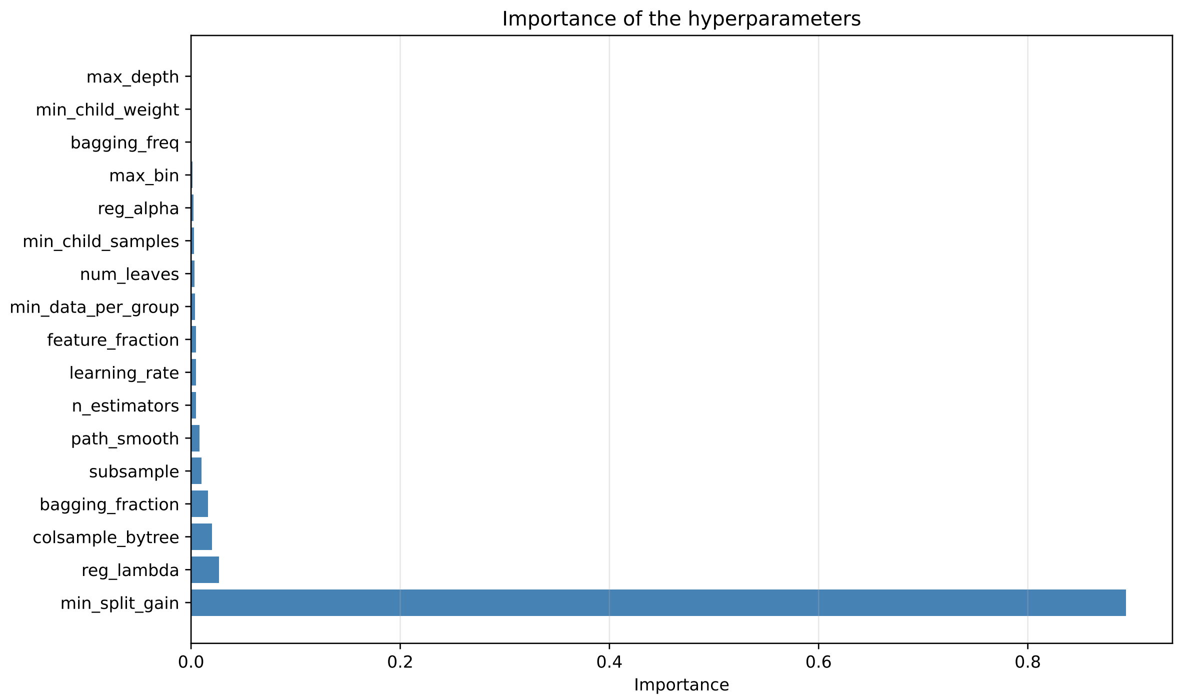
3.5 Model : LGBM

3.5.1 Gridsearch

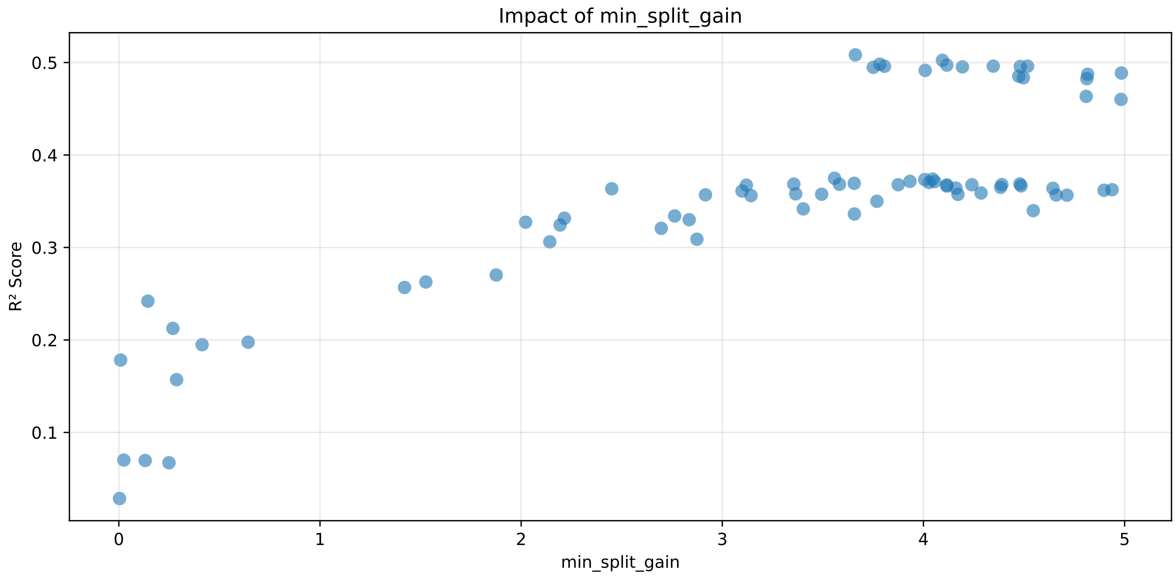
Distribution y_train vs y_test: y_train: mean=2.343, std=0.815, min=0.959, max=13.845 y_test: mean=2.357, std=0.767, min=0.973, max=7.277

==================== TOP Importance FEATURES ==================== feature importance r_umidity 54 r_temperature_fft_12h_fft_power3 41 r_CO2 37 r_NH3 15 r_NH3_fft_12h_fft_power3 14 r_NH3_fft_12h_fft_power1 11 r_umidity_fft_12h_fft_power1 11 r_temperature 10 r_THI 8 r_umidity_fft_12h_fft_power2 6 r_umidity_fft_12h_fft_power3 5 r_temperature_fft_12h_fft_period1_h 4 r_umidity_fft_12h_fft_f2_cph 4 r_temperature_fft_12h_fft_f3_cph 3 r_umidity_fft_12h_fft_period1_h 3 r_temperature_fft_12h_fft_power1 2 r_NH3_fft_12h_fft_power2 1 r_umidity_fft_12h_fft_f3_cph 1 r_NH3_fft_12h_fft_f1_cph 0 r_NH3_fft_12h_fft_period2_h 0 r_NH3_fft_12h_fft_f3_cph 0





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5333	0.5083	0.5376	0.0297	0.5083	0.5554	0.0471	1.0927	1.0493
2	0.5271	0.5024	0.5288	0.03	0.5024	0.5546	0.0522	1.1039	1.0492
3	0.5226	0.4982	0.5253	0.0339	0.4982	0.5494	0.0512	1.1028	1.0489
4	0.522	0.4972	0.526	0.0335	0.4972	0.547	0.0498	1.1002	1.0499
5	0.5186	0.4961	0.5249	0.0365	0.4961	0.5452	0.0491	1.099	1.0454
6	0.5201	0.496	0.5247	0.0326	0.496	0.5534	0.0573	1.1156	1.0486
7	0.5193	0.496	0.5247	0.0288	0.496	0.5501	0.0541	1.1091	1.0471
8	0.5182	0.4957	0.5239	0.0309	0.4957	0.5466	0.0509	1.1026	1.0454
9	0.5172	0.4953	0.5221	0.0357	0.4953	0.5511	0.0558	1.1127	1.0442
10	0.5187	0.4948	0.5217	0.0333	0.4948	0.545	0.0502	1.1014	1.0483
---	---	---	---	---	---	---	---	---	---
75	0.7524	0.6317	0.6668	0.0369	0.0286	0.6877	0.056	1.0886	1.191

Hyperparameters:

Rank 1: {'reg___n_estimators': 200, 'reg___max_depth': 8, 'reg___learning_rate': 0.0968280616748841, 'reg___num_leaves': 50, 'reg___subsample': 0.6132445626194459, 'reg___colsample_bytree': 0.6705361180279369, 'reg___min_child_samples': 46, 'reg___reg_alpha': 5.363658193359896, 'reg___reg_lambda': 41.148548212204226, 'reg___min_split_gain': 3.6615845927500907, 'reg___path_smooth': 0.3695405939495684, 'reg___min_child_weight': 0.05797702175383069, 'reg___feature_fraction': 0.8356073034517285, 'reg___bagging_fraction': 0.5575162602024801, 'reg___bagging_freq': 3, 'reg___max_bin': 238, 'reg___min_data_per_group': 149}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.04163622233195188, 'reg__num_leaves': 47, 'reg__subsample': 0.6485409534144699, 'reg__colsample_bytree': 0.5552852258761208, 'reg__min_child_samples': 47, 'reg__reg_alpha': 5.155901382239974, 'reg__reg_lambda': 46.03147236917649, 'reg__min_split_gain': 4.094263773424592, 'reg__path_smooth': 2.6509945502315375, 'reg__min_child_weight': 0.14995212290625165, 'reg__feature_fraction': 0.6837306322732171, 'reg__bagging_fraction': 0.6275611060079134, 'reg__bagging_freq': 6, 'reg__max_bin': 227, 'reg__min_data_per_group': 128}

Rank 3: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.03562628137269187, 'reg__num_leaves': 39, 'reg__subsample': 0.6117017054764973, 'reg__colsample_bytree': 0.5039247281976167, 'reg__min_child_samples': 25, 'reg__reg_alpha': 5.745952300989195, 'reg__reg_lambda': 42.01181545786103, 'reg__min_split_gain': 3.7830522407470903, 'reg__path_smooth': 3.421034252987278, 'reg__min_child_weight': 0.060965582788573454, 'reg__feature_fraction': 0.6550792521989042, 'reg__bagging_fraction': 0.5251861300240253, 'reg__bagging_freq': 7, 'reg__max_bin': 216, 'reg__min_data_per_group': 106}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.09765762540254616, 'reg__num_leaves': 53, 'reg__subsample': 0.6783517596210789, 'reg__colsample_bytree': 0.5900975446595691, 'reg__min_child_samples': 50, 'reg__reg_alpha': 4.468053801811618, 'reg__reg_lambda': 48.787477433030666, 'reg__min_split_gain': 4.1161516520334365, 'reg__path_smooth': 2.972263889788342, 'reg__min_child_weight': 0.2401192839860882, 'reg__feature_fraction': 0.7776101495113402, 'reg__bagging_fraction': 0.5528420552766191, 'reg__bagging_freq': 4, 'reg__max_bin': 247, 'reg__min_data_per_group': 152}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.024729570922073143, 'reg__num_leaves': 47, 'reg__subsample': 0.6541352752371019, 'reg__colsample_bytree': 0.8798840462321068, 'reg__min_child_samples': 48, 'reg__reg_alpha': 4.290957267808499, 'reg__reg_lambda': 48.56486850748776, 'reg__min_split_gain': 4.346893743209744, 'reg__path_smooth': 2.859349118809811, 'reg__min_child_weight': 1.6293697370399753, 'reg__feature_fraction': 0.7496645878239687, 'reg__bagging_fraction': 0.5562397214686136, 'reg__bagging_freq': 6, 'reg__max_bin': 255, 'reg__min_data_per_group': 138}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.024334957162149957, 'reg__num_leaves': 63, 'reg__subsample': 0.6251692815352282, 'reg__colsample_bytree': 0.508337965901674, 'reg__min_child_samples': 38, 'reg__reg_alpha': 3.4818458593354684, 'reg__reg_lambda': 29.685507451734715, 'reg__min_split_gain': 4.5177872103718535, 'reg__path_smooth': 4.84994725932542, 'reg__min_child_weight': 0.78126364834969, 'reg__feature_fraction': 0.784445037551302, 'reg__bagging_fraction': 0.5031973441835443, 'reg__bagging_freq': 5, 'reg__max_bin': 191, 'reg__min_data_per_group': 120}

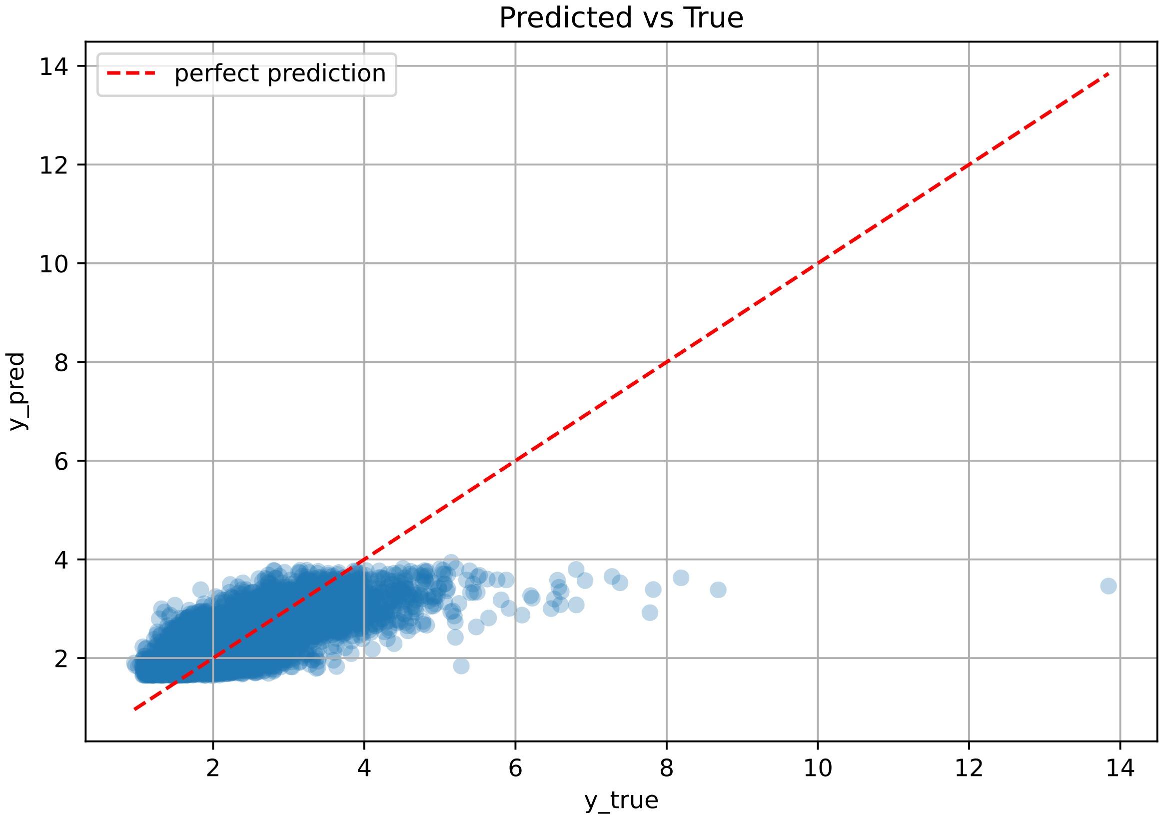
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.06873134015560313, 'reg__num_leaves': 54, 'reg__subsample': 0.6139505209199554, 'reg__colsample_bytree': 0.52614045037997, 'reg__min_child_samples': 45, 'reg__reg_alpha': 5.882659716890354, 'reg__reg_lambda': 41.41538107807264, 'reg__min_split_gain': 3.806306186779696, 'reg__path_smooth': 2.312077763561487, 'reg__min_child_weight': 0.06310141115431006, 'reg__feature_fraction': 0.8043700937409155, 'reg__bagging_fraction': 0.5269287057327372, 'reg__bagging_freq': 7, 'reg__max_bin': 211, 'reg__min_data_per_group': 108}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.0919736003886404, 'reg__num_leaves': 50, 'reg__subsample': 0.6508586818018065, 'reg__colsample_bytree': 0.6823617824652897, 'reg__min_child_samples': 46, 'reg__reg_alpha': 5.1875093004506985, 'reg__reg_lambda': 41.952284238369074, 'reg__min_split_gain': 4.4813569533652124, 'reg__path_smooth': 2.8952009039757614, 'reg__min_child_weight': 0.07055884491344726, 'reg__feature_fraction': 0.8394628181391559, 'reg__bagging_fraction': 0.5587528458624034, 'reg__bagging_freq': 3, 'reg__max_bin': 240, 'reg__min_data_per_group': 146}

Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.029538105238473753, 'reg__num_leaves': 47, 'reg__subsample': 0.6779885323484997, 'reg__colsample_bytree': 0.898524222046622, 'reg__min_child_samples': 49, 'reg__reg_alpha': 4.2465168423567174, 'reg__reg_lambda': 48.55494916841869, 'reg__min_split_gain': 4.193593756491124, 'reg__path_smooth': 2.724960768975515, 'reg__min_child_weight': 0.21218230481184014, 'reg__feature_fraction': 0.6818884190385224, 'reg__bagging_fraction': 0.571781129785453, 'reg__bagging_freq': 6, 'reg__max_bin': 246, 'reg__min_data_per_group': 135}

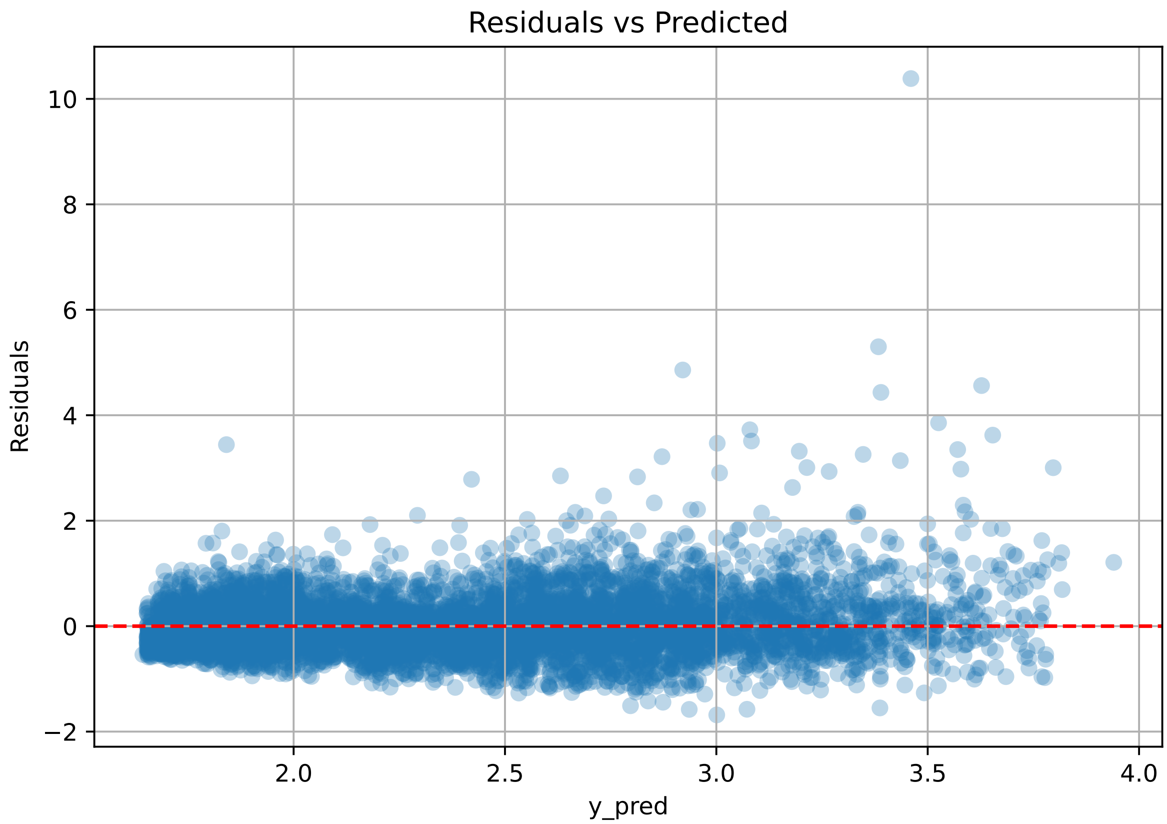
Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.05745137293257136, 'reg__num_leaves': 39, 'reg__subsample': 0.6119210362171392, 'reg__colsample_bytree': 0.5265583106732079, 'reg__min_child_samples': 46, 'reg__reg_alpha': 5.789661141110081, 'reg__reg_lambda': 41.95693861896865, 'reg__min_split_gain': 3.750509471355812, 'reg__path_smooth': 2.835651044449605, 'reg__min_child_weight': 0.06298996587191186, 'reg__feature_fraction': 0.6537593638092922, 'reg__bagging_fraction': 0.528957369214326, 'reg__bagging_freq': 7, 'reg__max_bin': 214, 'reg__min_data_per_group': 102}

3.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

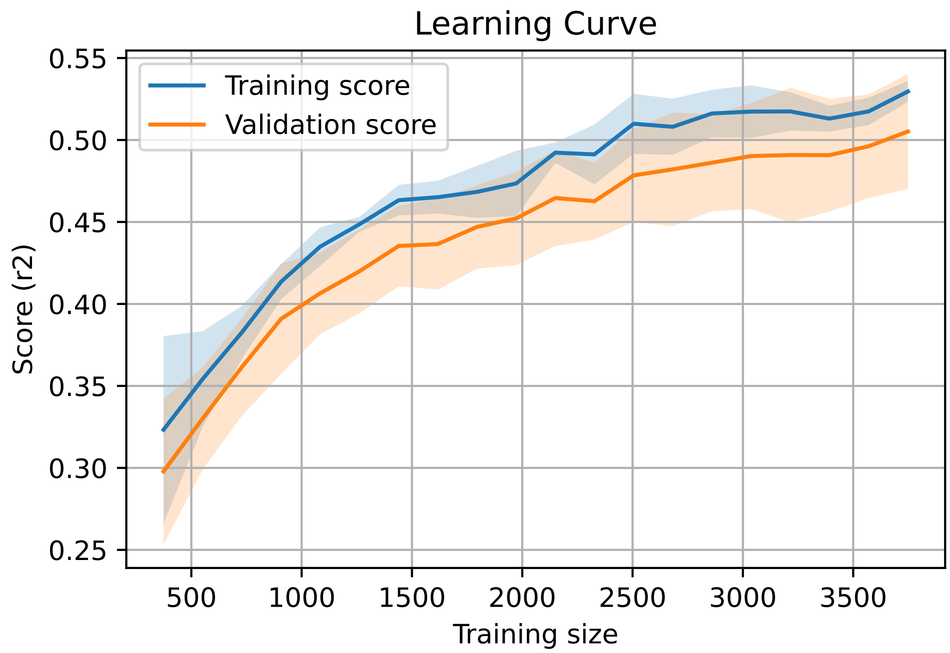
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7651, **Len(training)** : 5356,

Len(training_real) : 3750, **Len(test_of_training)** : 1606, **Len(test)** : 2295

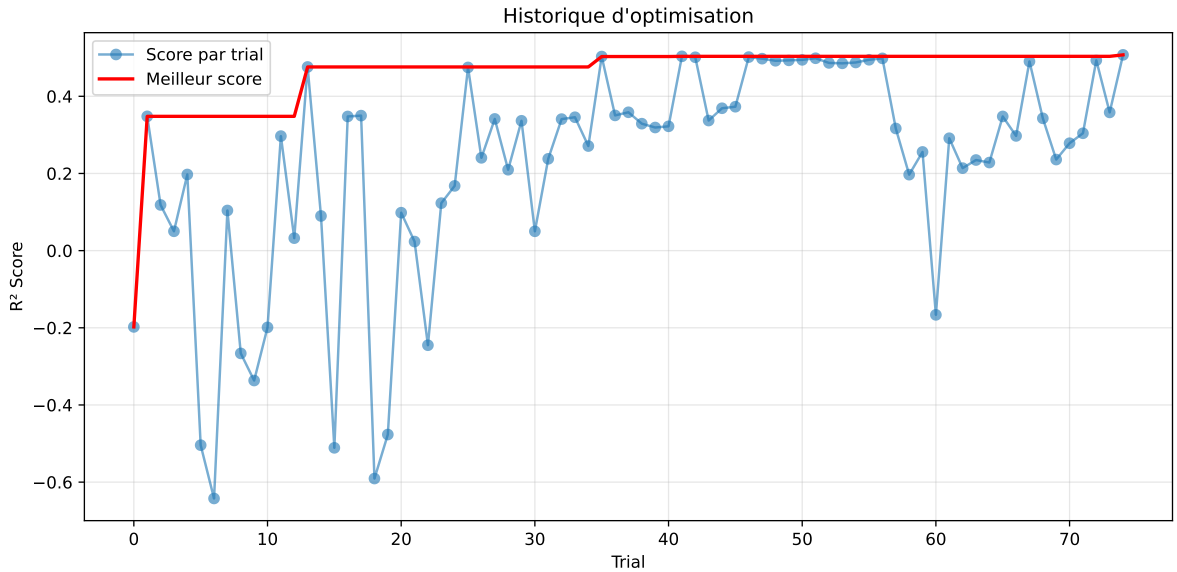


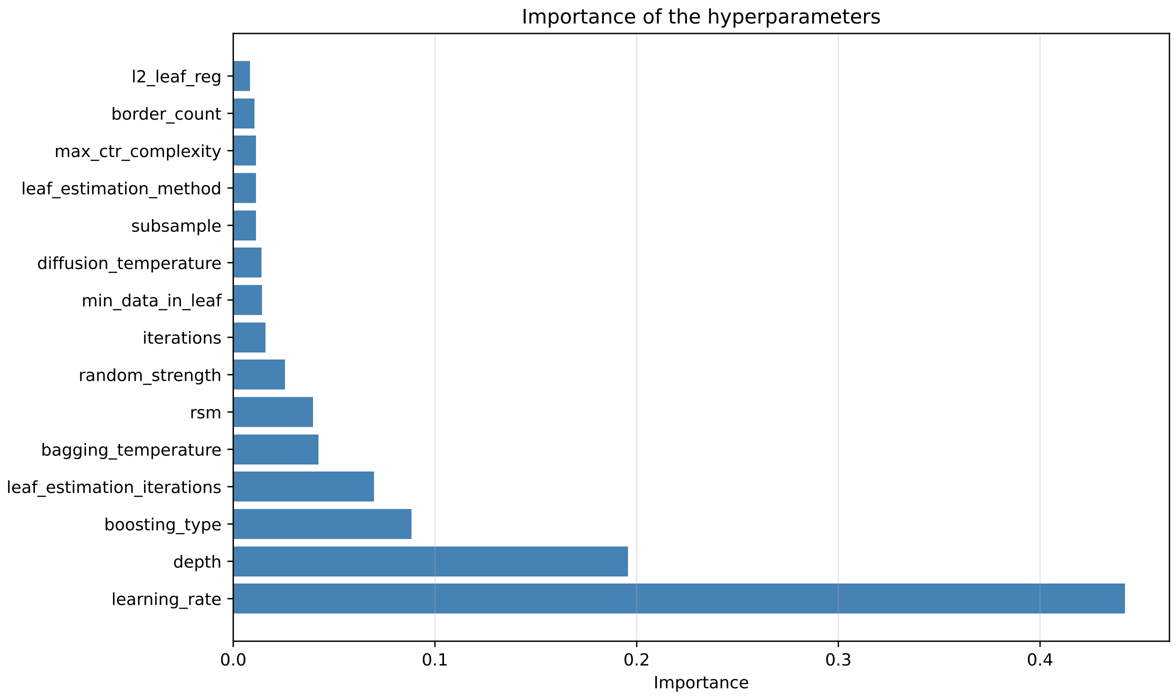
3.6 Model : CatBoost

3.6.1 Gridsearch

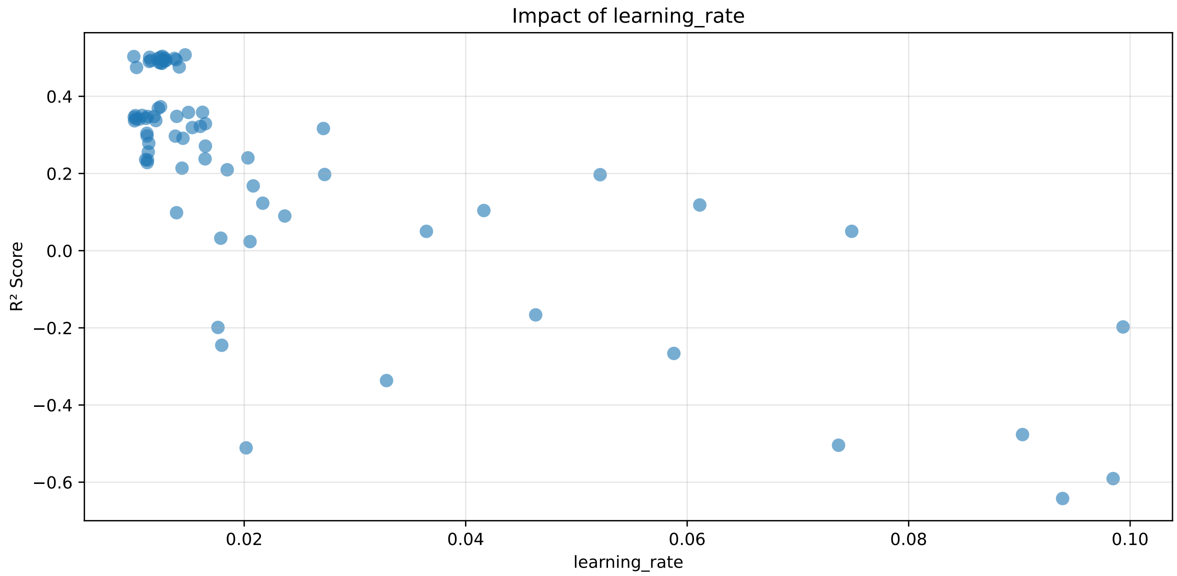
Distribution y_train vs y_test: y_train: mean=2.343, std=0.815, min=0.959, max=13.845 y_test: mean=2.357, std=0.767, min=0.973, max=7.277

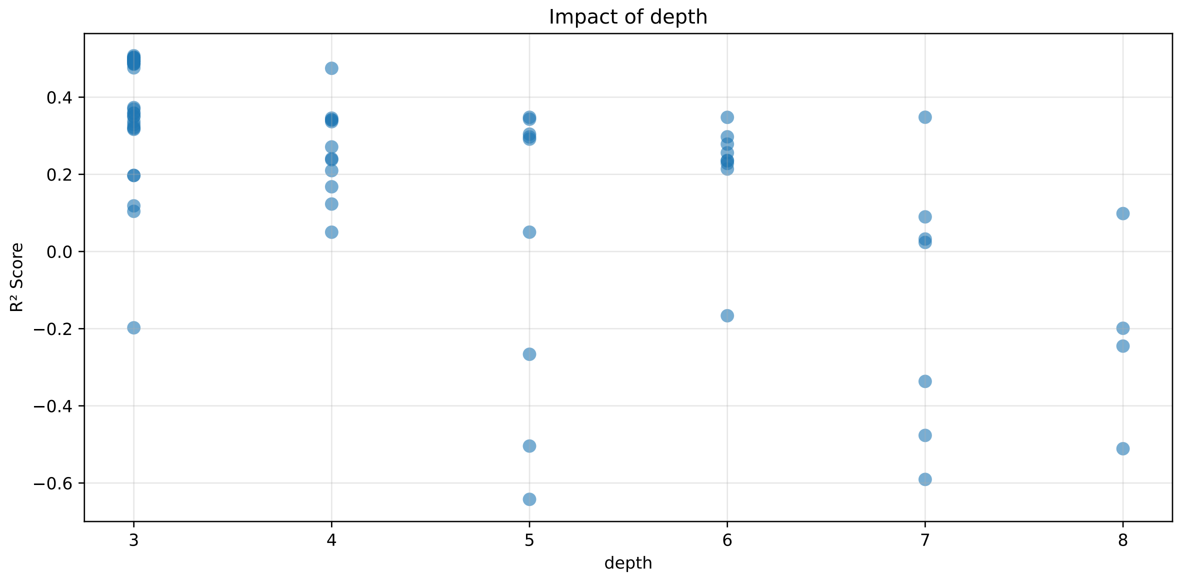
===================== TOP Importance FEATURES ===================== feature importance r_umidity 45.230806 r_temperature 11.856969 r_NH3 9.692330 r_temperature_fft_12h_fft_power3 9.536277 r_CO2 4.611830 r_THI 4.195434 r_umidity_fft_12h_fft_power1 1.985682 r_NH3_fft_12h_fft_power1 1.942192 r_temperature_fft_12h_fft_power1 1.645594 r_NH3_fft_12h_fft_power3 1.602358 r_umidity_fft_12h_fft_power2 1.570133 r_NH3_fft_12h_fft_power2 1.469874 r_umidity_fft_12h_fft_power3 1.293389 r_temperature_fft_12h_fft_period1_h 0.550586 r_umidity_fft_12h_fft_f2_cph 0.522587 r_temperature_fft_12h_fft_f3_cph 0.520228 r_umidity_fft_12h_fft_period1_h 0.478926 r_umidity_fft_12h_fft_f3_cph 0.464209 r_NH3_fft_12h_fft_f1_cph 0.388941 r_NH3_fft_12h_fft_period2_h 0.271114 r_NH3_fft_12h_fft_f3_cph 0.170540

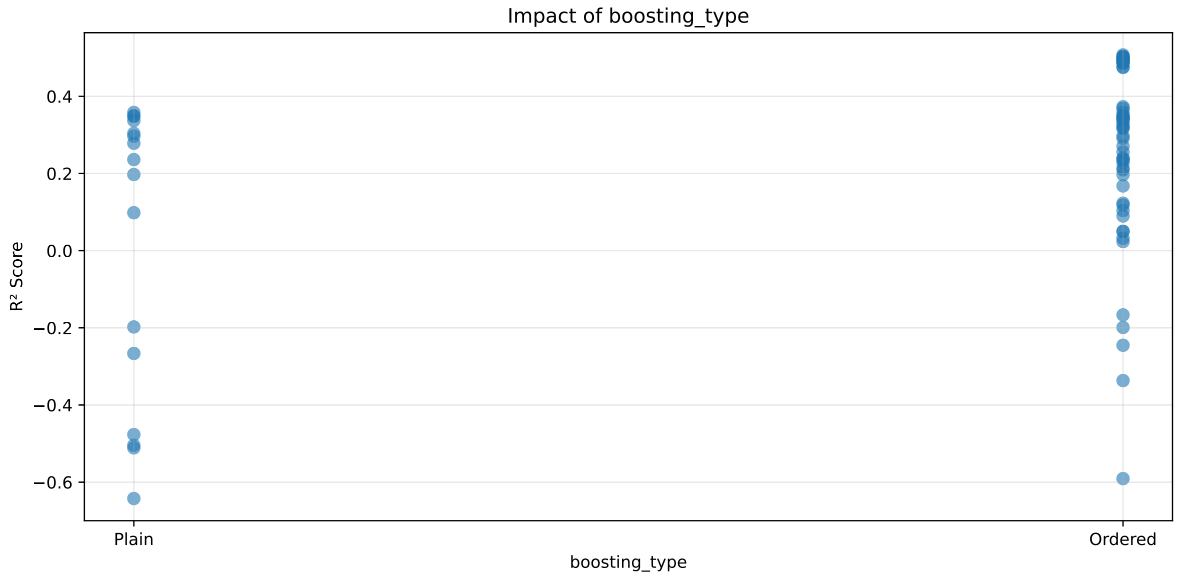


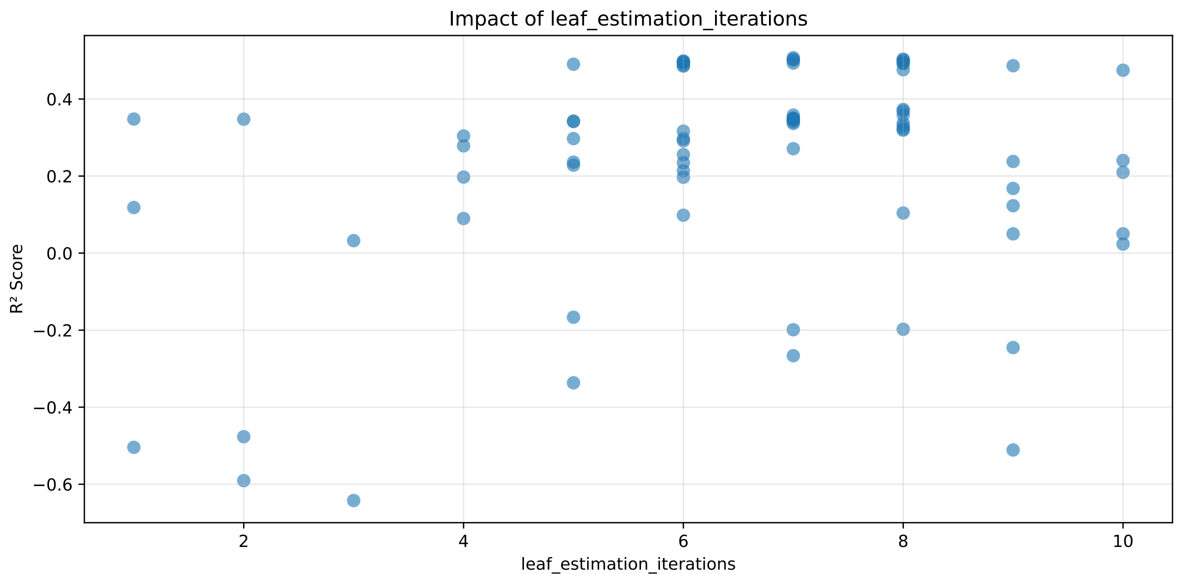


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'boosting_type', 'leaf_estimation_iterations']


Impact of learning_rate
R² Score
learning_rate
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.02
0.04
0.06
0.08
0.10

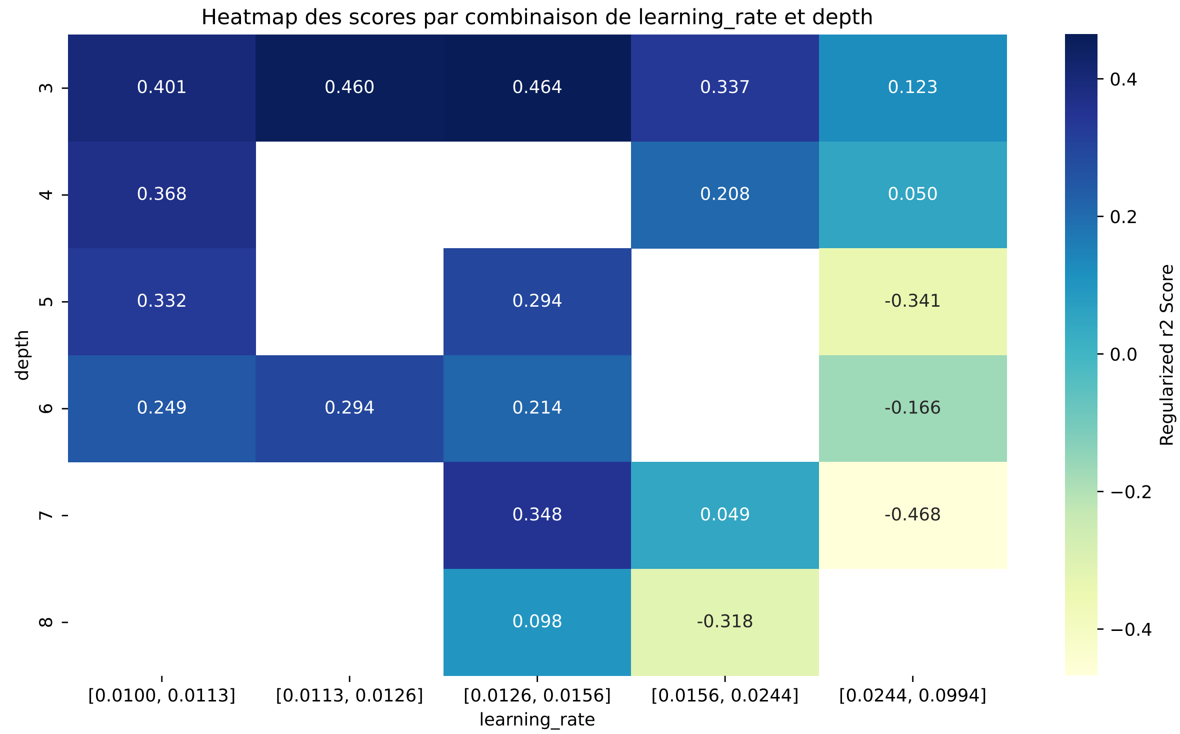

Impact of depth
R² Score
depth
0.4
0.2
0.0
−0.2
−0.4
−0.6
3
4
5
6
7
8






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5317	0.5074	0.5365	0.032	0.5074	0.5473	0.0399	1.0787	1.0478
2	0.5265	0.5036	0.5329	0.0313	0.5036	0.5446	0.041	1.0815	1.0456
3	0.5246	0.5031	0.5329	0.0324	0.5031	0.5423	0.0392	1.078	1.0429
4	0.5236	0.5016	0.5298	0.0324	0.5016	0.544	0.0424	1.0844	1.0438
5	0.5246	0.5012	0.5323	0.0312	0.5012	0.5449	0.0437	1.0872	1.0468
6	0.5167	0.4986	0.529	0.0325	0.4986	0.5352	0.0366	1.0734	1.0362
7	0.5193	0.4983	0.5291	0.0314	0.4983	0.5405	0.0422	1.0847	1.0421
8	0.5175	0.4975	0.526	0.0319	0.4975	0.5358	0.0383	1.077	1.0401
9	0.5129	0.4947	0.5224	0.031	0.4947	0.5315	0.0369	1.0745	1.0368
10	0.5116	0.4945	0.5236	0.0322	0.4945	0.5299	0.0355	1.0717	1.0347
---	---	---	---	---	---	---	---	---	---
75	0.9536	0.6876	0.7302	0.0398	-0.6423	0.754	0.0664	1.0966	1.3868

Hyperparameters:

Rank 1: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.014667442911812486, ’reg__l2_leaf_reg’: 30.907825670262497, ’reg__bagging_temperature’: 1.8871627479191577, ’reg__random_strength’: 3.9794695863127947, ’reg__subsample’: 0.8789237918916646, ’reg__min_data_in_leaf’: 25, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.7140402569951348, ’reg__border_count’: 52, ’reg__leaf_estimation_method’: ‘Newton’, 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 3121.1135414618807}

Rank 2: {'reg___iterations': 600, 'reg___depth': 3, 'reg___learning_rate': 0.012625313964879132, 'reg___l2_leaf_reg': 22.921521541709193, 'reg___bagging_temperature': 3.1437912933006706, 'reg___random_strength': 2.797609819921191, 'reg___subsample': 0.8660156670356318, 'reg___min_data_in_leaf': 36, 'reg___leaf_estimation_iterations': 8, 'reg___rsm': 0.6672420207216168, 'reg___border_count': 45, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 1242.9646086081877}

Rank 3: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.01003428417549171, 'reg___l2_leaf_reg': 22.63114832967421, 'reg___bagging_temperature': 3.466735631346225, 'reg___random_strength': 2.682760646759418, 'reg___subsample': 0.866012495548768, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6611204245683134, 'reg___border_count': 65, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 1738.6250644418149}

Rank 4: {'reg___iterations': 600, 'reg___depth': 3, 'reg___learning_rate': 0.012461956771211428, 'reg___l2_leaf_reg': 48.71643740673409, 'reg___bagging_temperature': 2.947585940461539, 'reg___random_strength': 2.883663349836425, 'reg___subsample': 0.8739799043351328, 'reg___min_data_in_leaf': 26, 'reg___leaf_estimation_iterations': 8, 'reg___rsm': 0.7422252774305045, 'reg___border_count': 47, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 2308.635310110983}

Rank 5: {'reg___iterations': 600, 'reg___depth': 3, 'reg___learning_rate': 0.011459649214497115, 'reg___l2_leaf_reg': 22.220673254721873, 'reg___bagging_temperature': 4.69749643833301, 'reg___random_strength': 2.6887821643826726, 'reg___subsample': 0.862778812730607, 'reg___min_data_in_leaf': 26, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6634894943837829, 'reg___border_count': 126, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 2402.515995967583}

Rank 6: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.012834272545680134, 'reg___l2_leaf_reg': 28.08499519634293, 'reg___bagging_temperature': 3.015283781344319, 'reg___random_strength': 2.9899528143850045, 'reg___subsample': 0.8386813976524478, 'reg___min_data_in_leaf': 34, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.7279856803021704, 'reg___border_count': 61, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 4264.918125059337}

Rank 7: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.013698245413456716, 'reg___l2_leaf_reg': 42.68649686081776, 'reg___bagging_temperature': 2.859083852738856, 'reg___random_strength': 2.9381960348498084, 'reg___subsample': 0.8312805424519285,

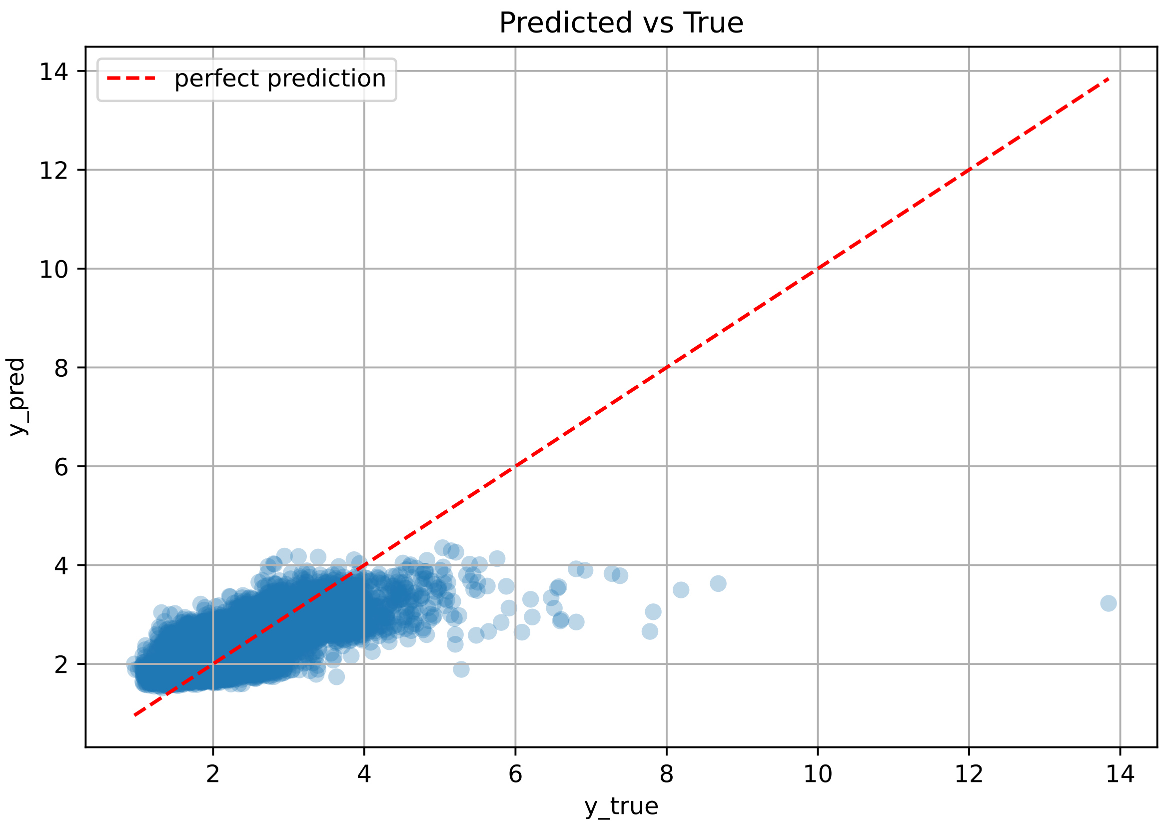
’reg__min_data_in_leaf’: 32, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.7501487595734669, ’reg__border_count’: 60, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 4021.3789846882437}

Rank 8: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.012188512862512872, ’reg__l2_leaf_reg’: 48.80186566211637, ’reg__bagging_temperature’: 3.108917529180762, ’reg__random_strength’: 2.9221451743368996, ’reg__subsample’: 0.8757011956703643, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 8, ’reg__rsm’: 0.5719612746270899, ’reg__border_count’: 43, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 4, ’reg__diffusion_temperature’: 2623.8863108002715}

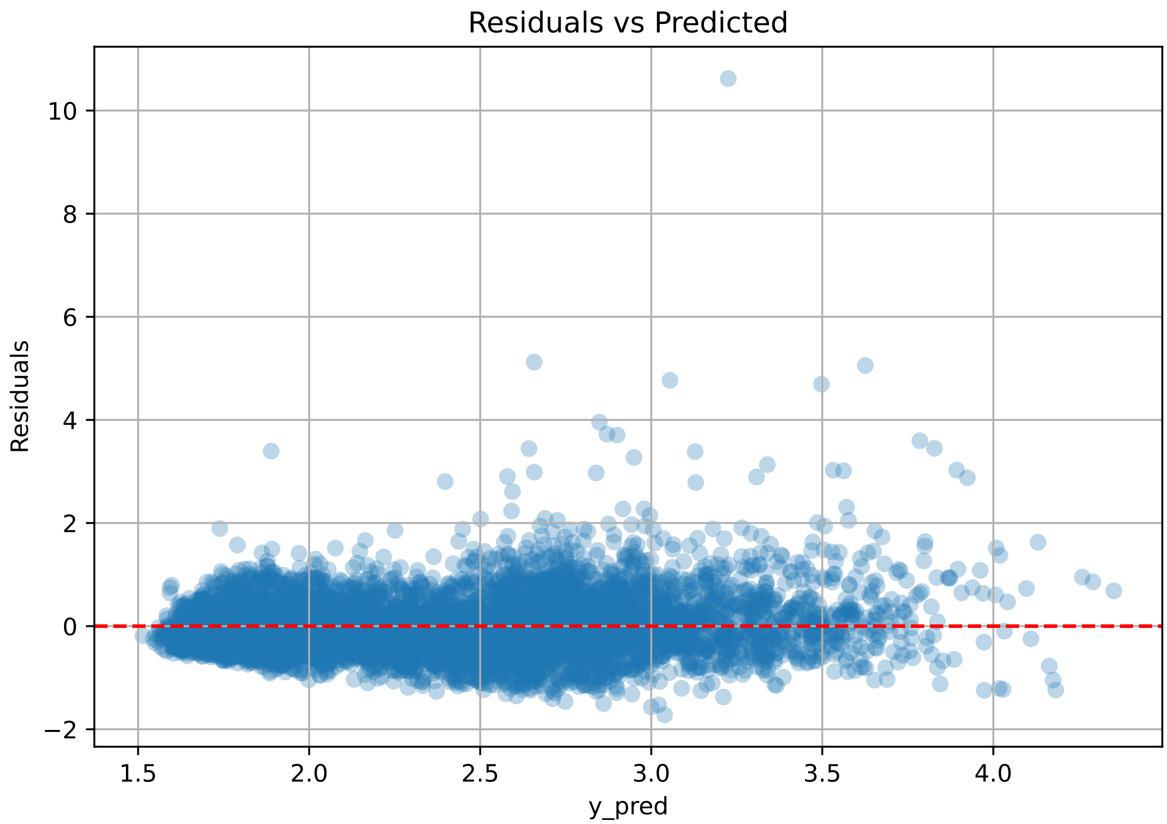
Rank 9: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.013841337625289763, ’reg__l2_leaf_reg’: 43.200217670052005, ’reg__bagging_temperature’: 2.824513656667862, ’reg__random_strength’: 3.1865275570347547, ’reg__subsample’: 0.8328207127984188, ’reg__min_data_in_leaf’: 33, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.7329866111944711, ’reg__border_count’: 32, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 929.572652822523}

Rank 10: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.012617891842723706, ’reg__l2_leaf_reg’: 28.42183292661867, ’reg__bagging_temperature’: 3.069616754396232, ’reg__random_strength’: 2.9978451151450938, ’reg__subsample’: 0.8339099363073661, ’reg__min_data_in_leaf’: 34, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.735966829588123, ’reg__border_count’: 62, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 958.8672974232029}

3.6.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

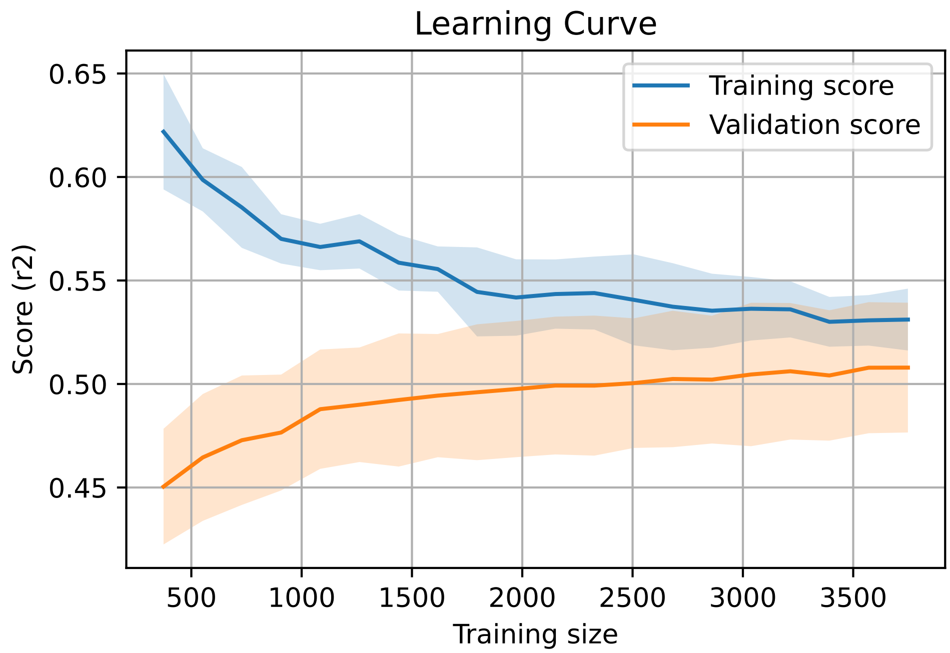
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7651, **Len(training)** : 5356,

Len(training_real) : 3750, **Len(test_of_training)** : 1606, **Len(test)** : 2295



3.7 Sumup-table

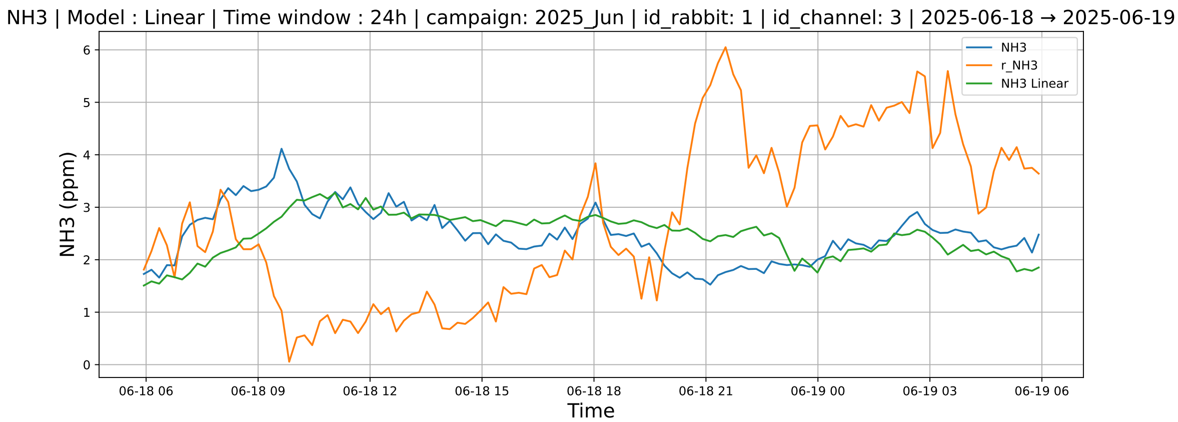
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.407	0.408	0.029	0.409	0.41	0.003	1.007	0.999	nan
LGBM	0.533	0.508	0.03	0.538	0.555	0.047	1.093	1.049	0.5083
CatBoost	0.532	0.507	0.032	0.536	0.547	0.04	1.079	1.048	0.5074

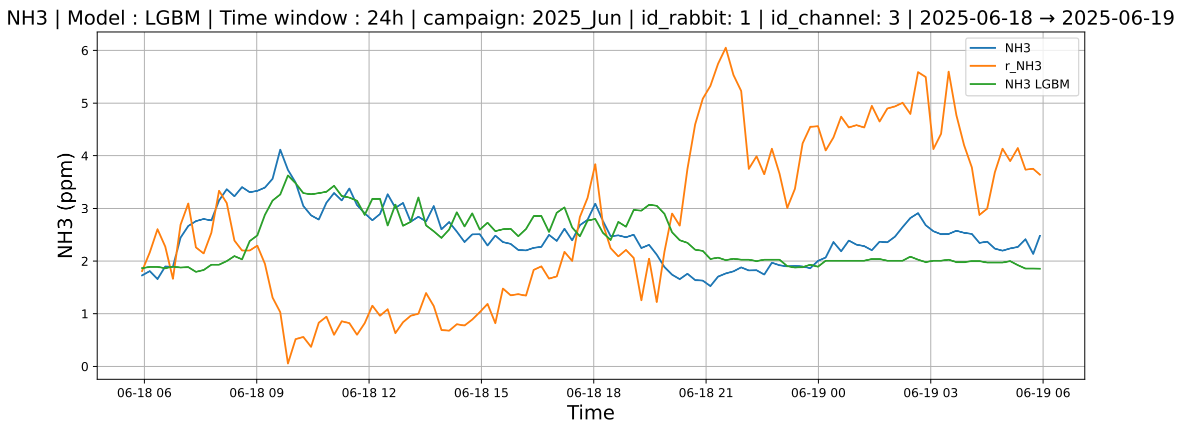
4 Plot

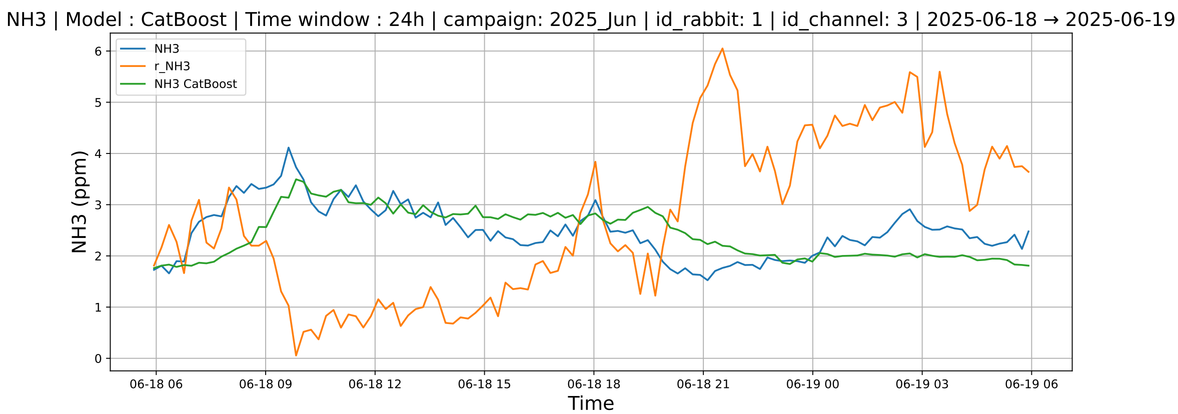
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

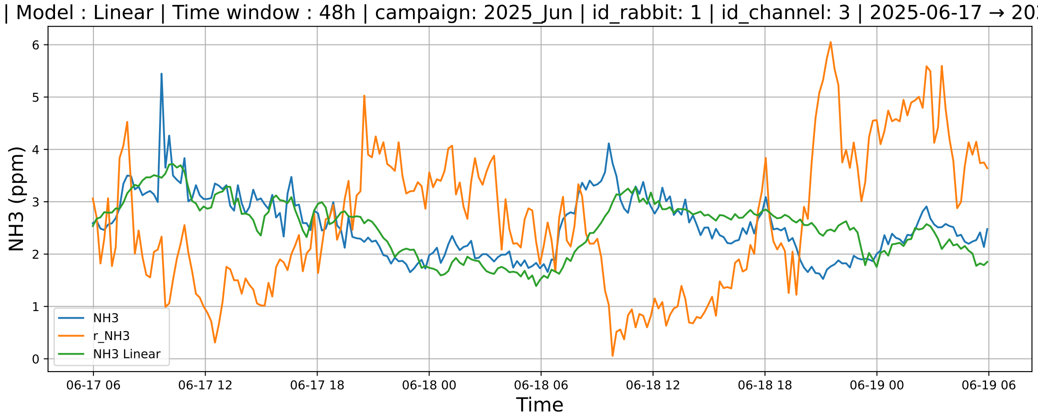




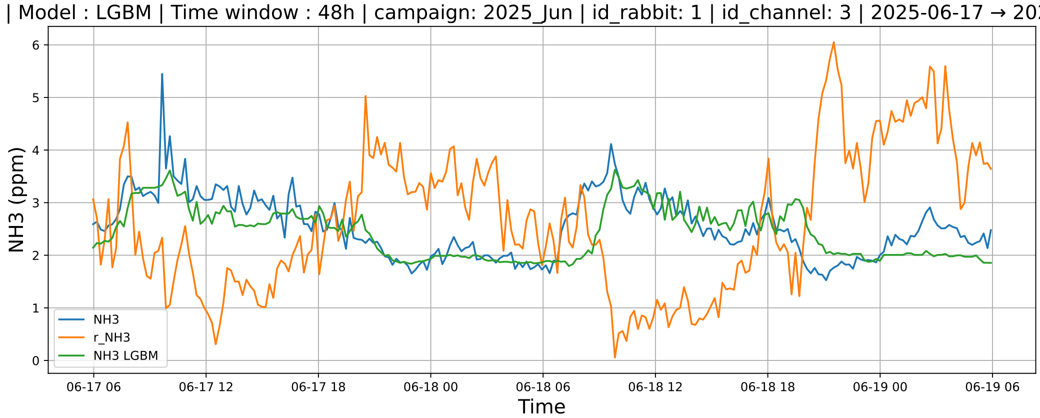


4.1.1.2 Time window : 48

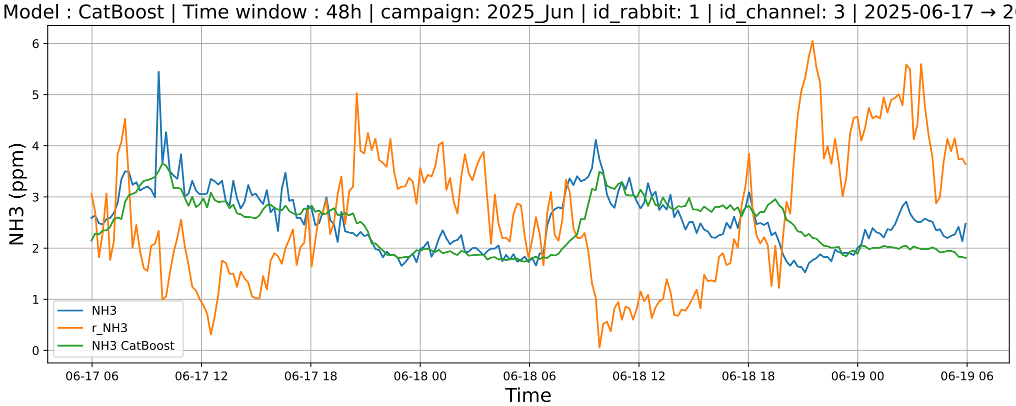
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

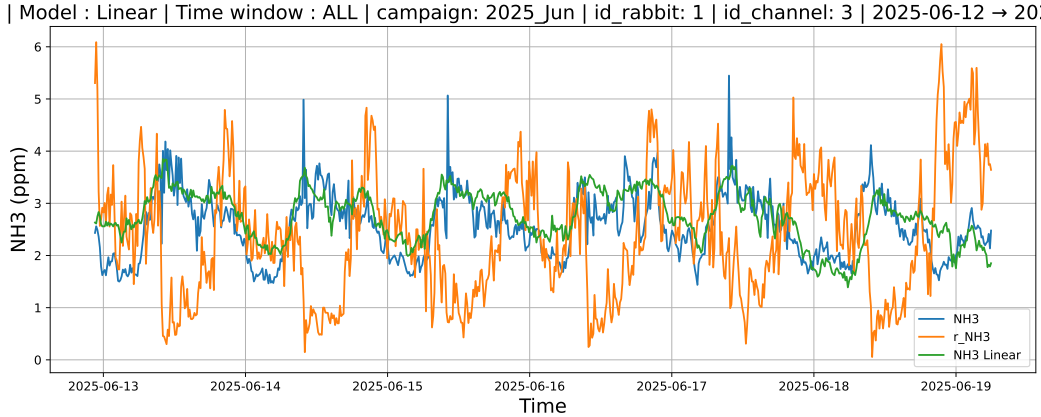


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

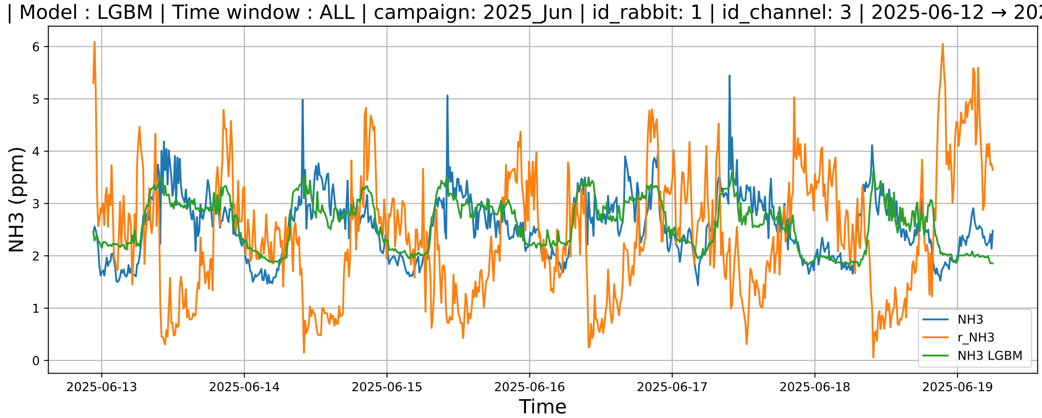


4.1.1.3 Time window : ALL

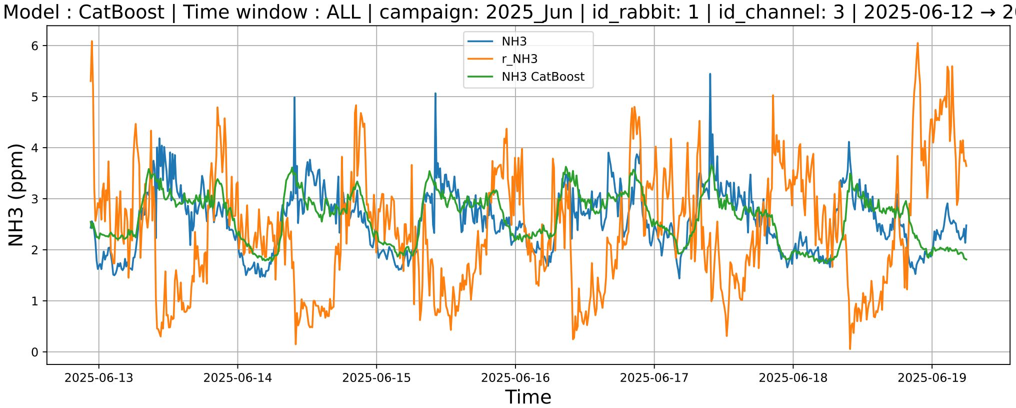
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

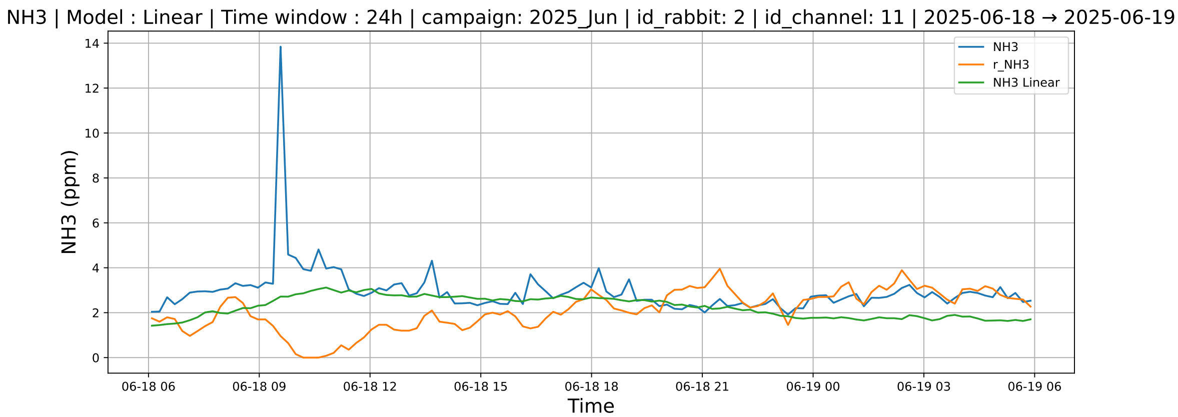


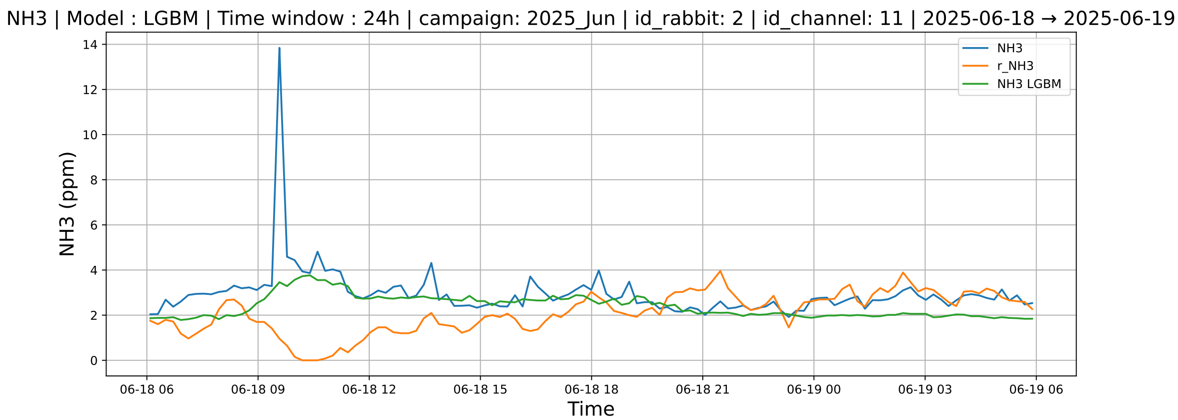
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

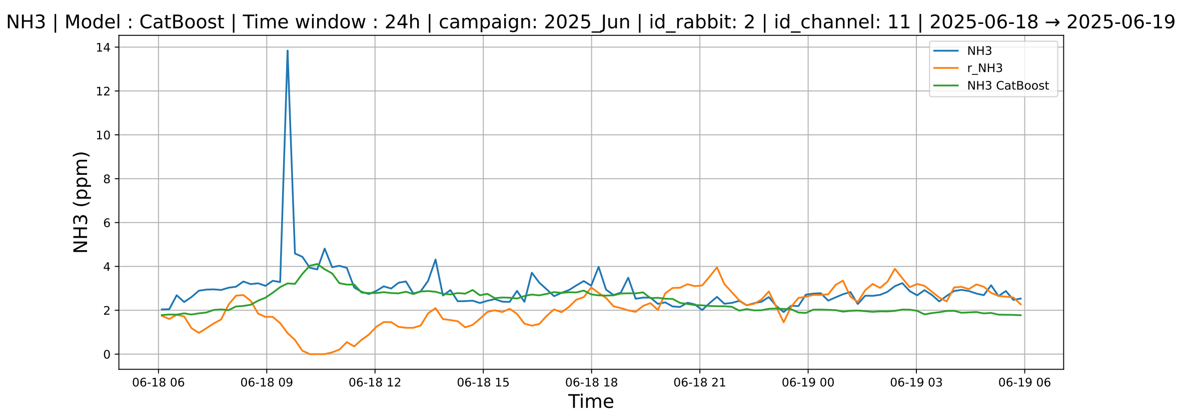


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

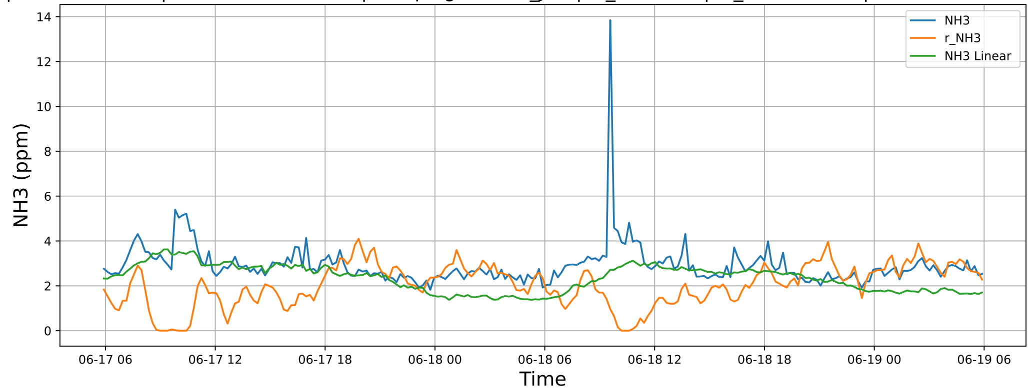




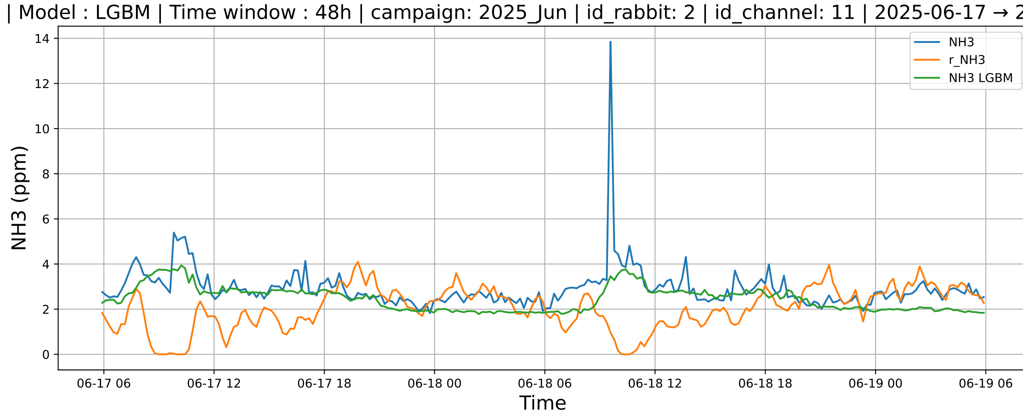


4.1.2.2 Time window : 48

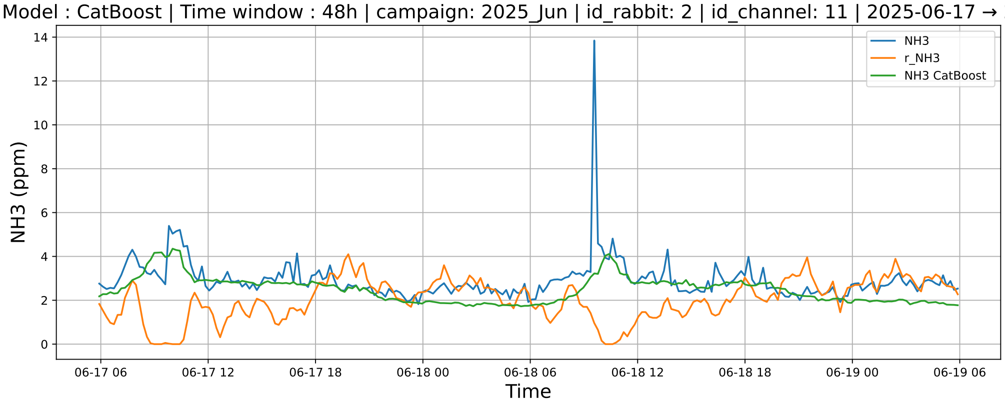
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

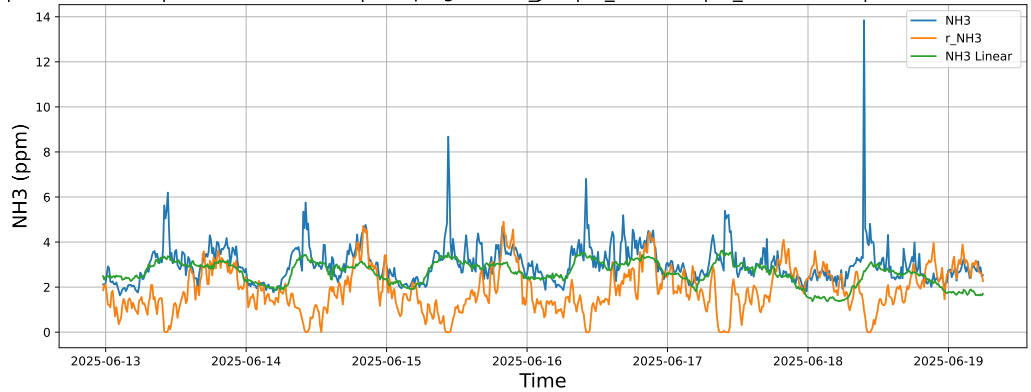


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

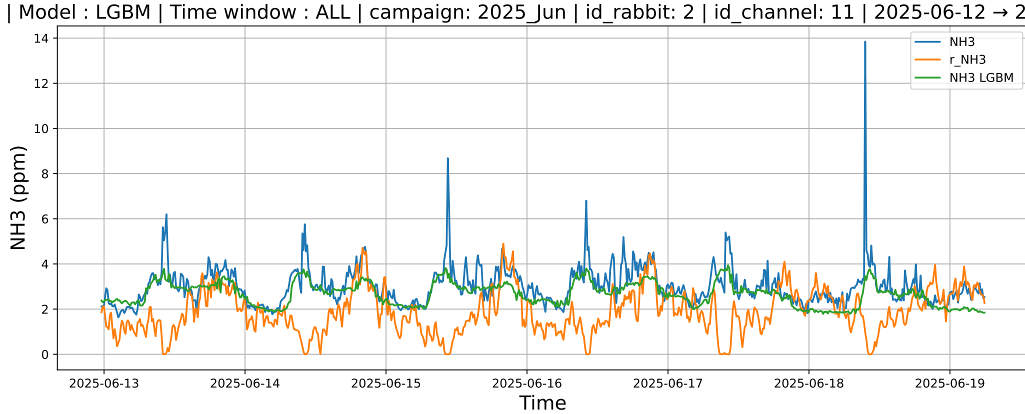


4.1.2.3 Time window : ALL

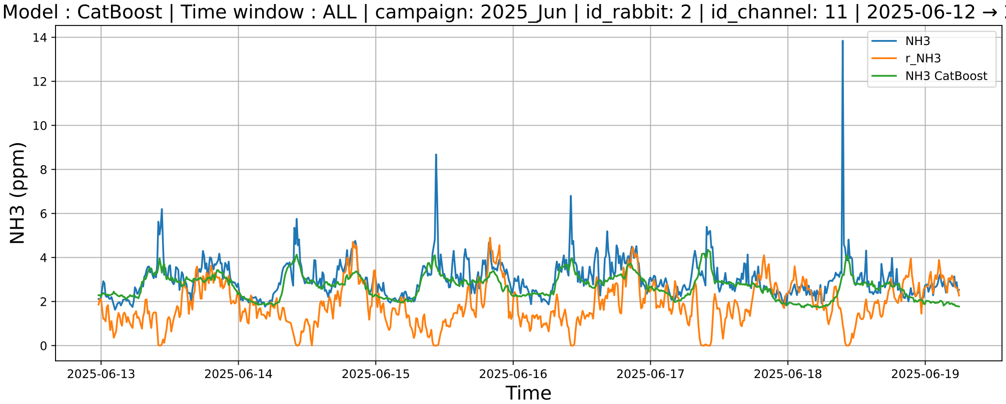
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

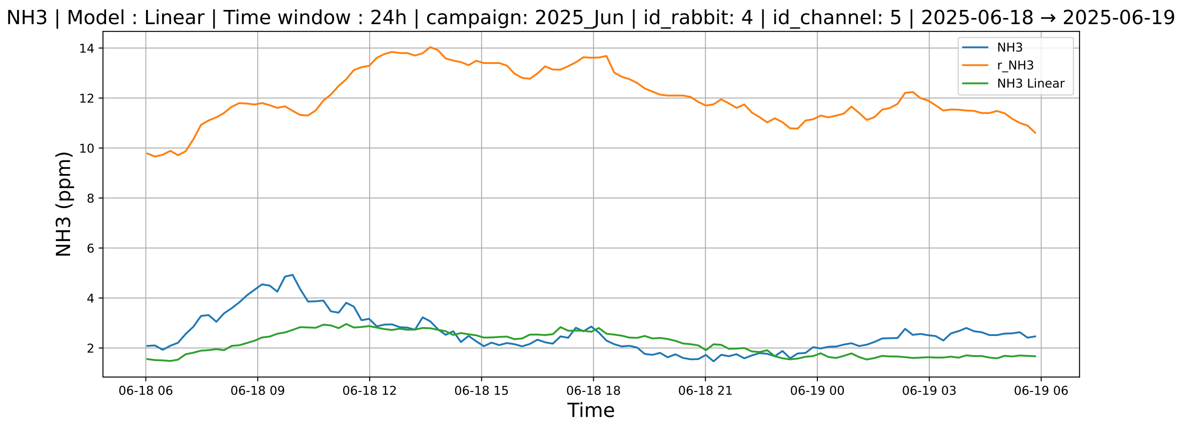


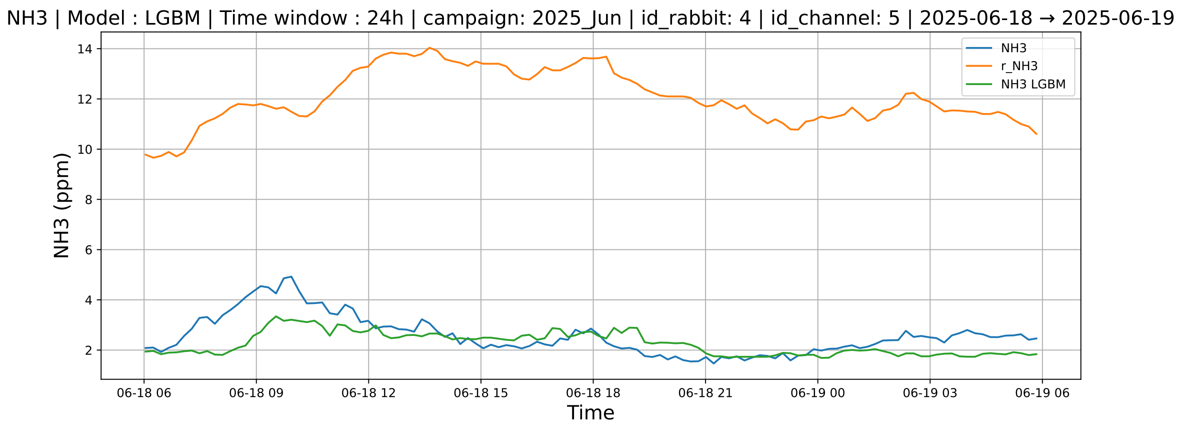
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

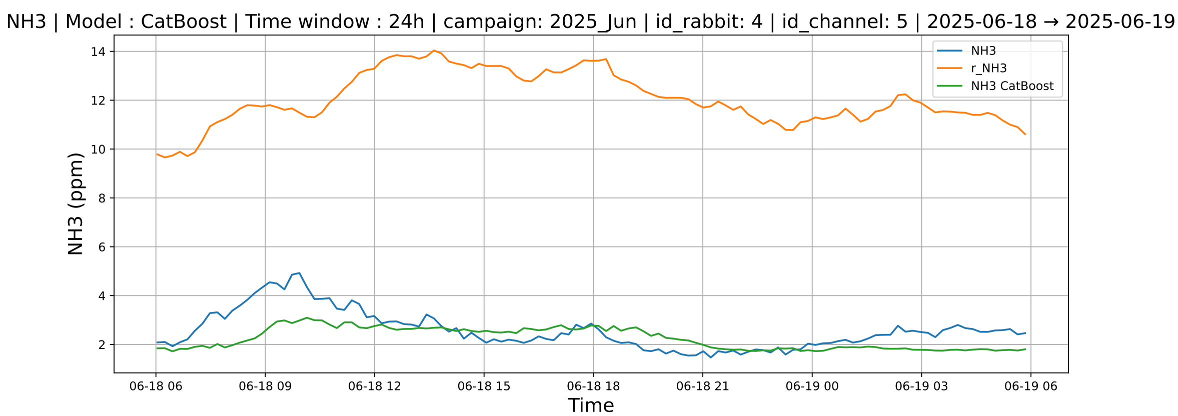


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

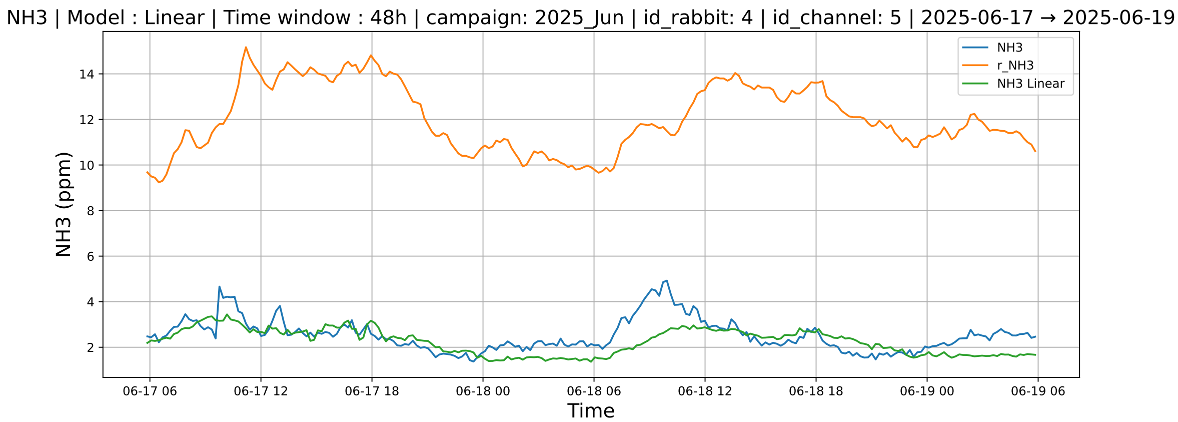
4.1.3.1 Time window : 24

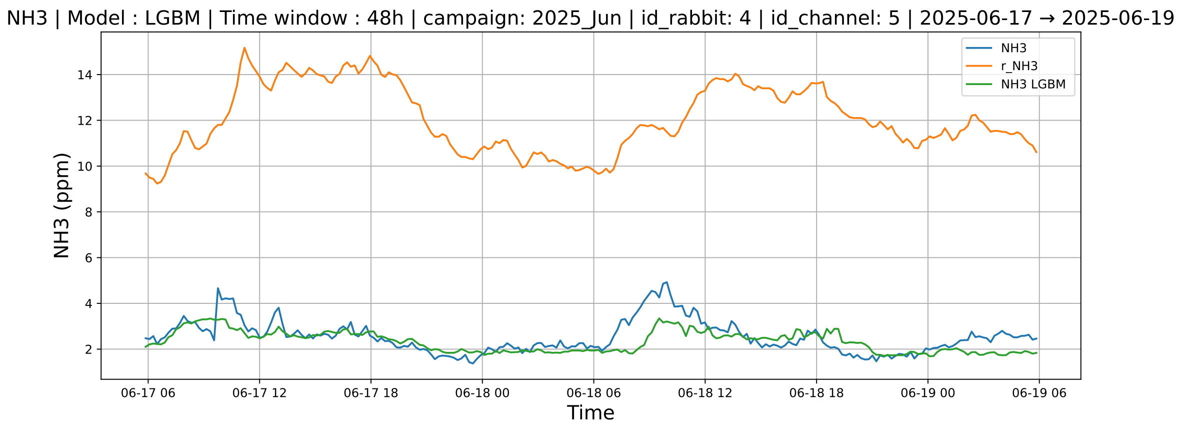


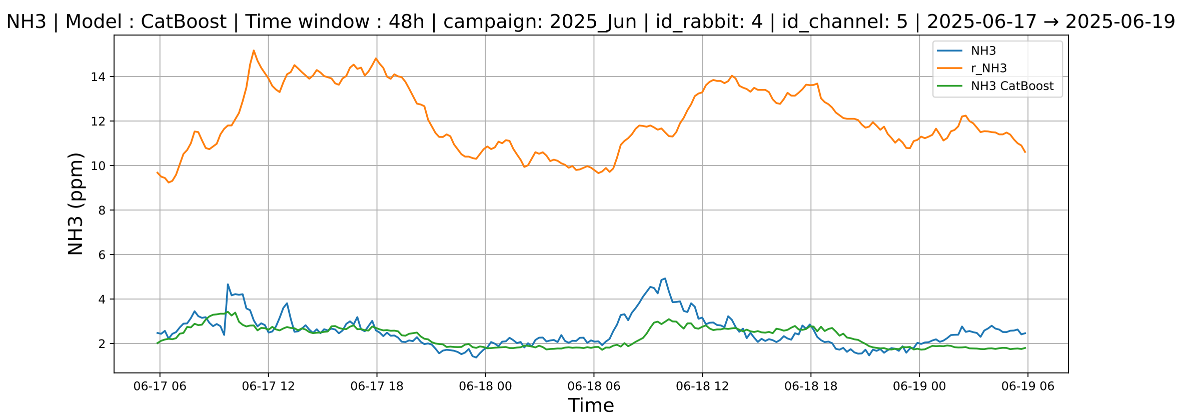




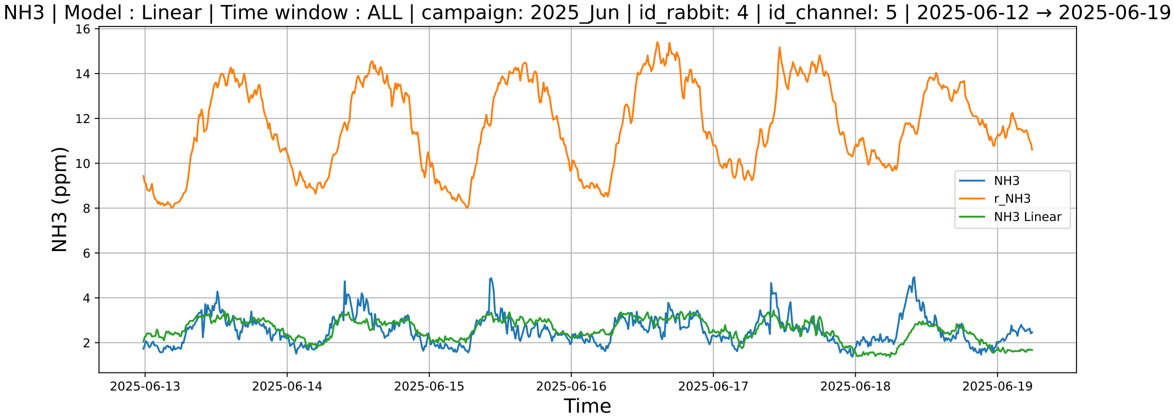
4.1.3.2 Time window : 48

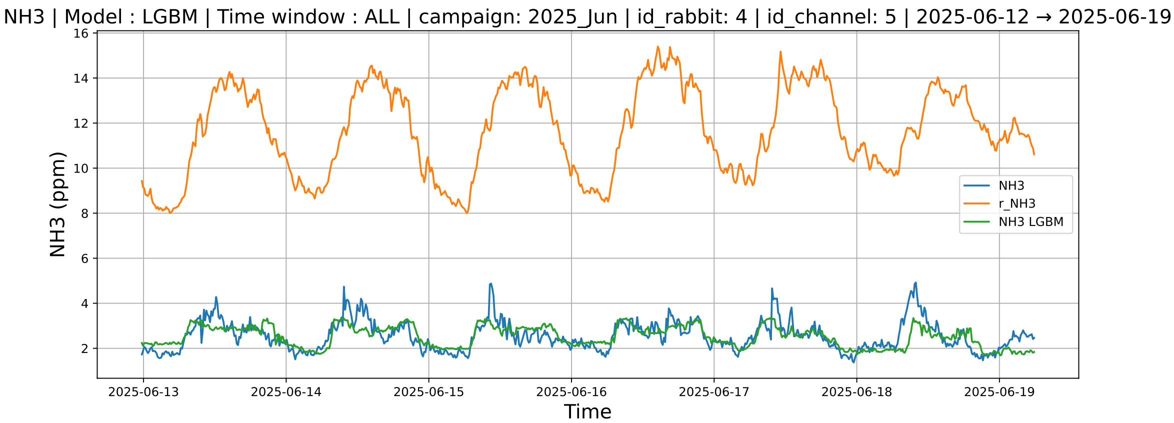


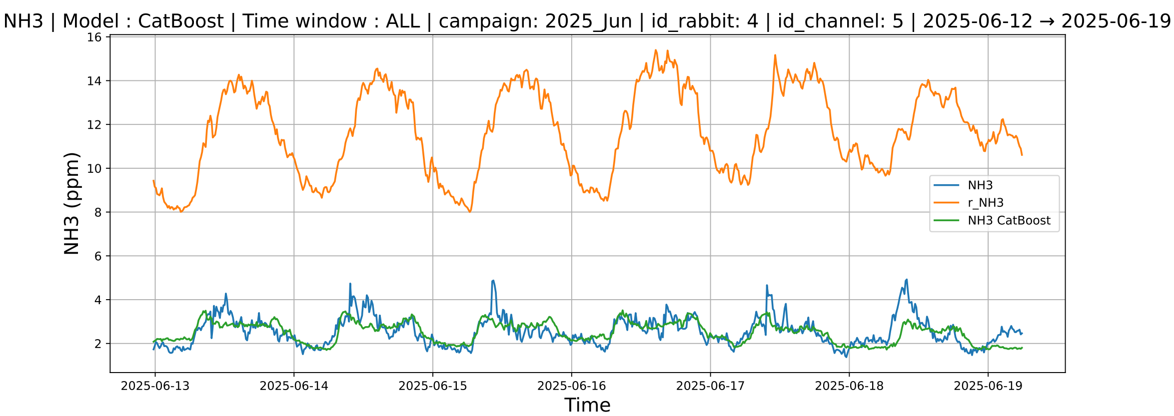




4.1.3.3 Time window : ALL

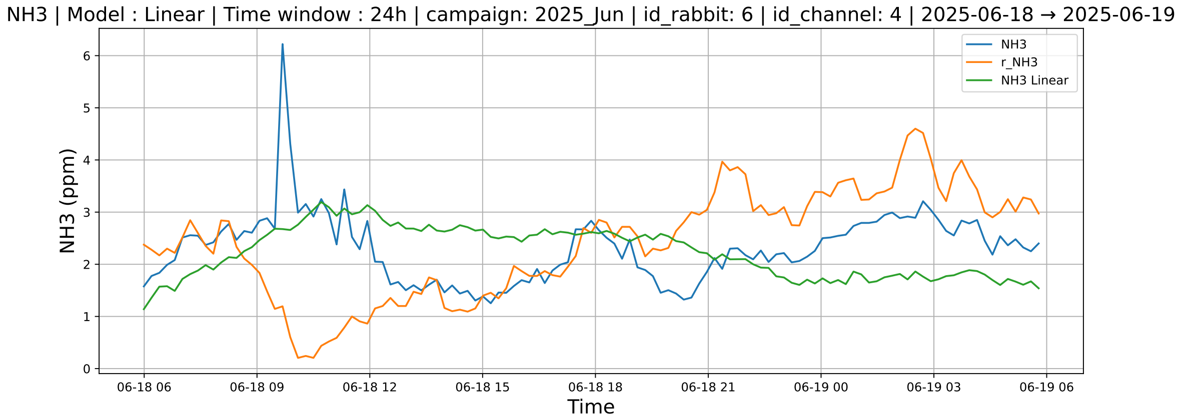


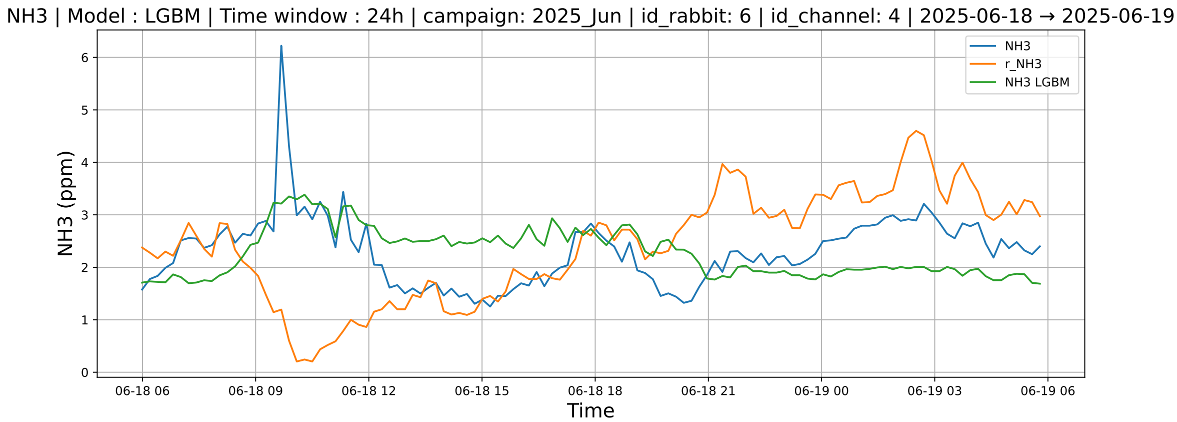


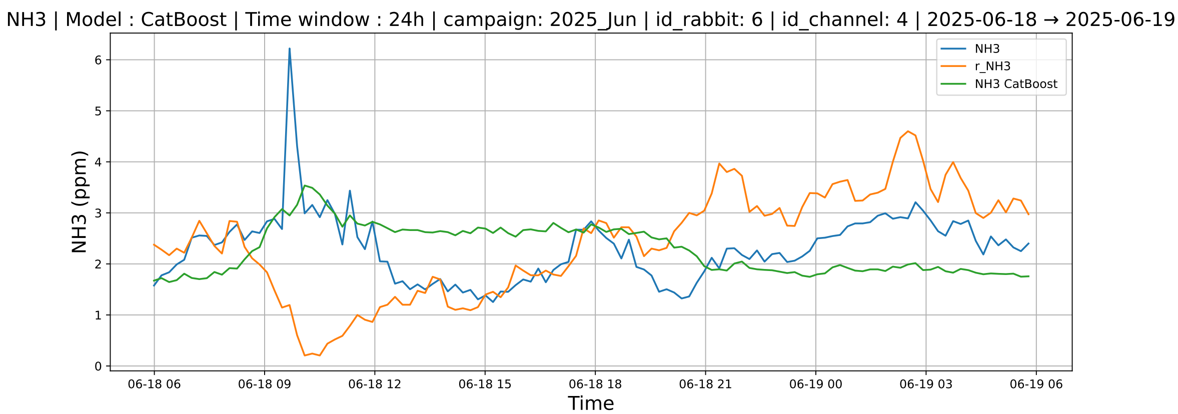


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

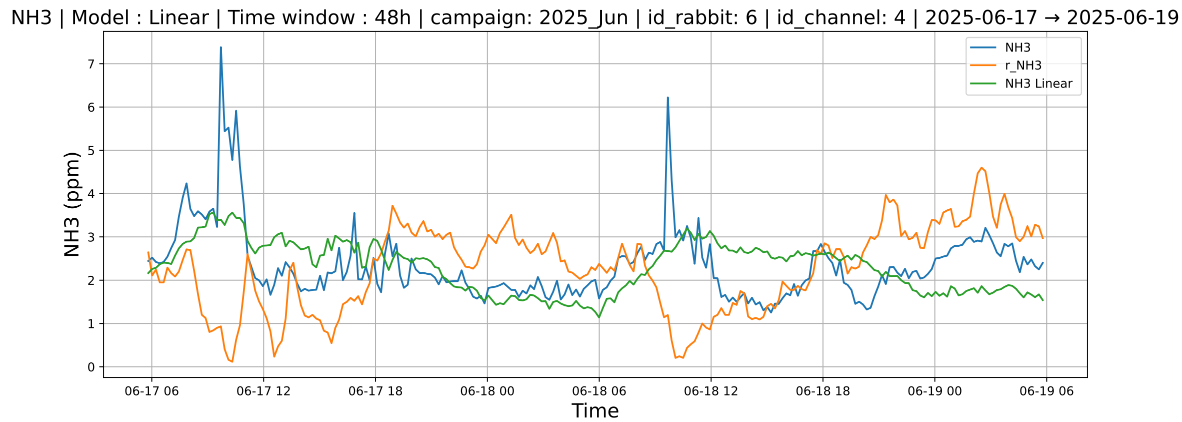
4.1.4.1 Time window : 24

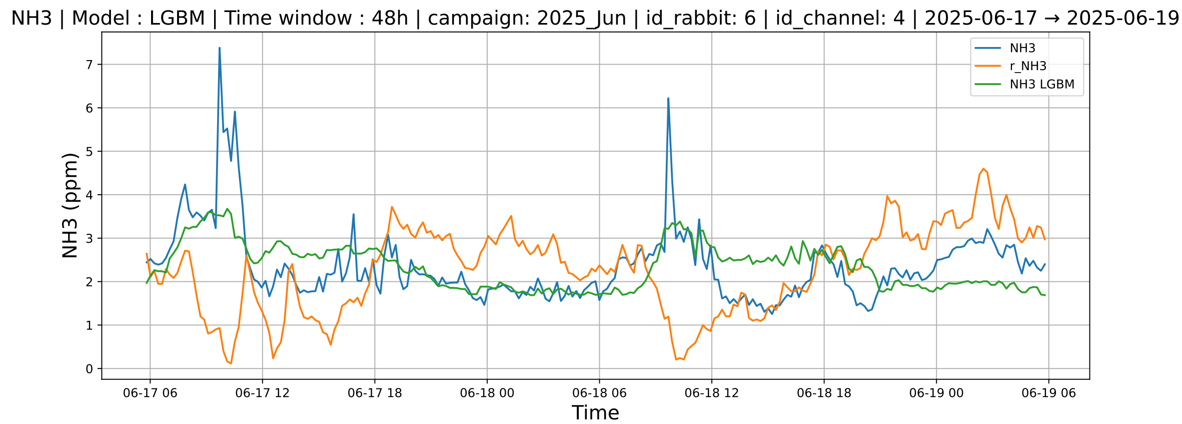


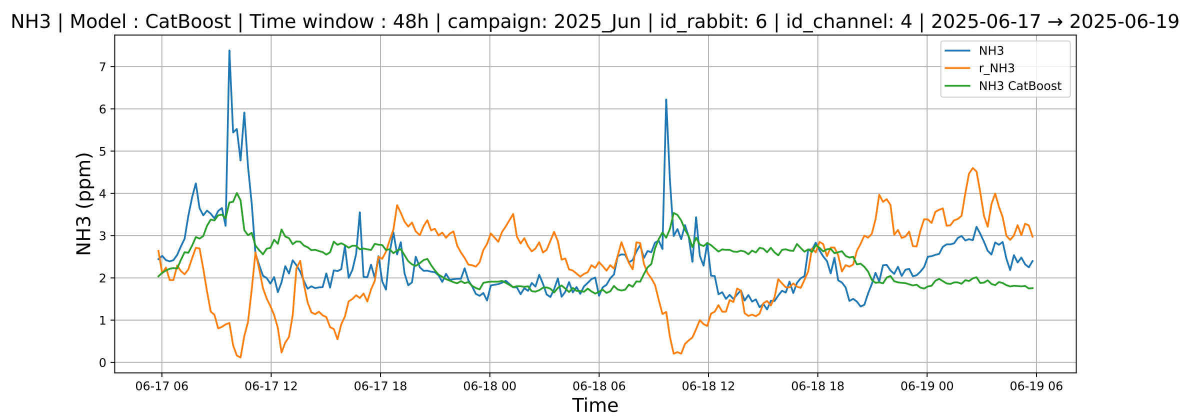




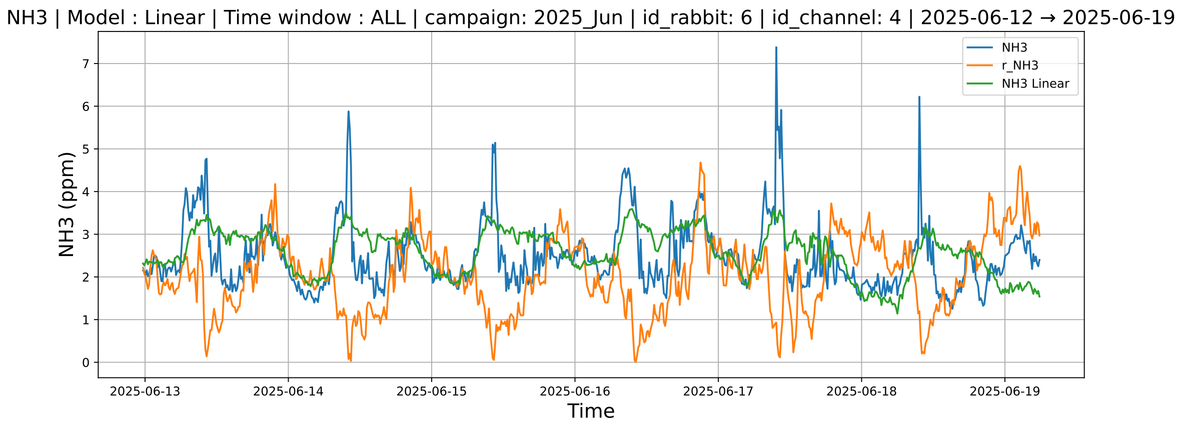
4.1.4.2 Time window : 48

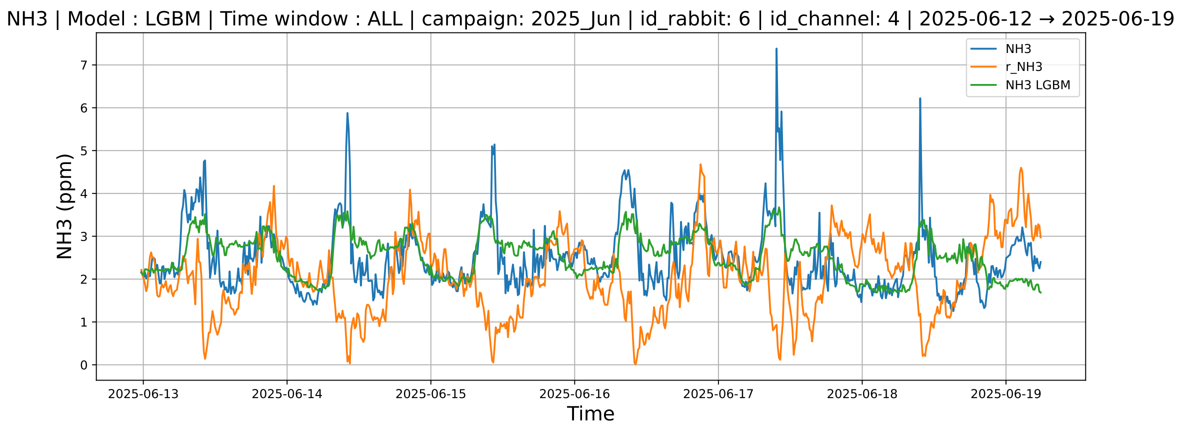


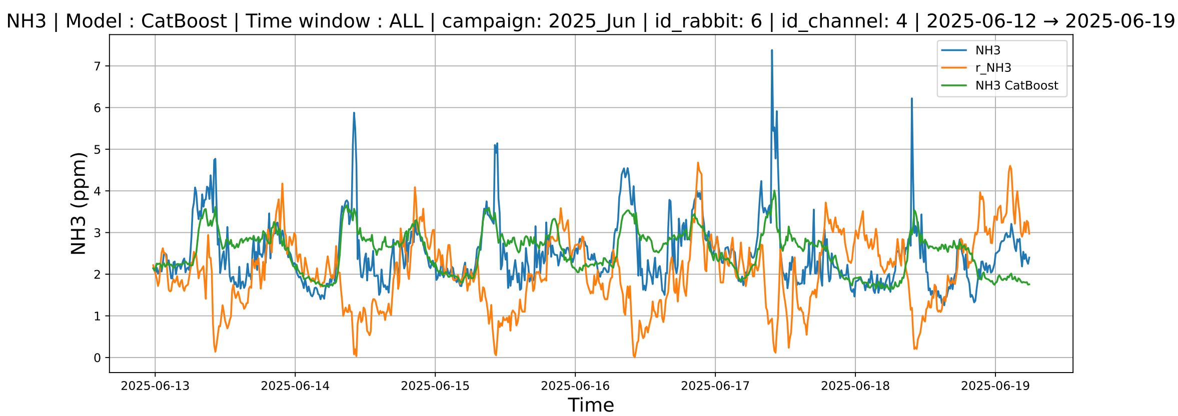




4.1.4.3 Time window : ALL



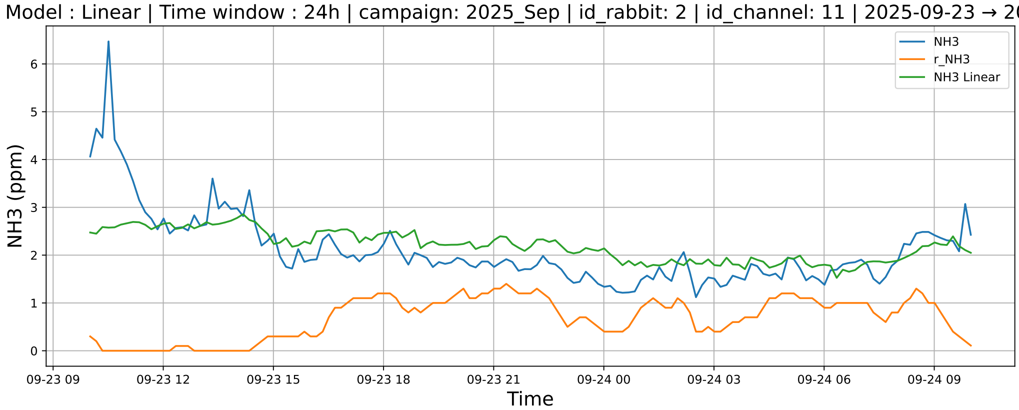




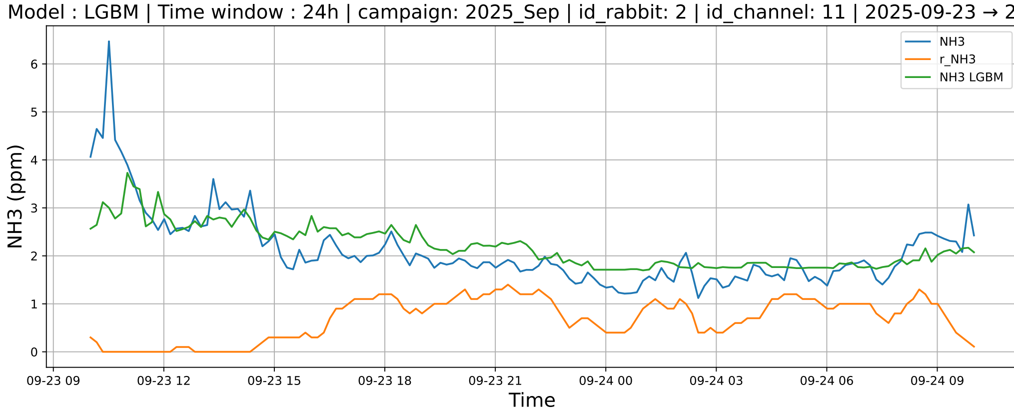
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

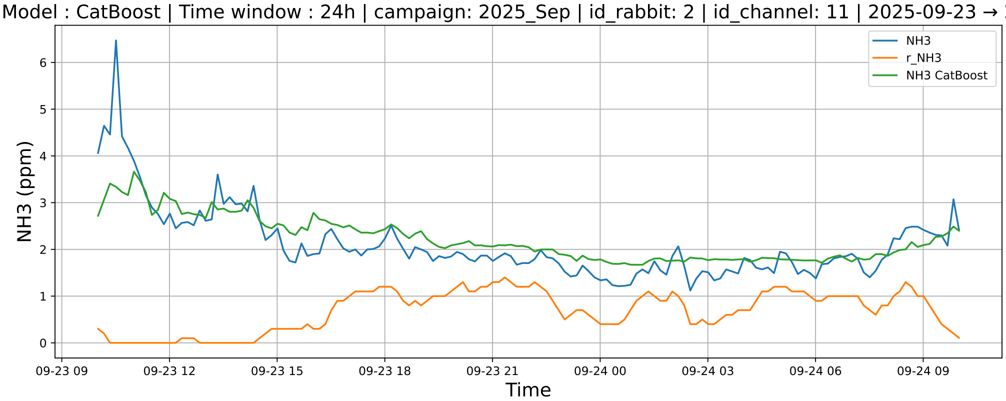
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



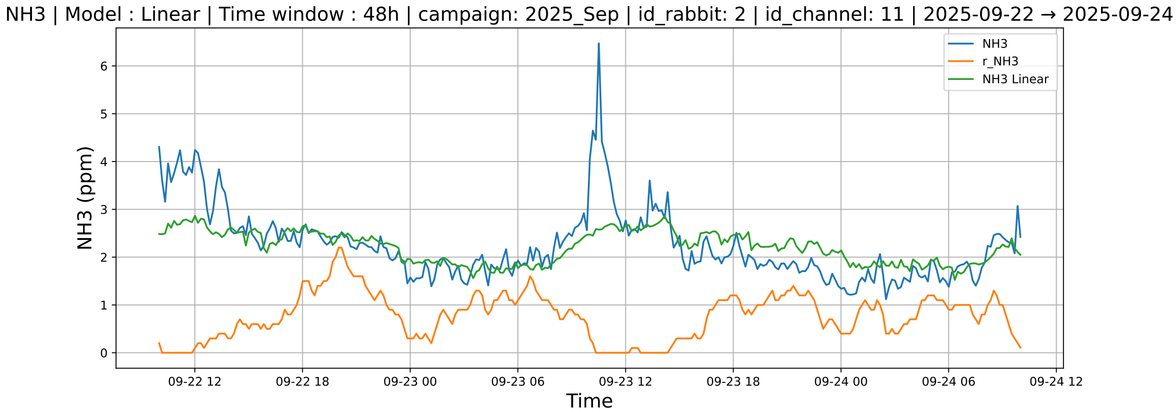
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

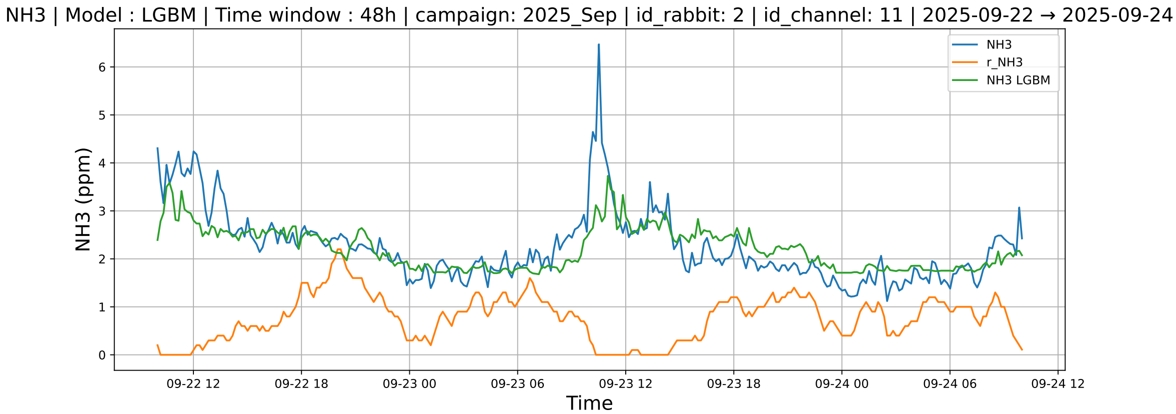


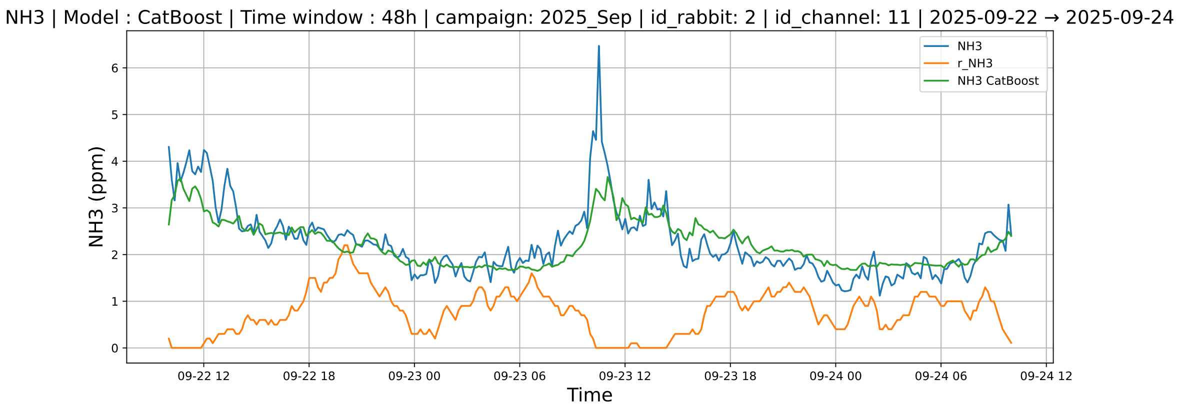
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



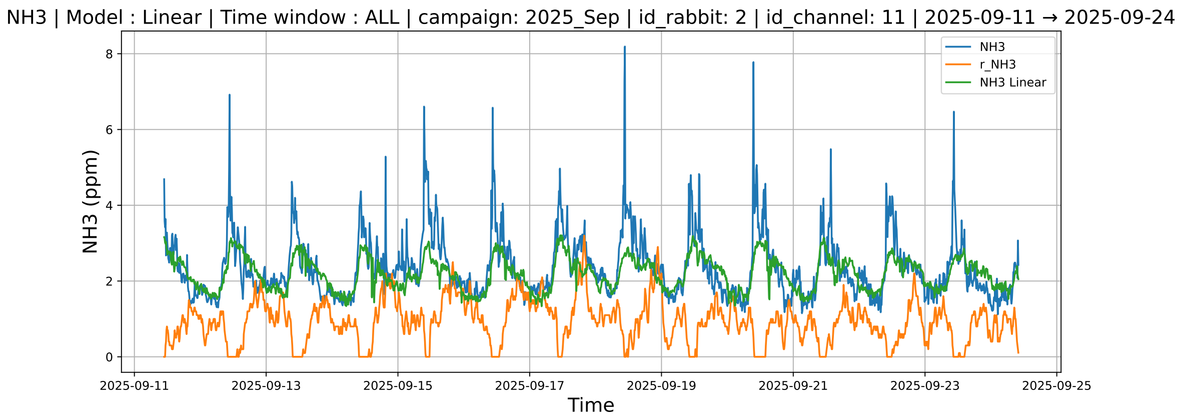
4.1.5.2 Time window : 48

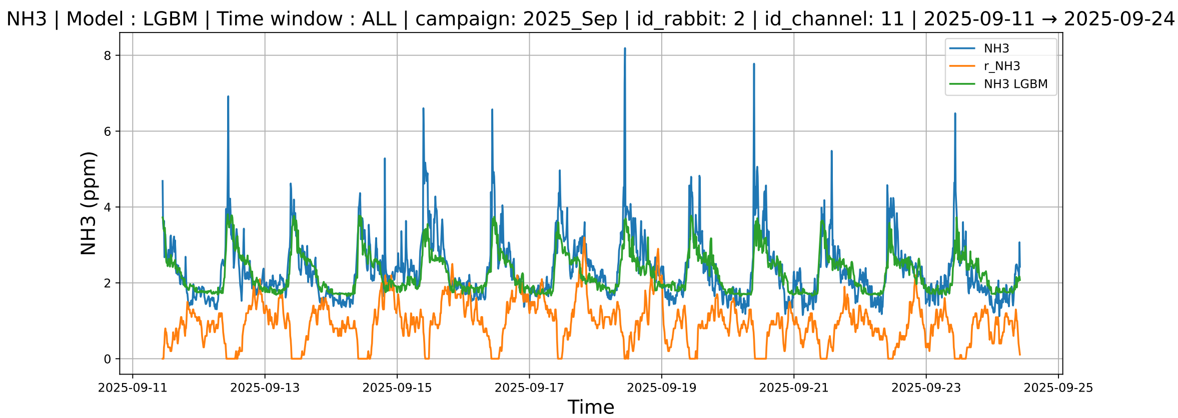


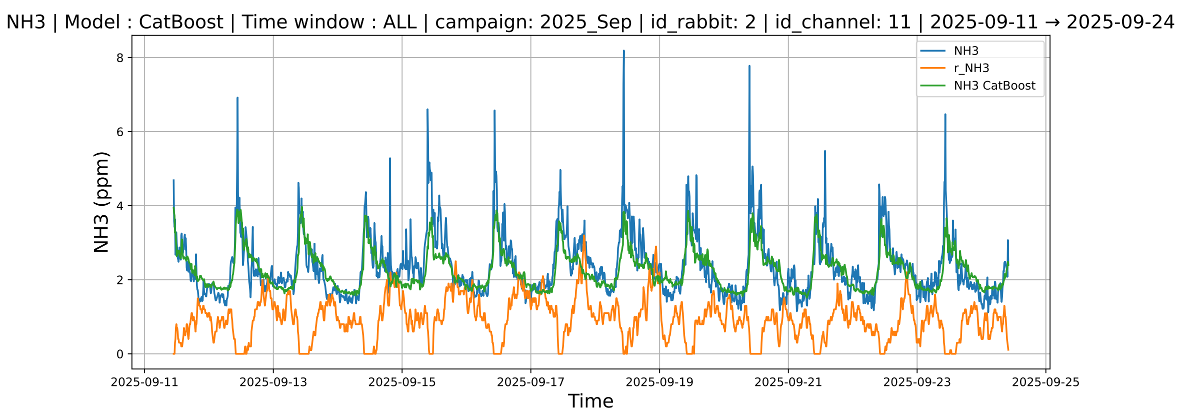




4.1.5.3 Time window : ALL



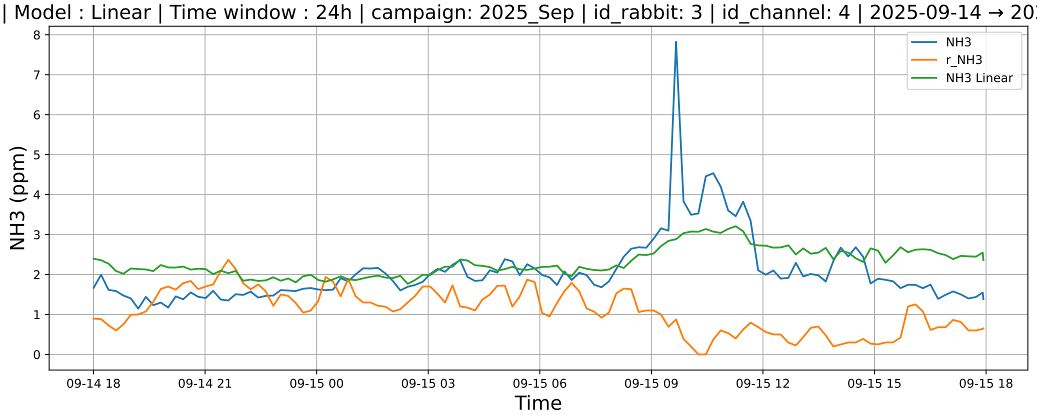




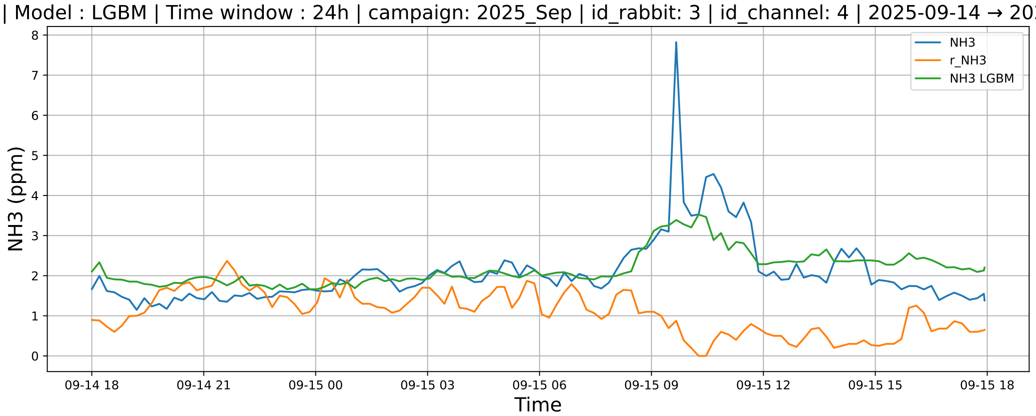
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

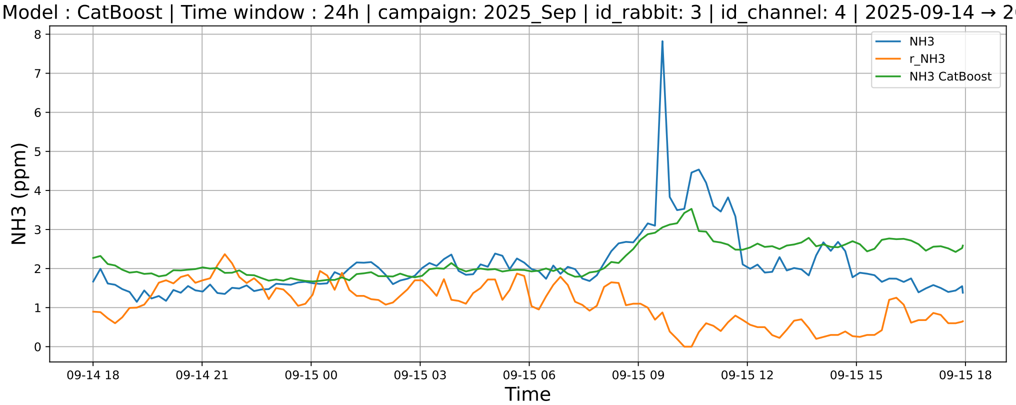
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



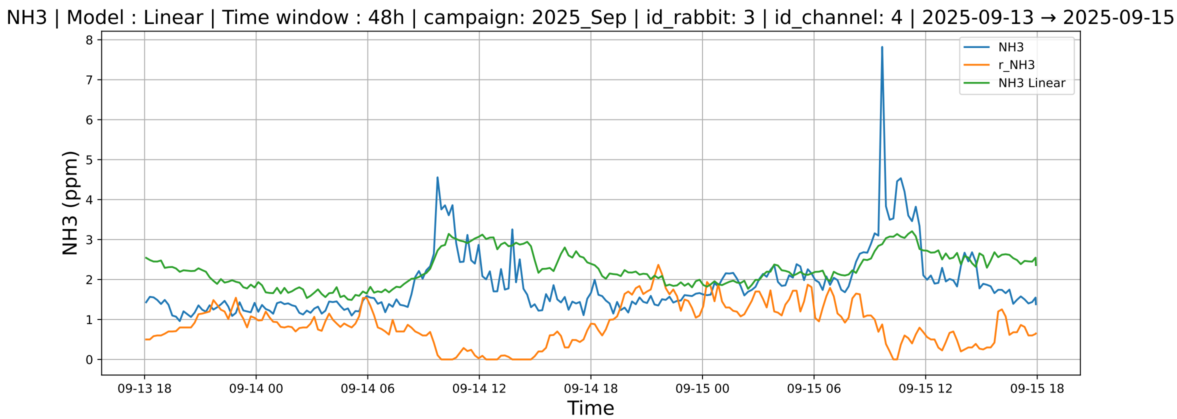
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

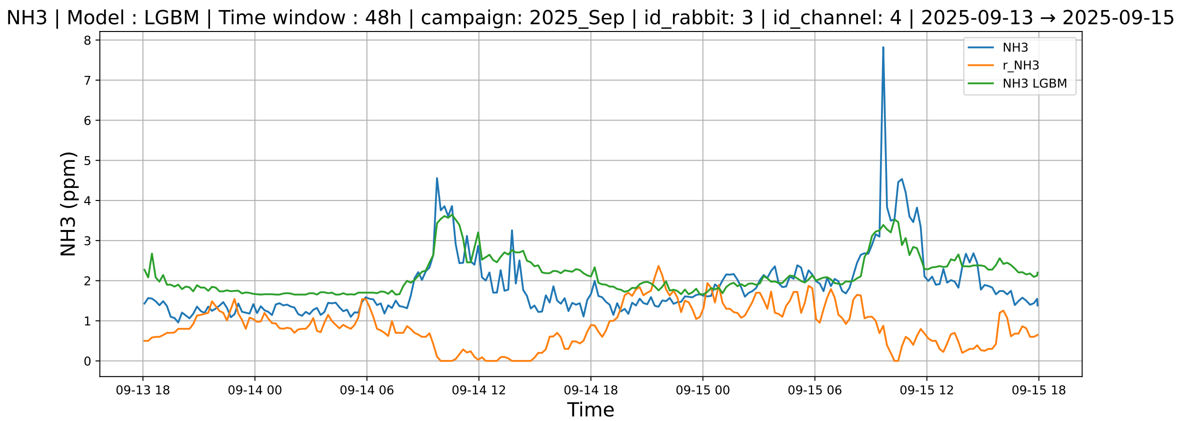


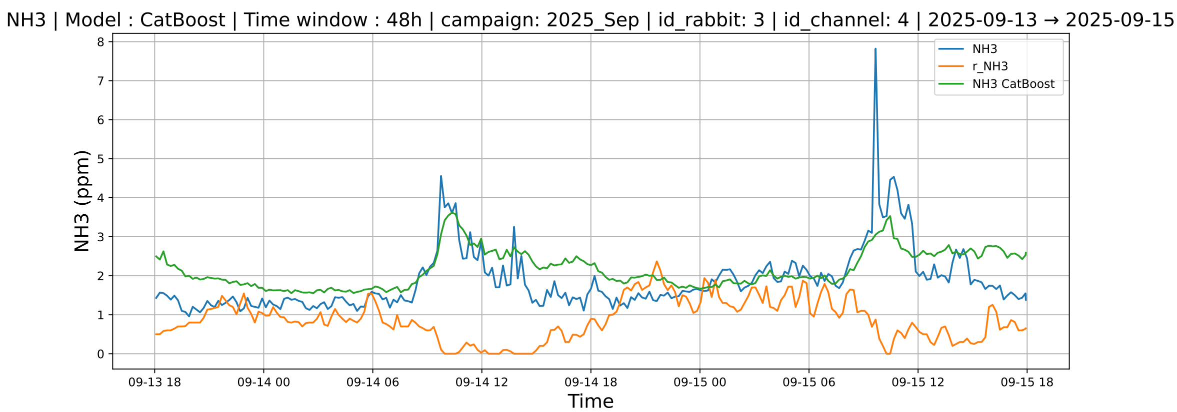
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



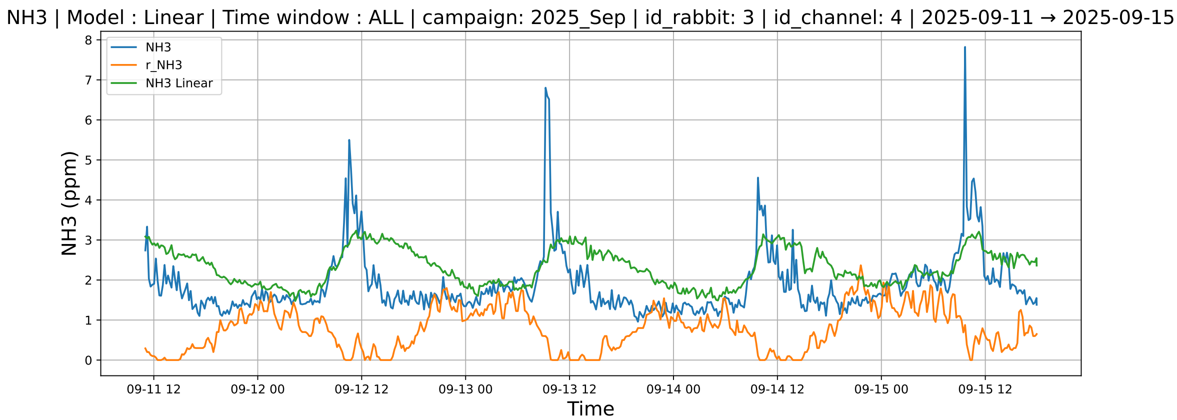
4.1.6.2 Time window : 48

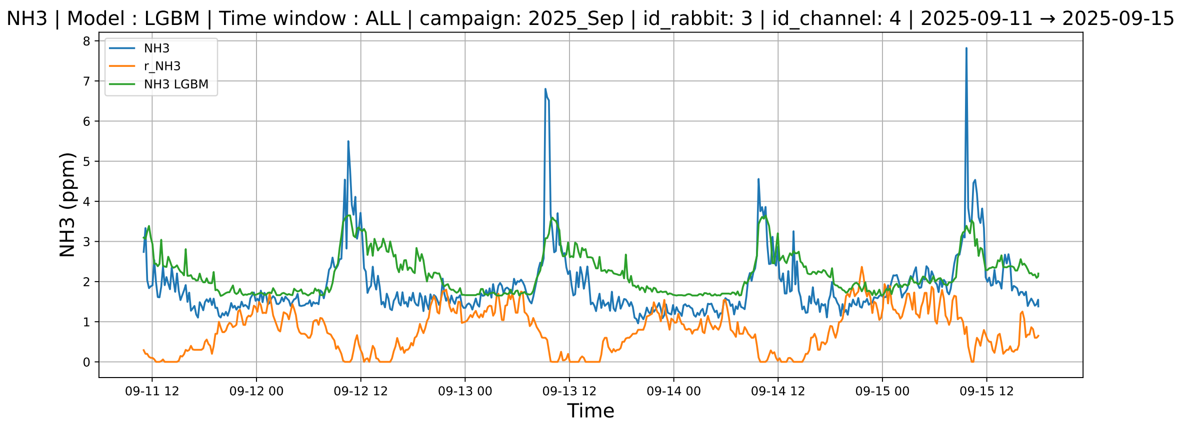


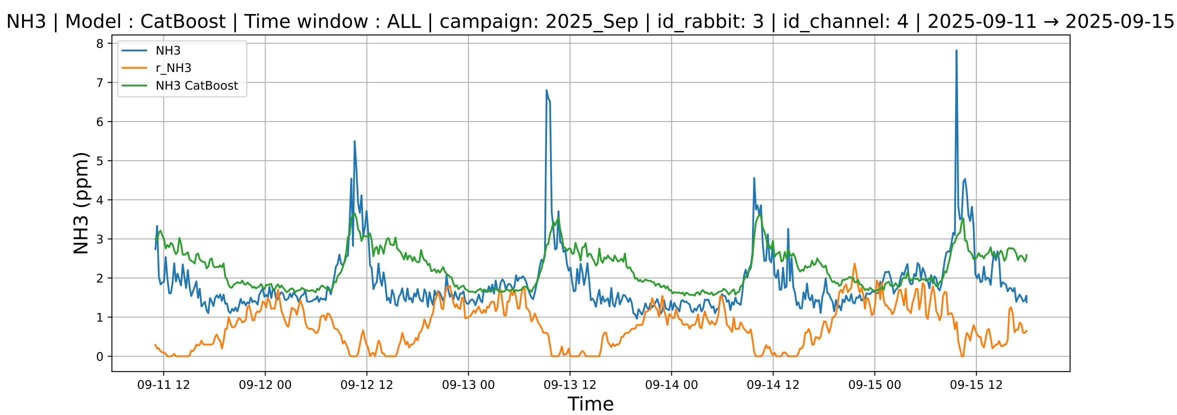




4.1.6.3 Time window : ALL

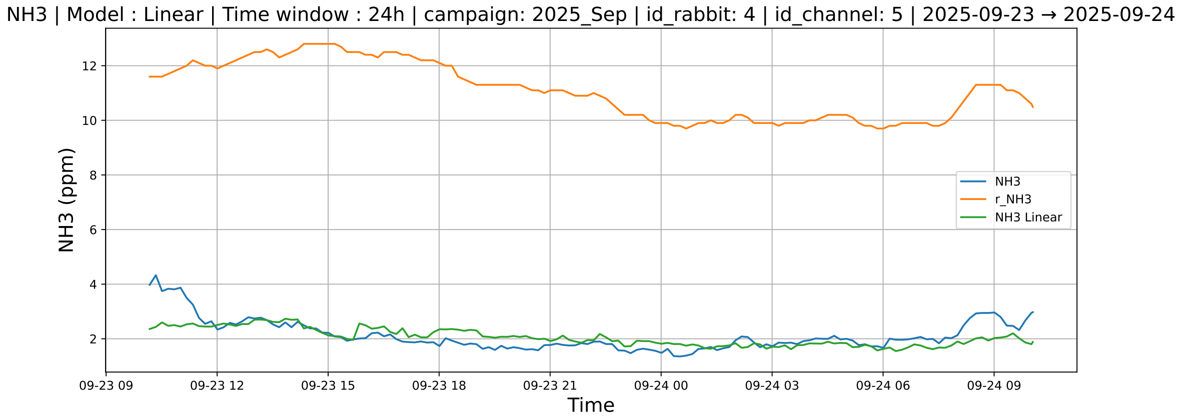


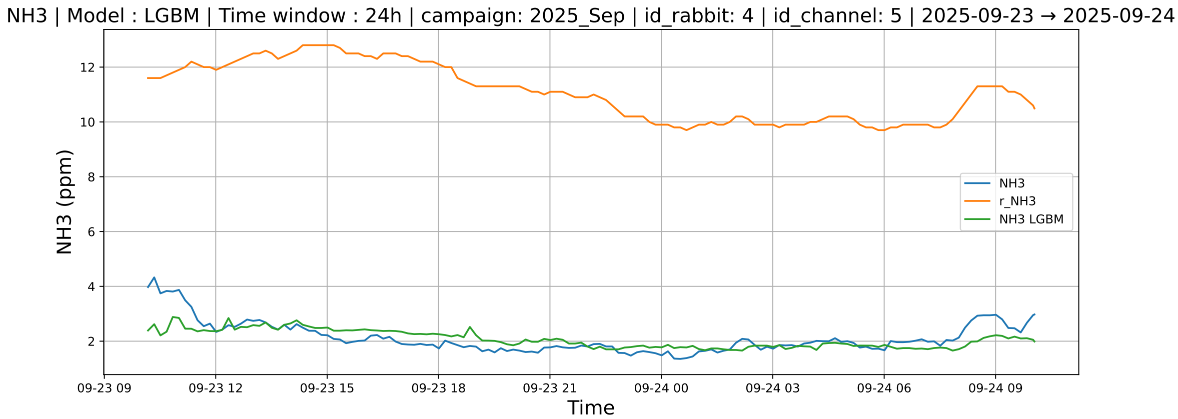


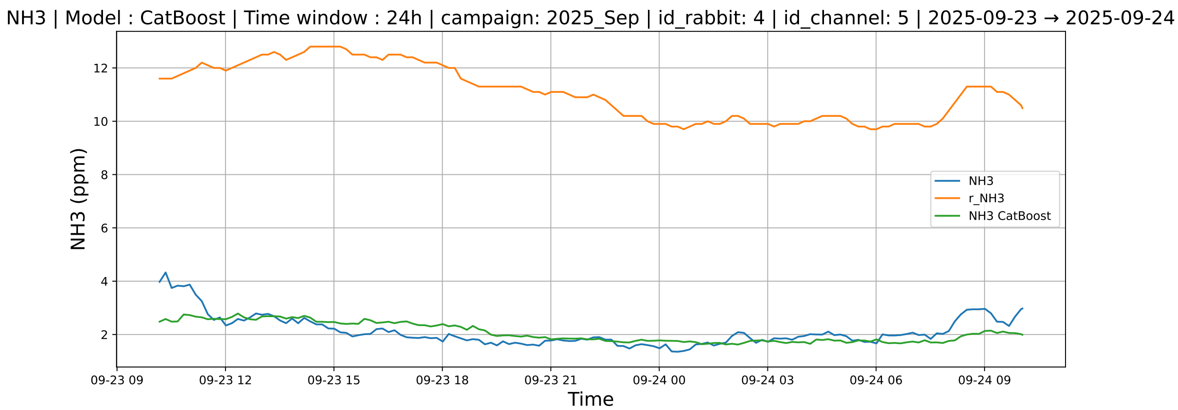


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

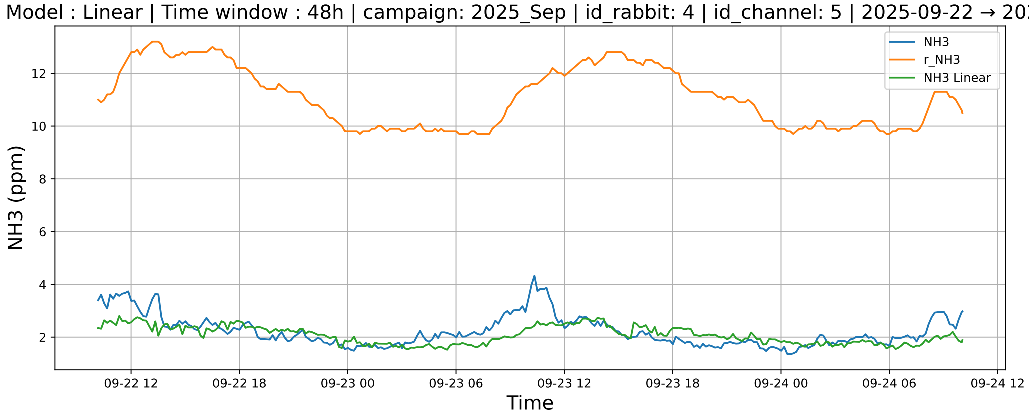




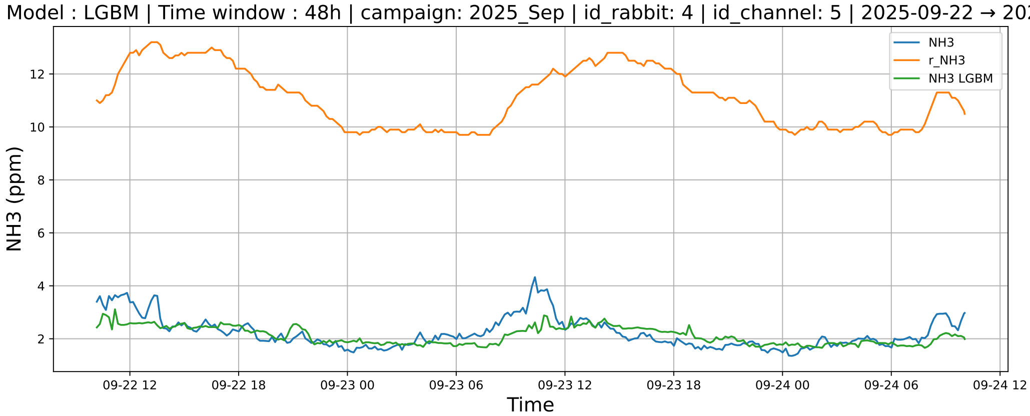


4.1.7.2 Time window : 48

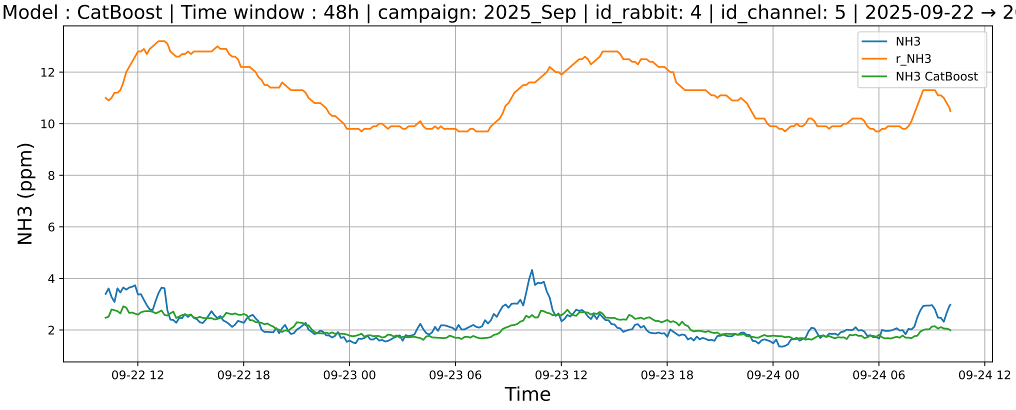
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



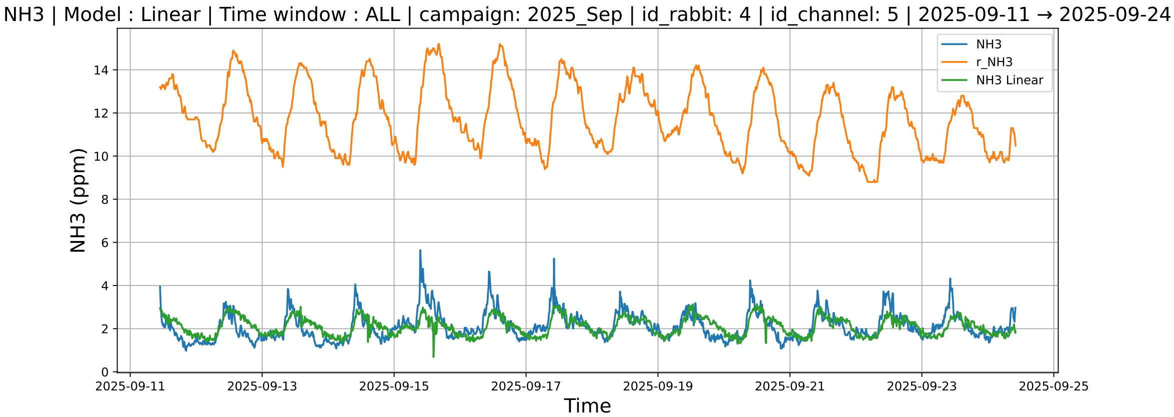
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

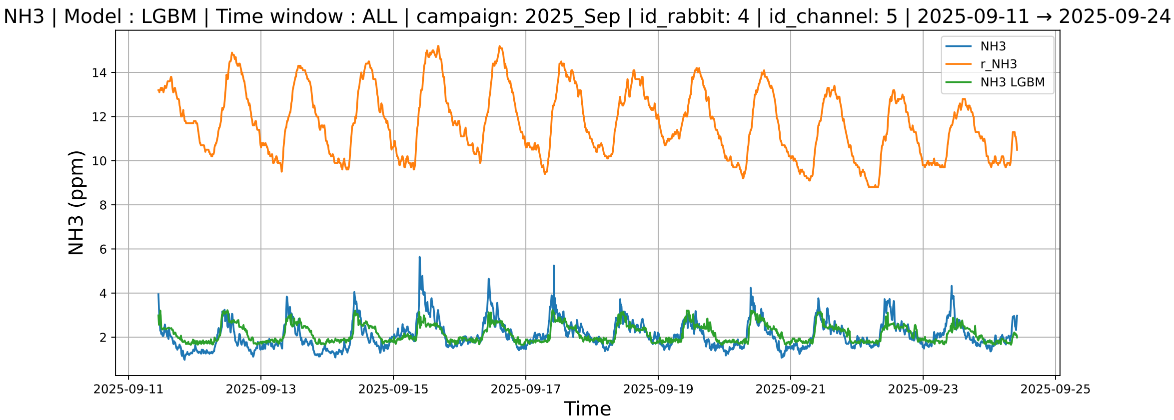


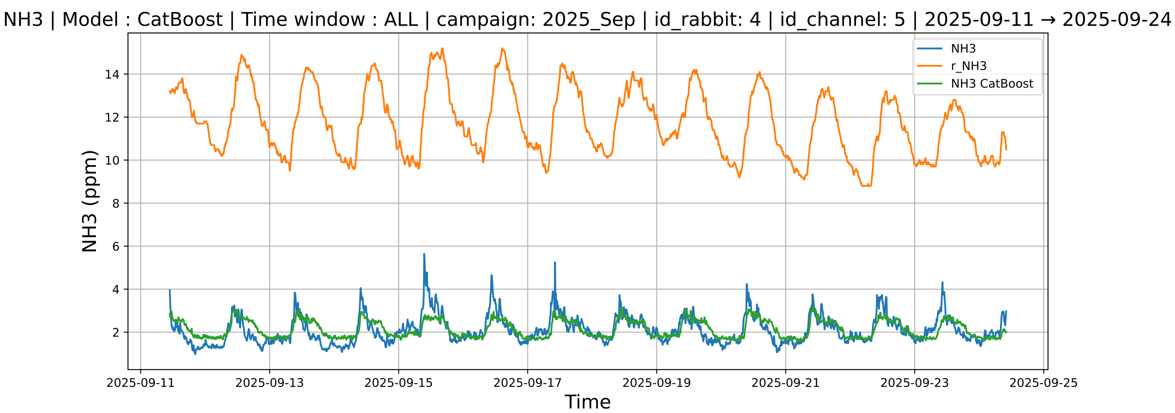
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.1.7.3 Time window : ALL

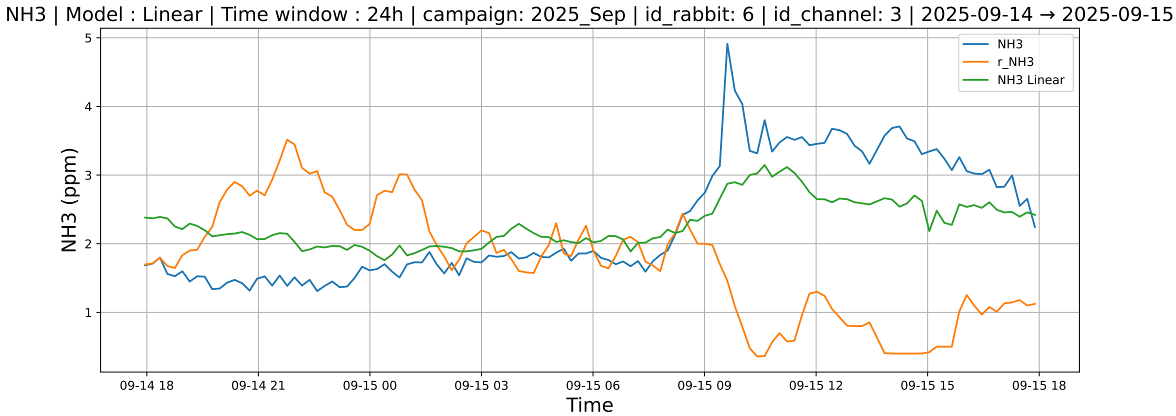


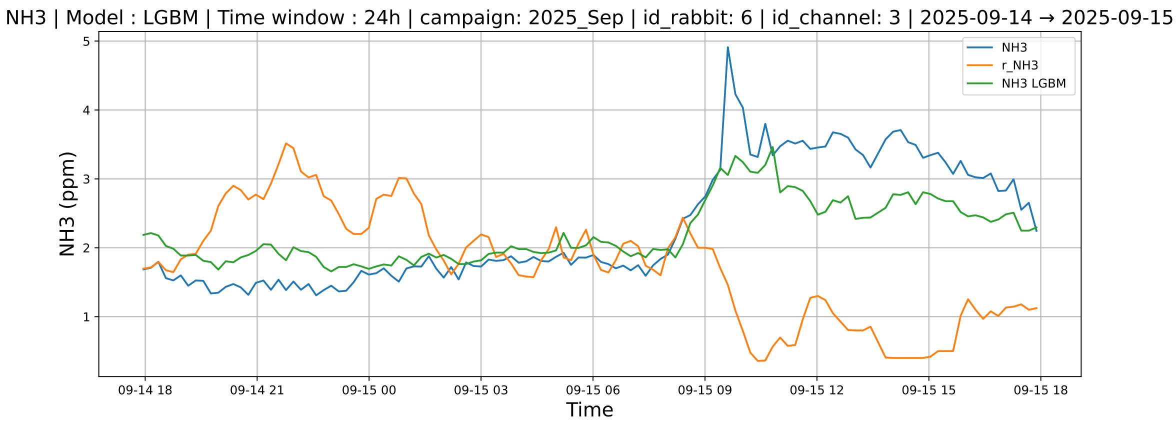


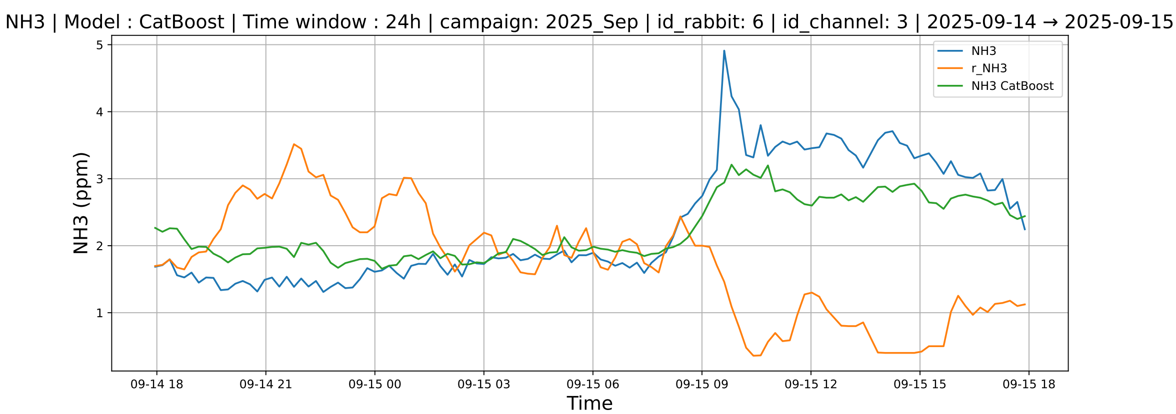


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

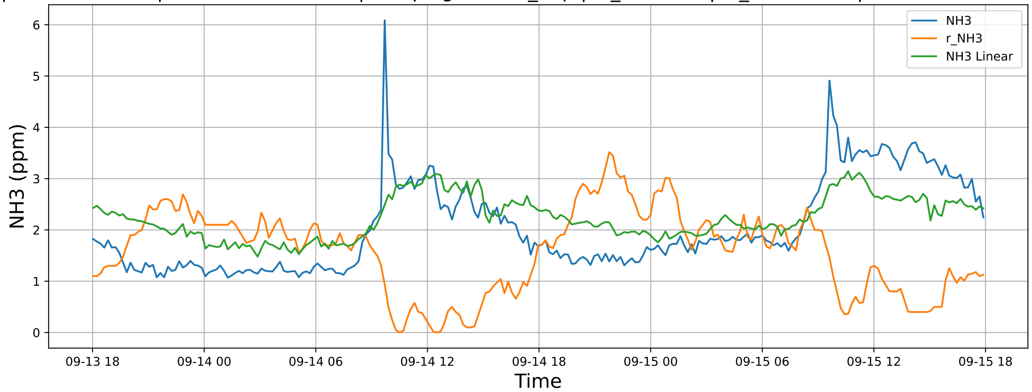




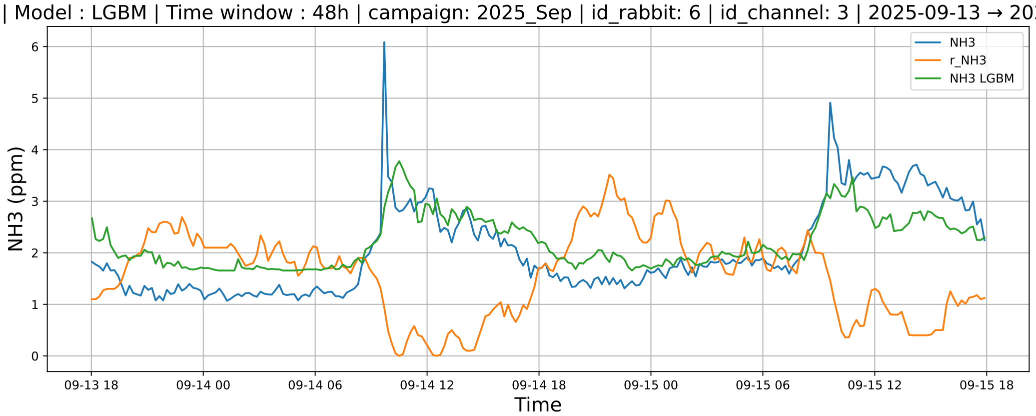


4.1.8.2 Time window : 48

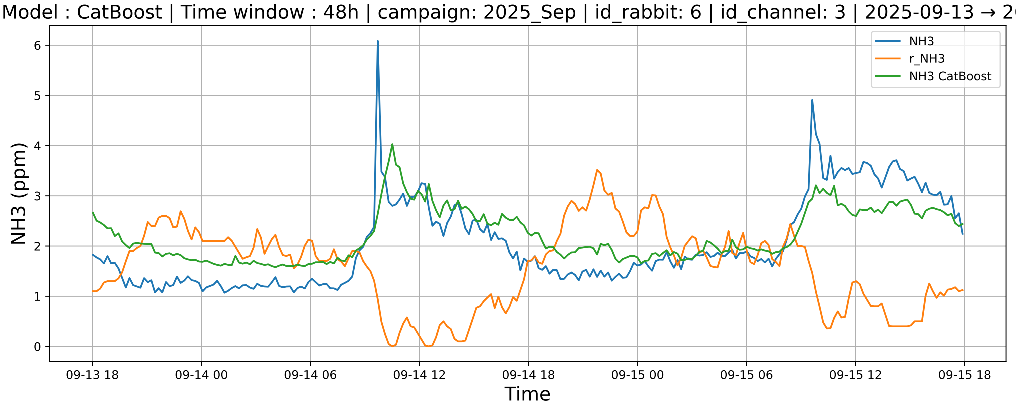
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

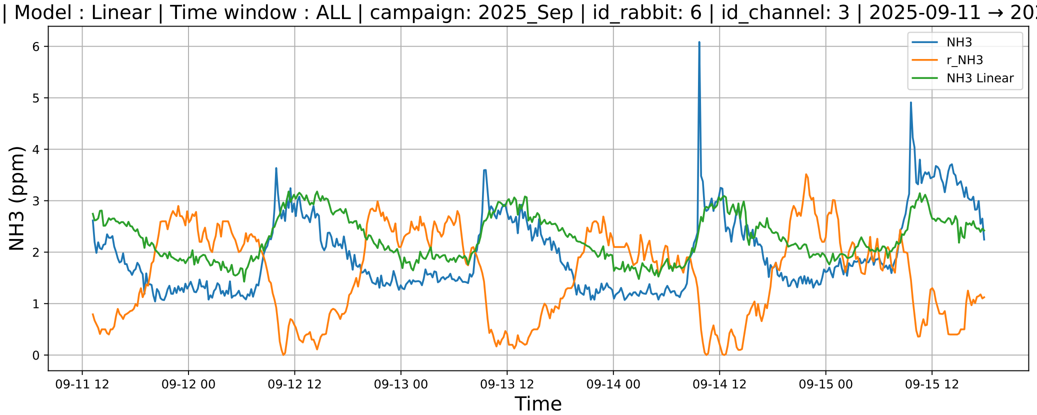


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

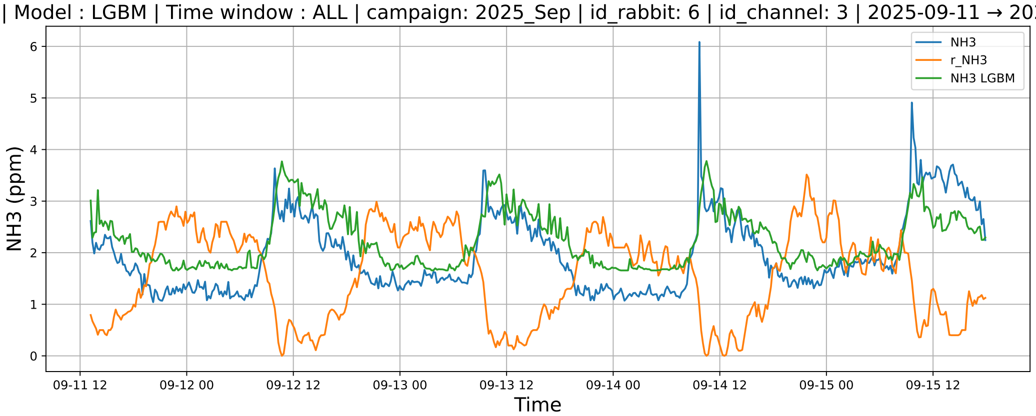


4.1.8.3 Time window : ALL

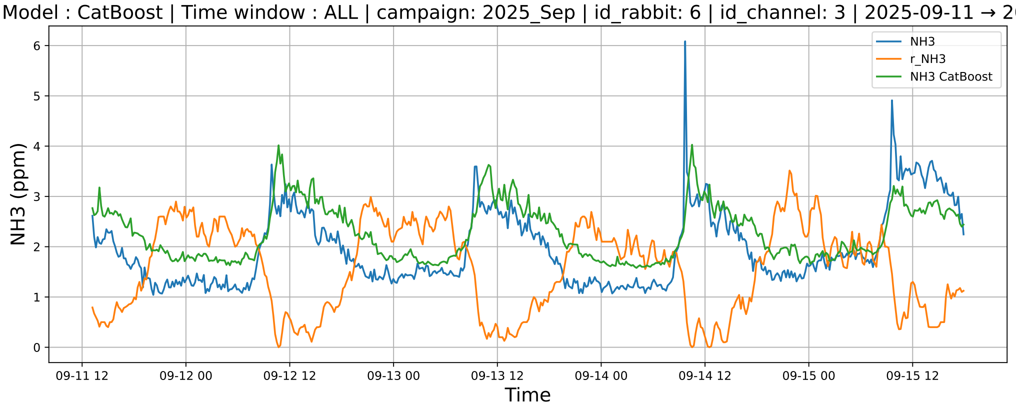
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



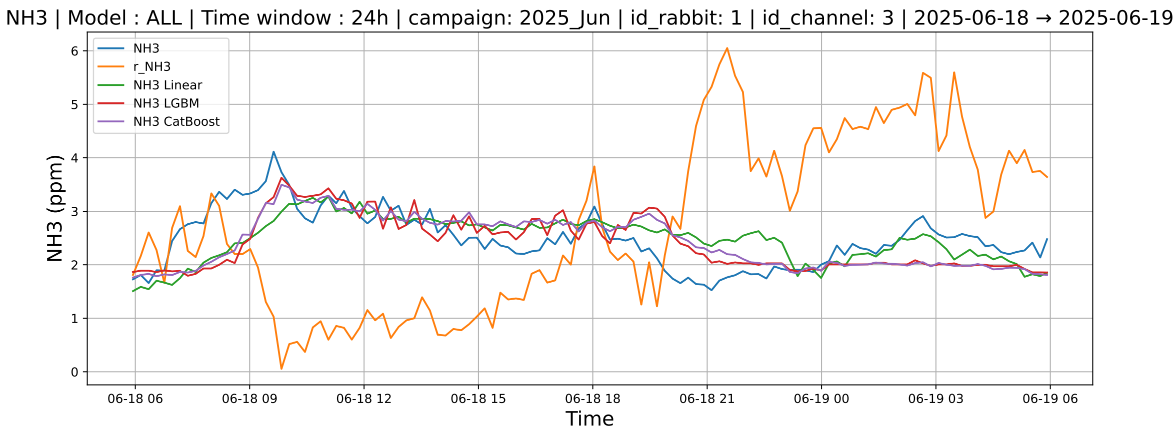
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



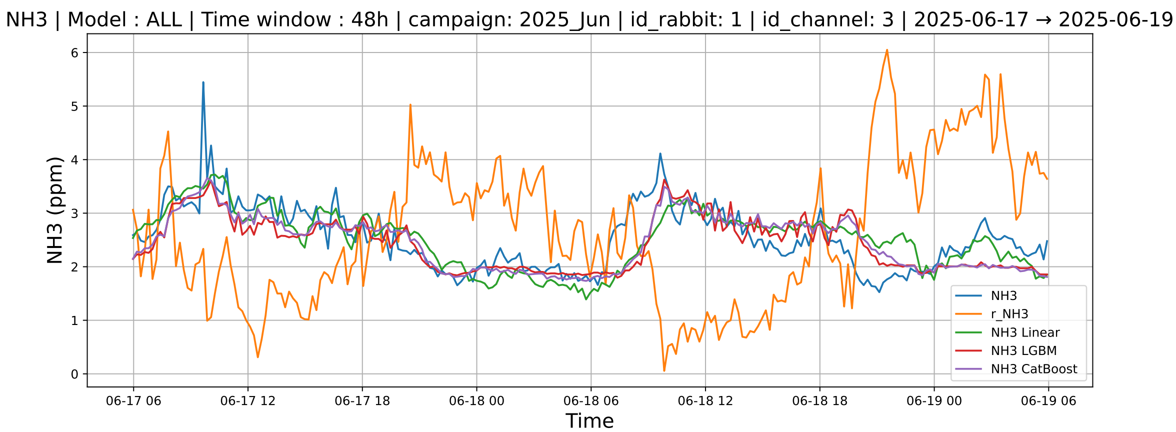
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

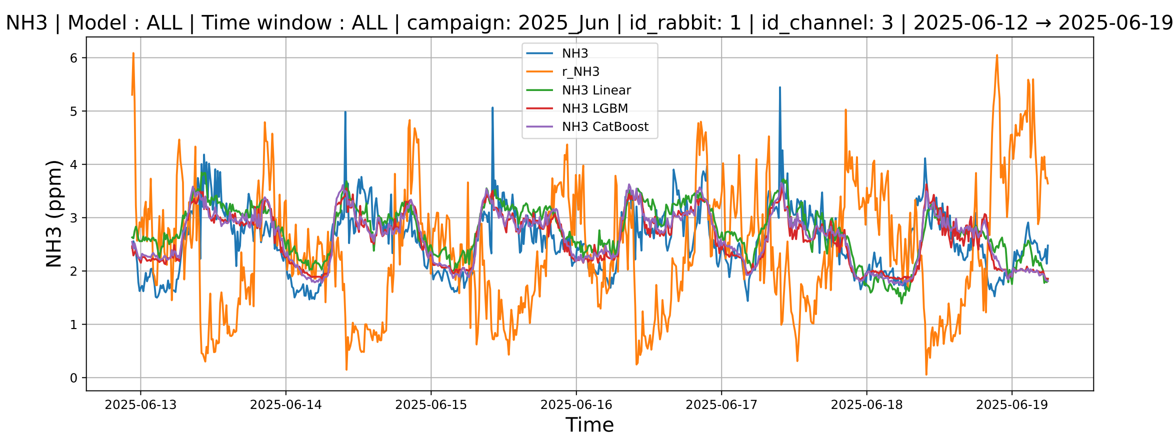
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48



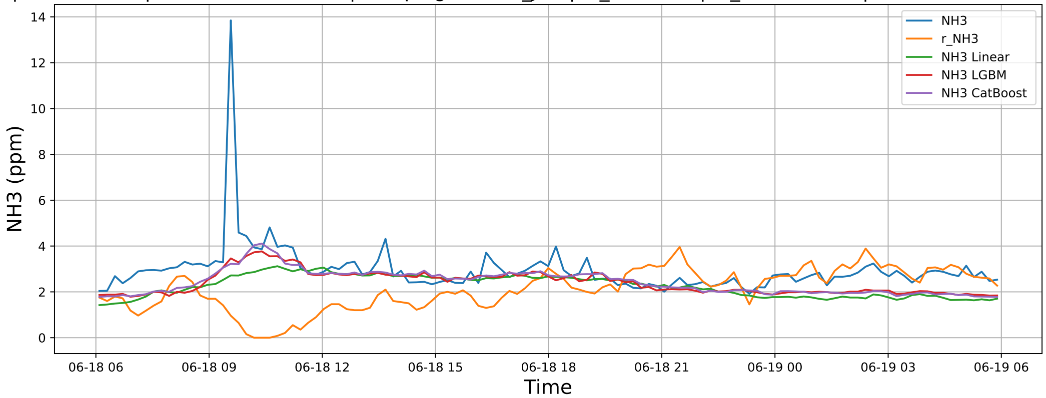
4.2.1.3 Time window : ALL



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

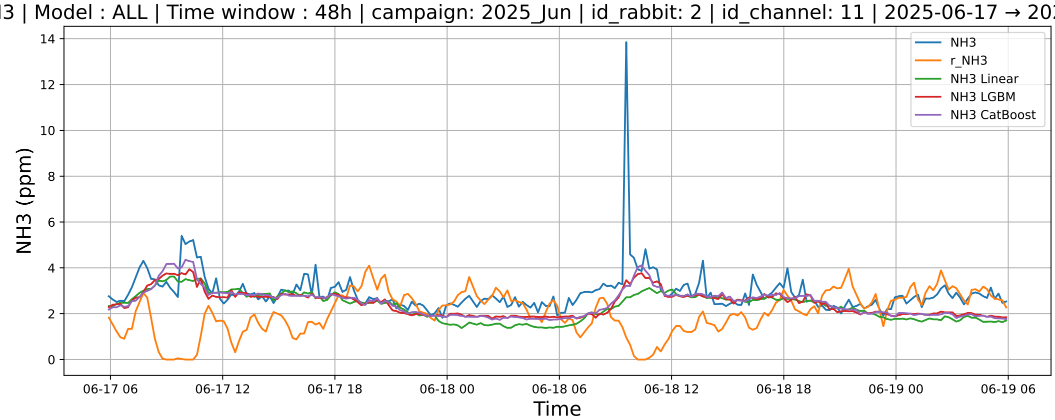
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

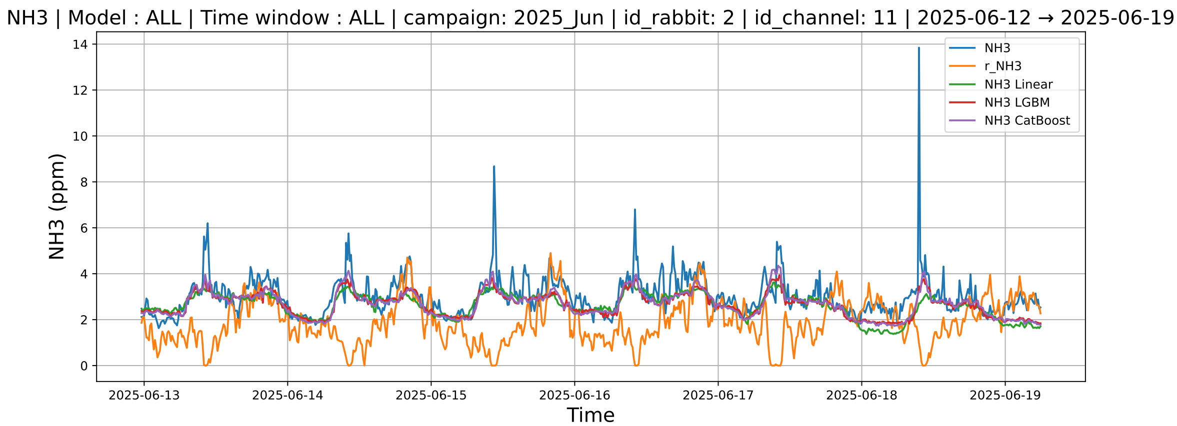


4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

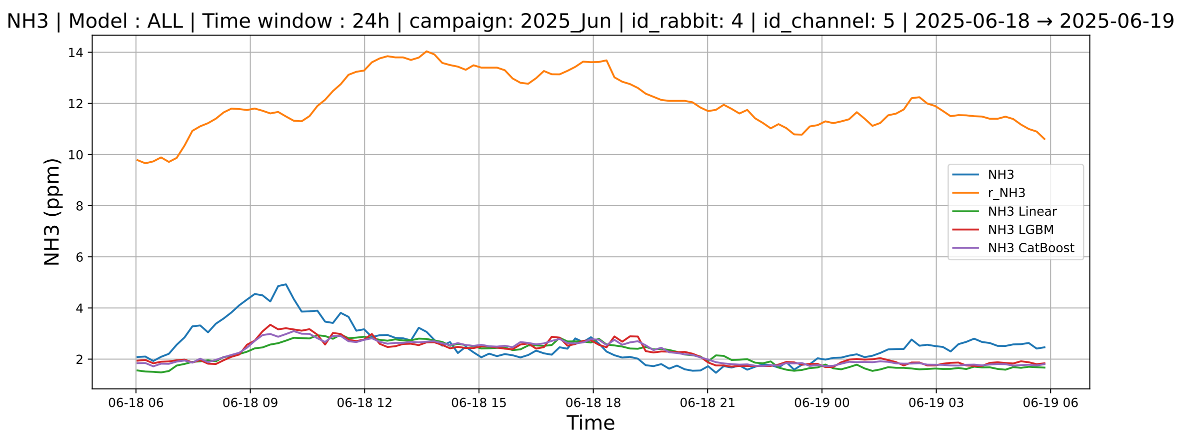


4.2.2.3 Time window : ALL

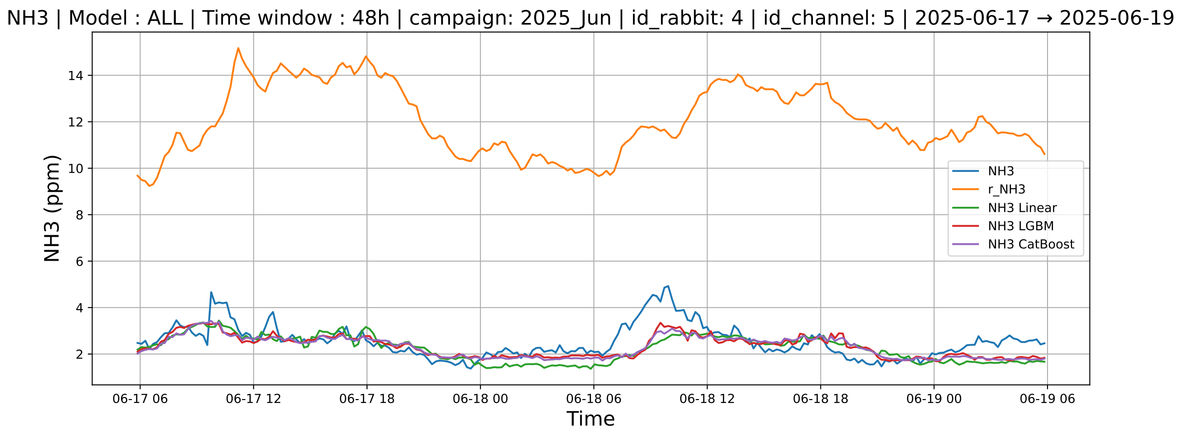


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

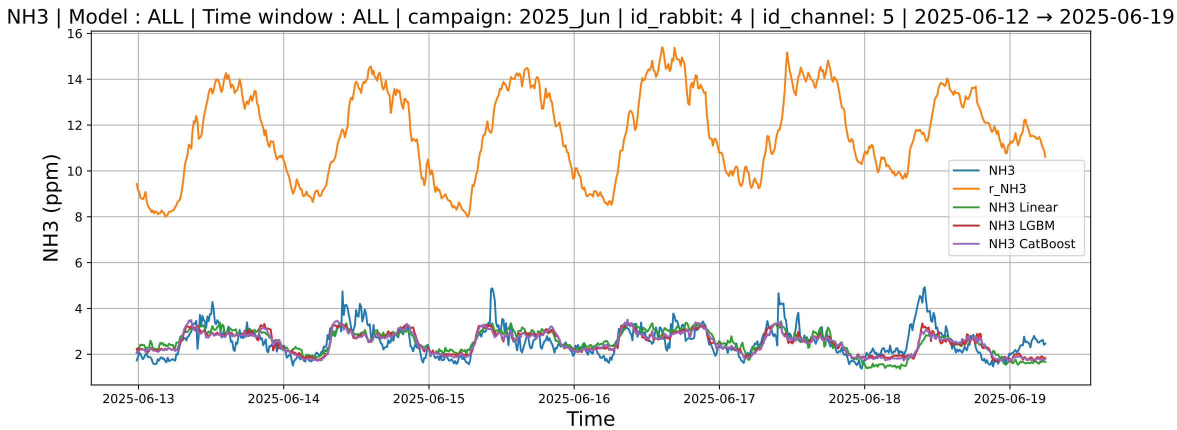
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

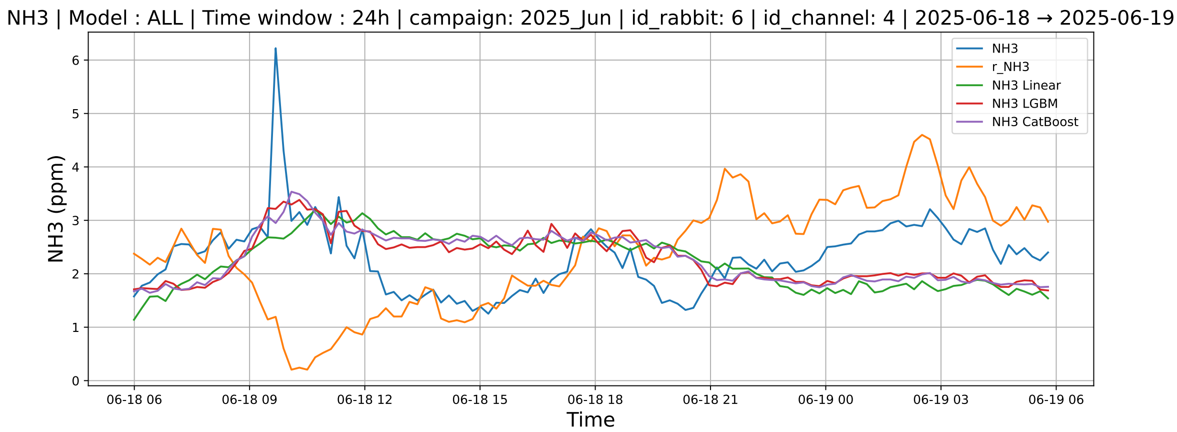


4.2.3.3 Time window : ALL

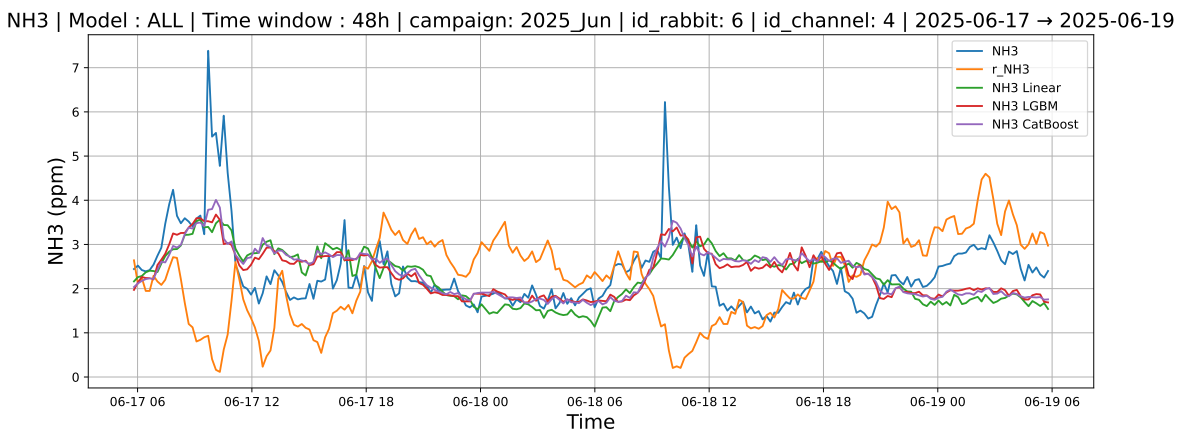


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

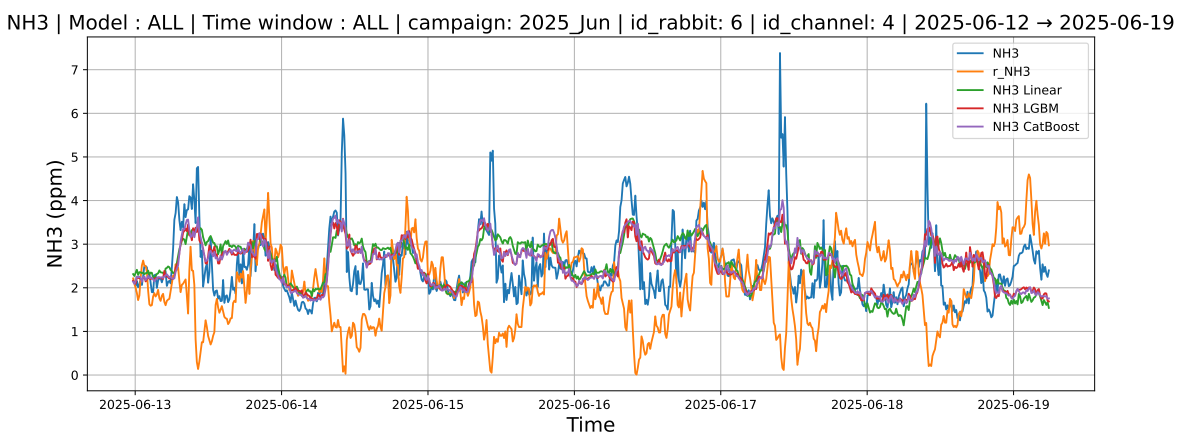
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



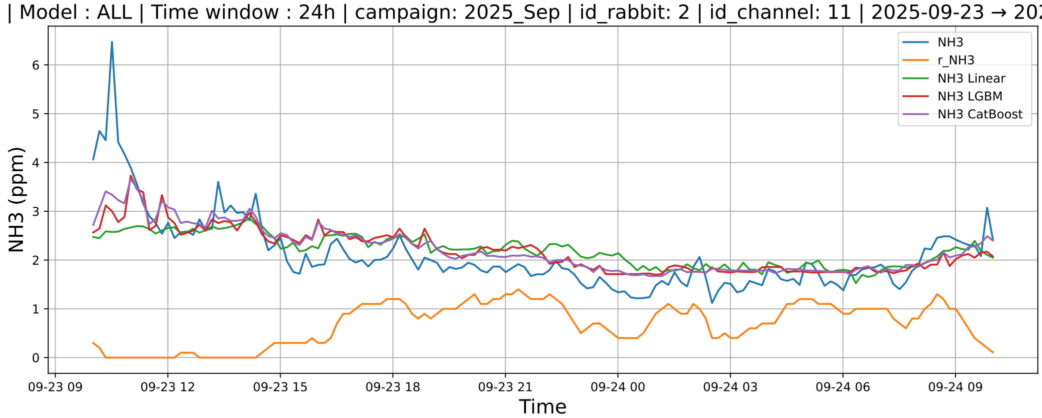
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

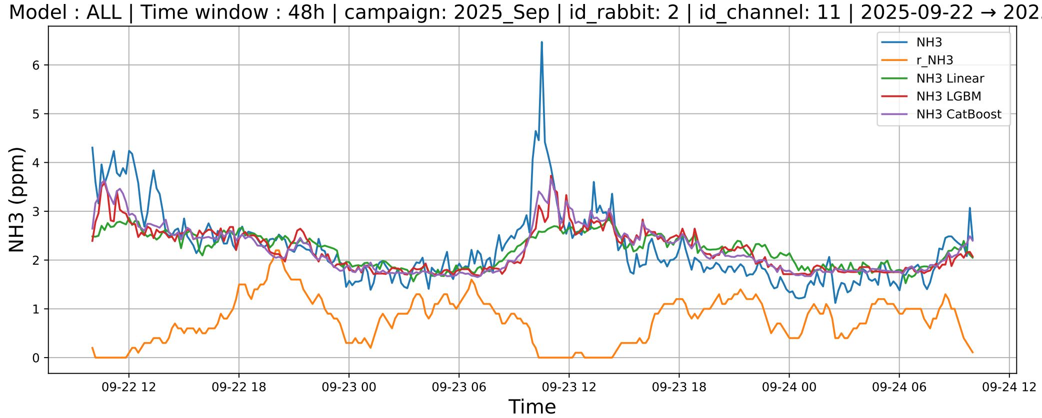
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

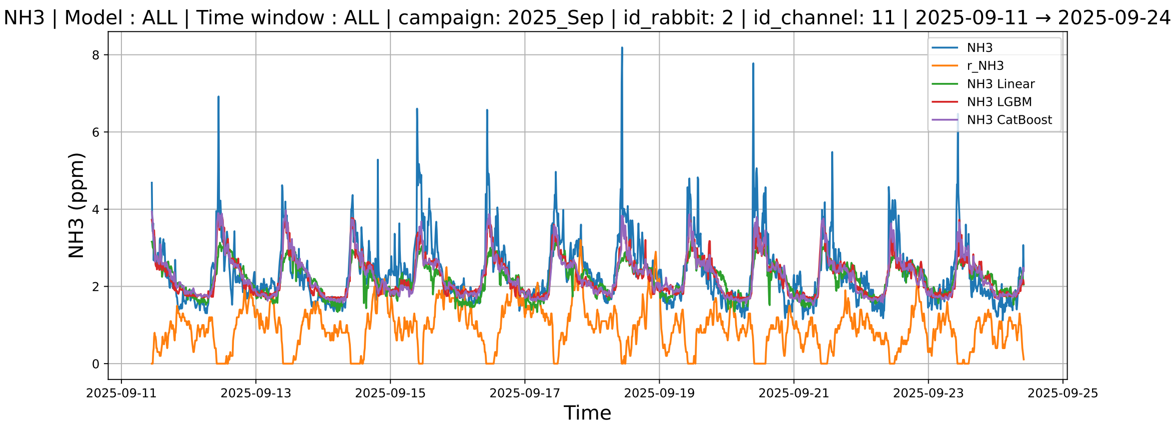


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

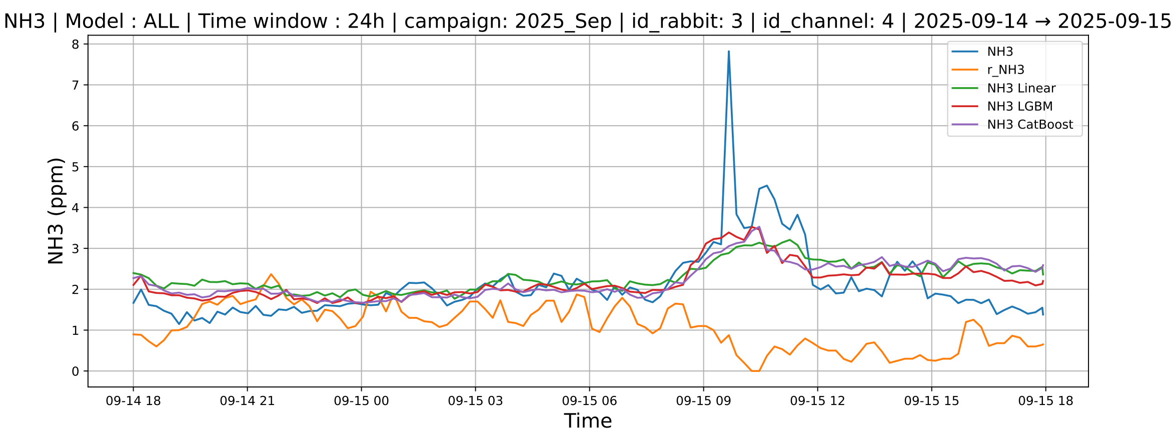


4.2.5.3 Time window : ALL

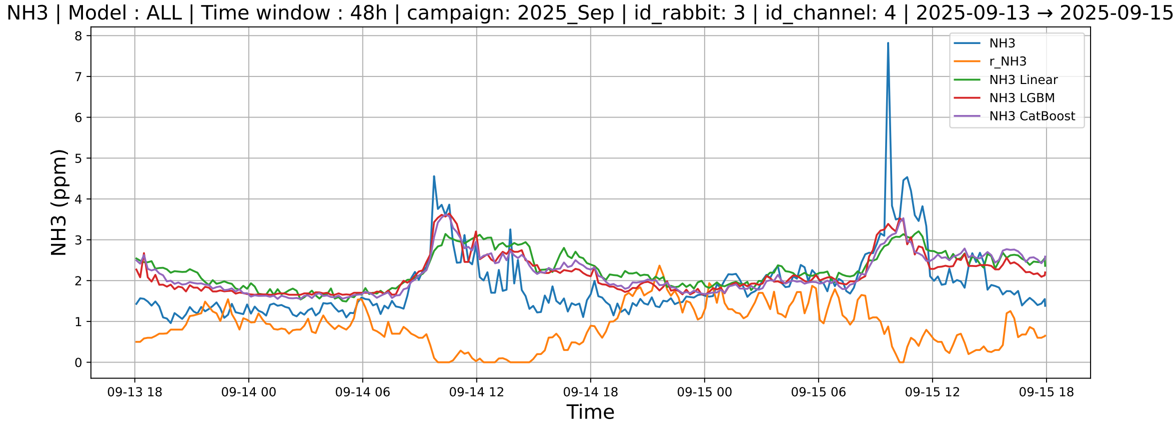


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

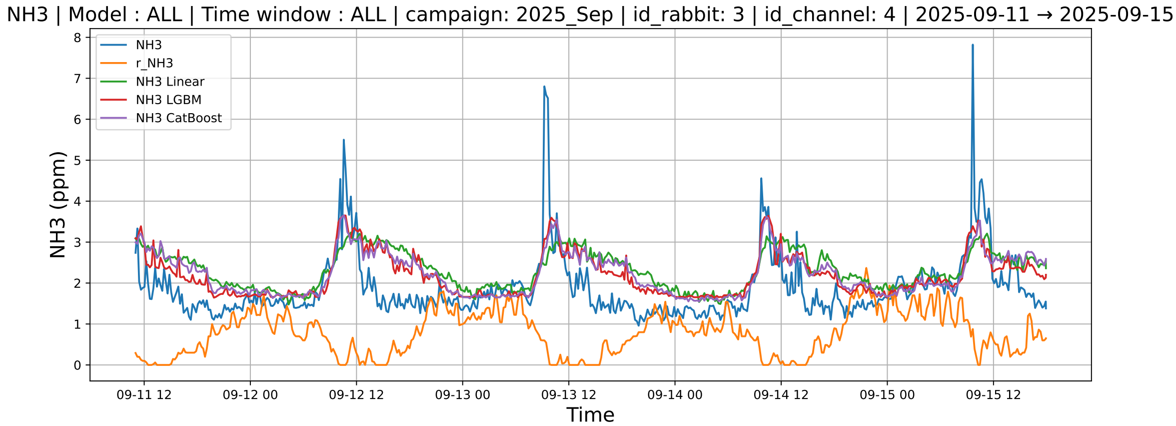
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48

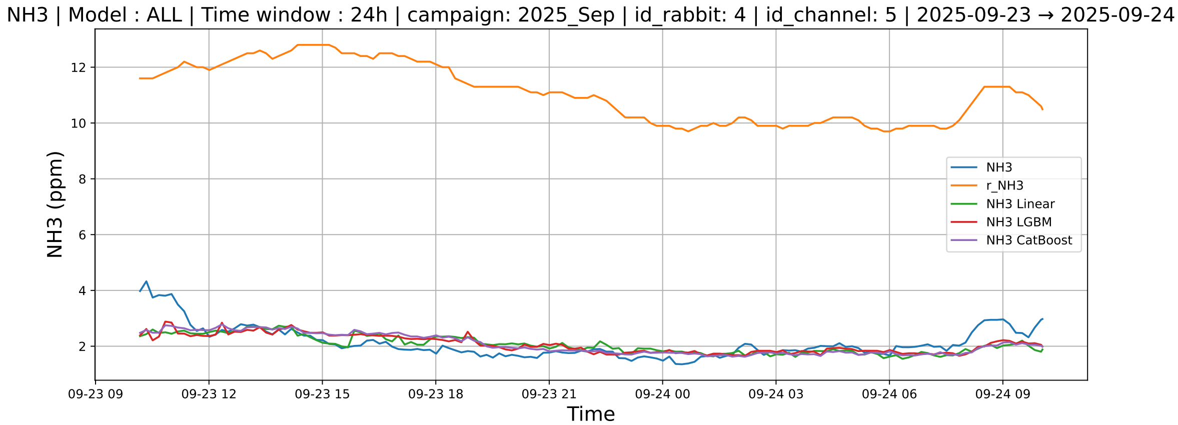


4.2.6.3 Time window : ALL

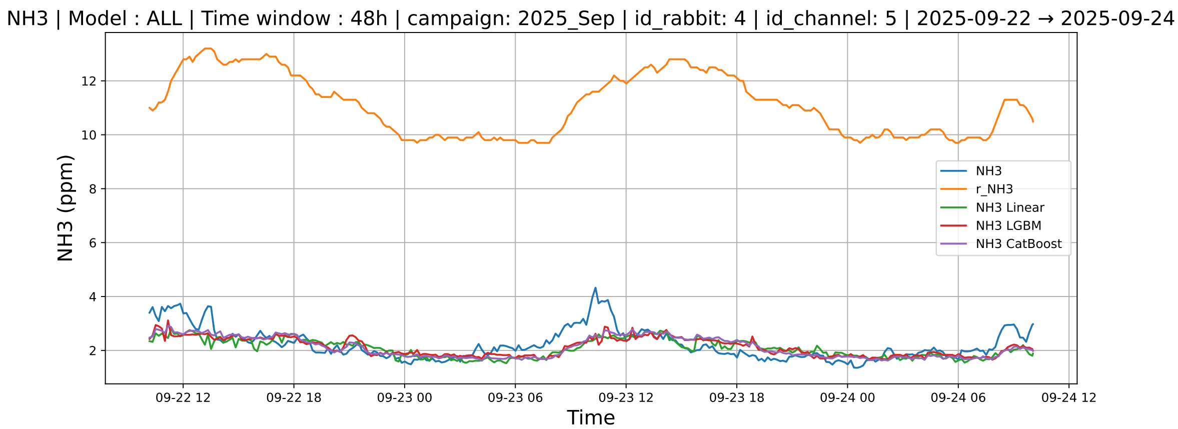


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

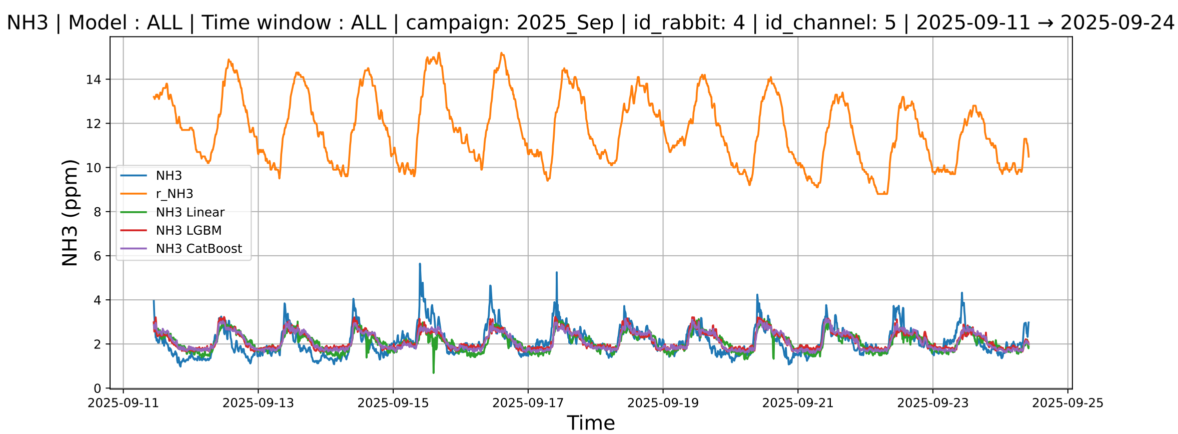
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48

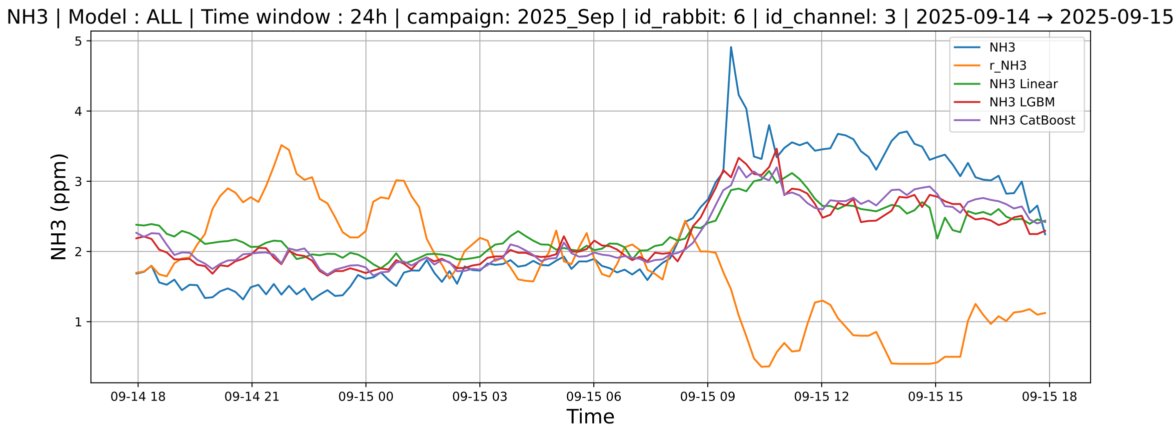


4.2.7.3 Time window : ALL

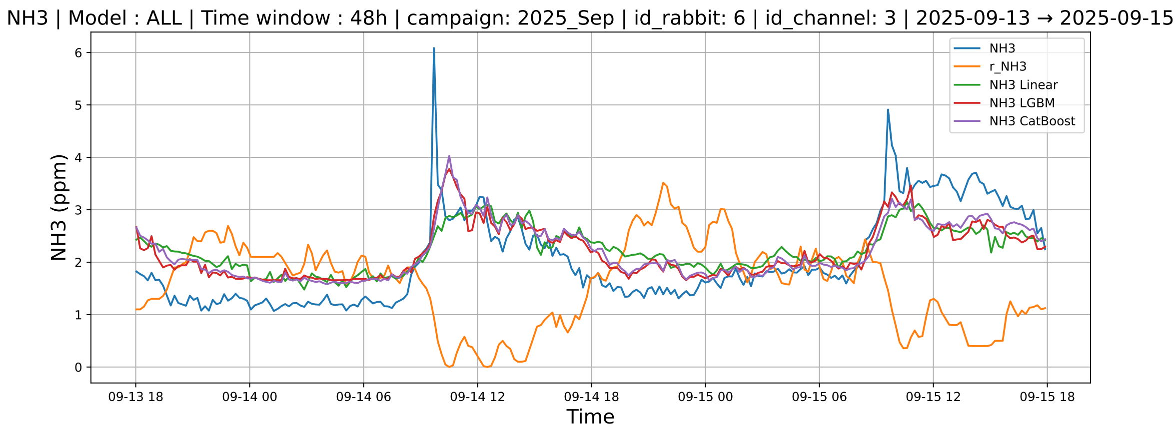


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

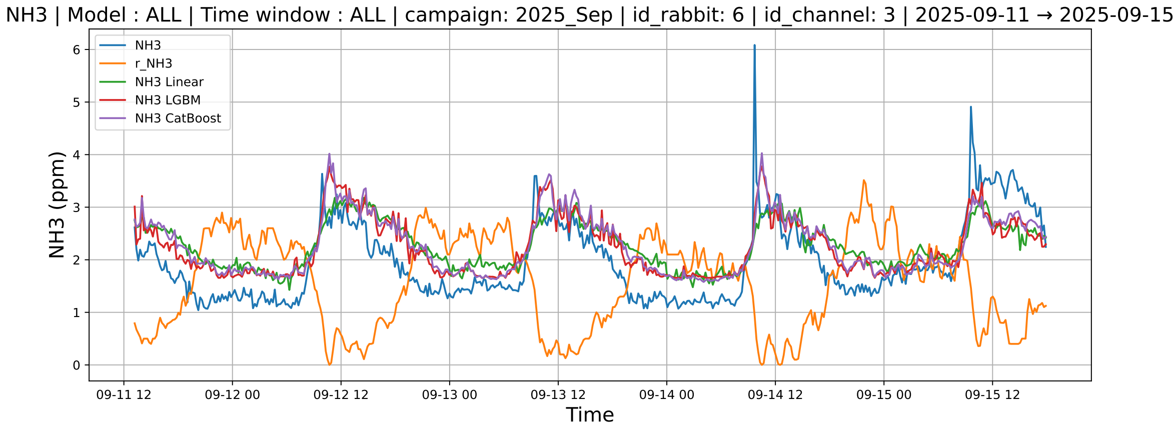
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12
Time